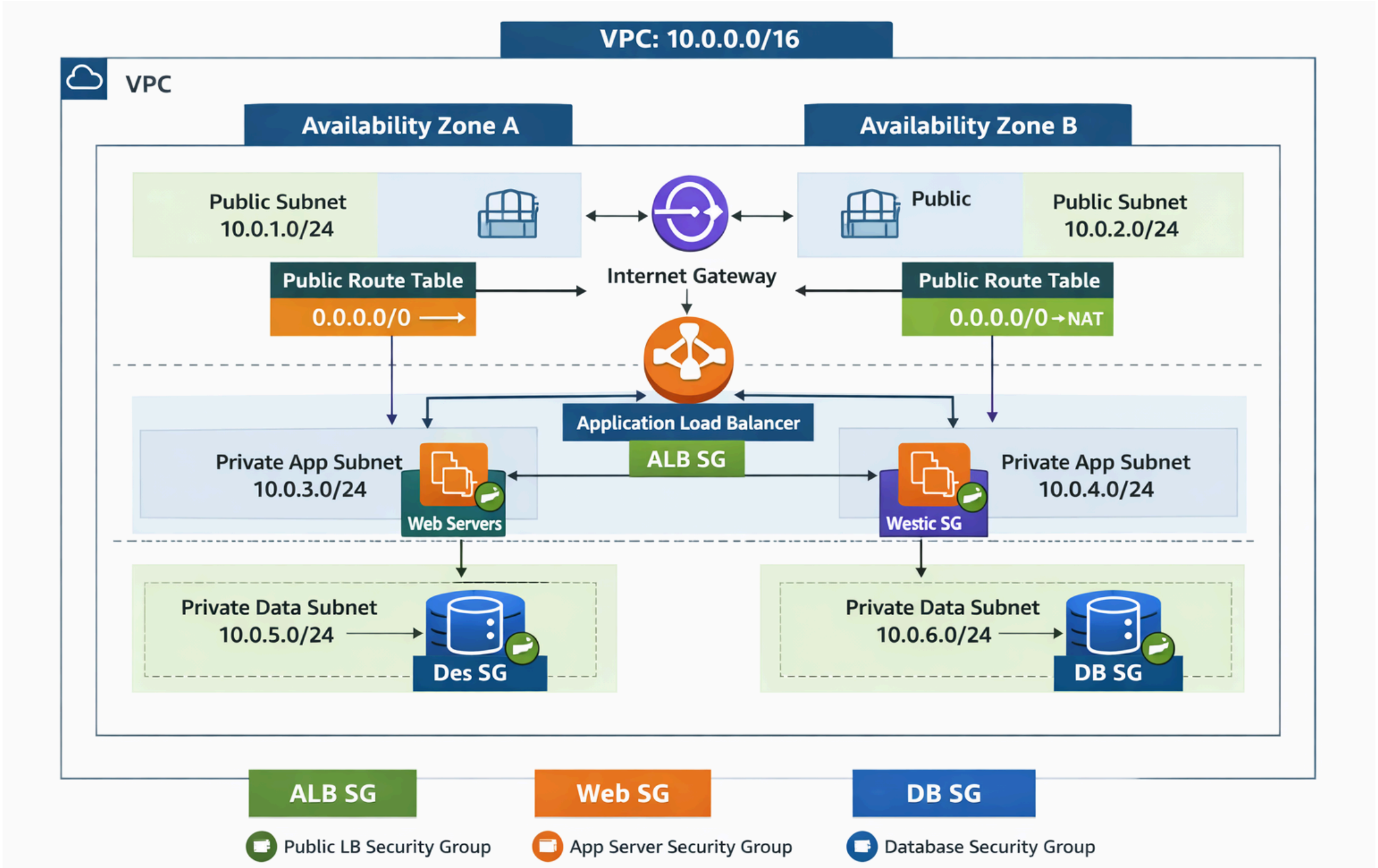
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AWS Architecture of the project



Description of the project

🏗️ The Architectural Design: "The Resilient Stack"

We aren't just building one subnet; we are building a redundant system across two **Availability Zones (AZs)** so that if one data center fails, the application stays online.

The Blueprint

- **VPC:** `10.0.0.0/16`
- **Public Subnets (AZ-A & AZ-B):** For the Application Load Balancer (ALB).
- **Private App Subnets (AZ-A & AZ-B):** For the EC2 Web Servers (The Logic).
- **Private Data Subnets (AZ-A & AZ-B):** For the RDS Database (The Data).

🛠️ Step 1: Building the Infrastructure (The "Skeleton")

1. Create the VPC

- **CIDR:** `10.0.0.0/16`
- **DNS Hostnames:** Enabled (Required for some AWS services to communicate).

2. Create 6 Subnets

To ensure high availability, you must pair them across two AZs:

- **Public-1** (`10.0.1.0/24`) in AZ-A | **Public-2** (`10.0.2.0/24`) in AZ-B
- **Private-App-1** (`10.0.3.0/24`) in AZ-A | **Private-App-2** (`10.0.4.0/24`) in AZ-B
- **Private-Data-1** (`10.0.5.0/24`) in AZ-A | **Private-Data-2** (`10.0.6.0/24`) in AZ-B

🛠️ Step 2: Routing Logic (The "Nervous System")

Architects separate traffic using multiple Route Tables.

1. **Internet Gateway (IGW):** Attach to the VPC.
2. **Public Route Table:** * **Route:** `0.0.0.0/0` -> `IGW`.
    - **Association:** Associate with **both** Public Subnets.
3. **NAT Gateway (High Availability):**
    - Deploy **one NAT Gateway** in Public-1.

    - *Note for the Exam:* In a real production environment, you'd put one in each AZ for redundancy.
4. **Private Route Table:** * **Route:** `0.0.0.0/0` -> `NAT Gateway`.
    - **Association:** Associate with the **App** and **Data** subnets.

🛠️ Step 3: Security & Load Balancing (The "Defensive Layers")

1. The Security Group "Chain"

This is a key Architect concept: **Security Groups should reference other Security Groups, not IP addresses.**

- **ALB-SG:** Allow `HTTP (80)` from `0.0.0.0/0`.
- **Web-Server-SG:** Allow `HTTP (80)` only from **ALB-SG**.
- **DB-SG:** Allow `SQL (3306)` only from **Web-Server-SG**.

Step 4: Application Load Balancer (ALB)

- **Scheme:** Internet-facing.
- **Listeners:** Port 80.
- **Subnets:** Select **both** Public Subnets.
- **Target Group:** Point to your EC2 instances in the **Private App Subnets**

---.---

Step 5: The Resilience Upgrade

This phase transforms the architecture into a **self-healing, elastic system** capable of recovering from application failures and scaling during high-traffic events.

- **Launch Template:** Define a "golden image" with User Data containing a `timeout.php` script (70s sleep) to simulate app hangs and `stress-ng` for load testing.
- **Auto Scaling Group (ASG):** Deploy instances across **Private App Subnets** and switch to **ELB Health Checks**. This ensures the ASG replaces nodes returning **504 Gateway Timeouts**, as standard EC2 checks only monitor hardware power.
- **Target Tracking Policy:** Implement a scaling policy (e.g., 50% CPU threshold) to automatically increase instance count when the ALB faces heavy web requests.
- **Resilience Test:** Trigger `/timeout.php` to force a 504 error; verify the ASG detects the unhealthy node and provisions a fresh replacement.
- **Stress Test:** Use `stress-ng` to spike CPU usage; observe CloudWatch alarms triggering the ASG to "scale out," adding extra instances to maintain performance under load.

.🛠️ Step 6: The Data Tier (The "Vault")
Deploy a highly available, redundant database layer that is completely isolated from the public internet:

- DB Subnet Group: Group your two Private Data Subnets (10.0.5.0/24 and 10.0.6.0/24) together in the RDS console.
- RDS Deployment: Launch a Multi-AZ RDS Instance (such as MySQL or PostgreSQL). Placing the primary database in one data subnet and the standby in the other ensures resilience against data center failures.
- Security Group Referencing: Apply the Private Database SG and configure the inbound rules to accept traffic on port 3306 exclusively from the Private App SG. This locks out the public internet and follows the principle of least privilege.

Step 7: Enhanced Security with Secrets Manager

- **Store the RDS Secret**: Create a "Credentials for RDS database" secret in AWS Secrets Manager.
- **Update IAM Policy**: Attach a policy to your existing **SSM Session Manager IAM Role** (used in Step 5) that allows `secretsmanager:GetSecretValue` for that specific secret.

Performed the following actions:

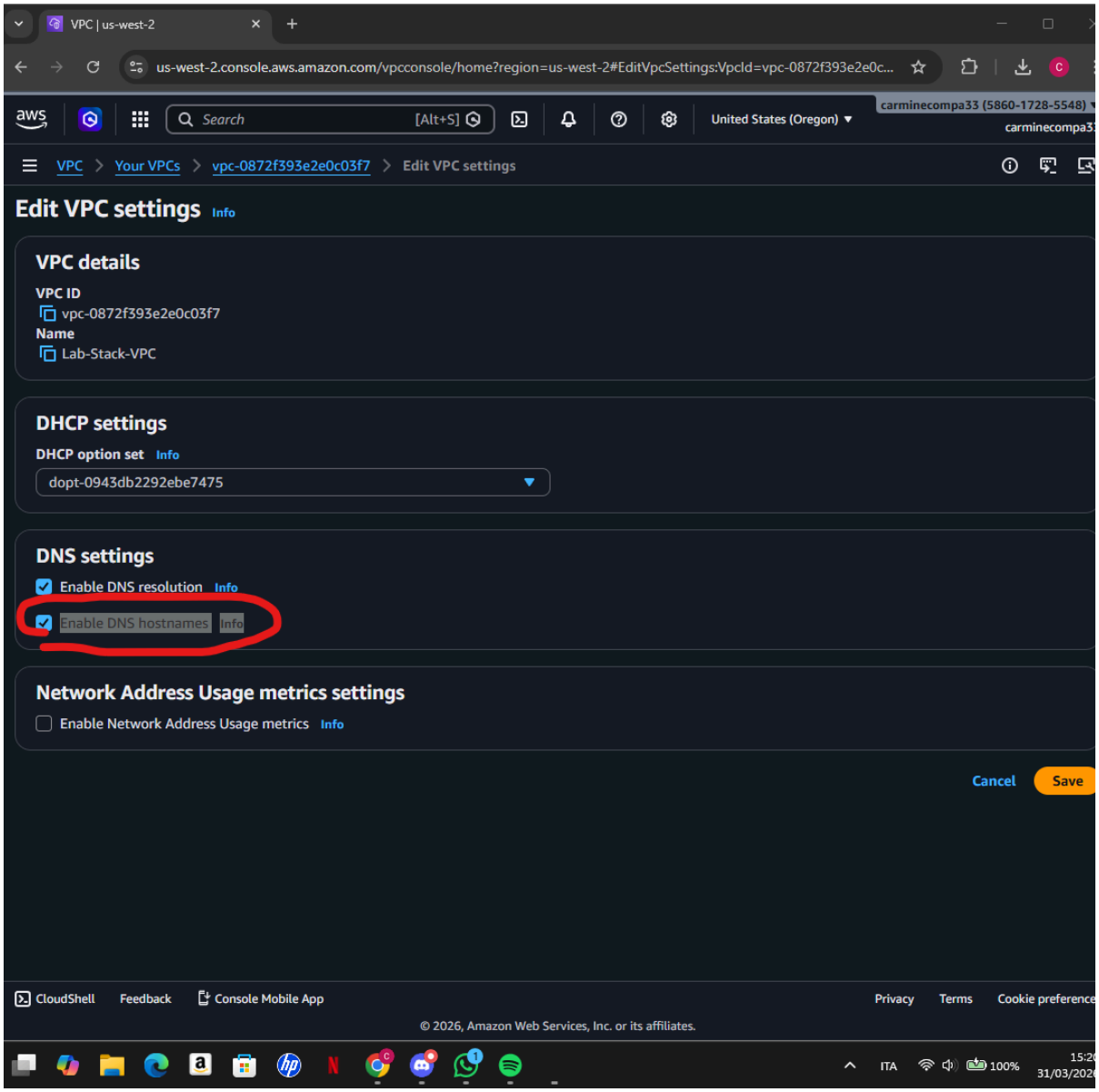
- Created an AWS-managed RDS secret.
- Attached a `secretsmanager:GetSecretValue` inline policy to the `AppServerSSMRole`.
- Updated the **Launch Template** and **ASG** to ensure all instances inherit this identity.

Step 8: save AWS Infrastructure with CloudFormation - IaC generator

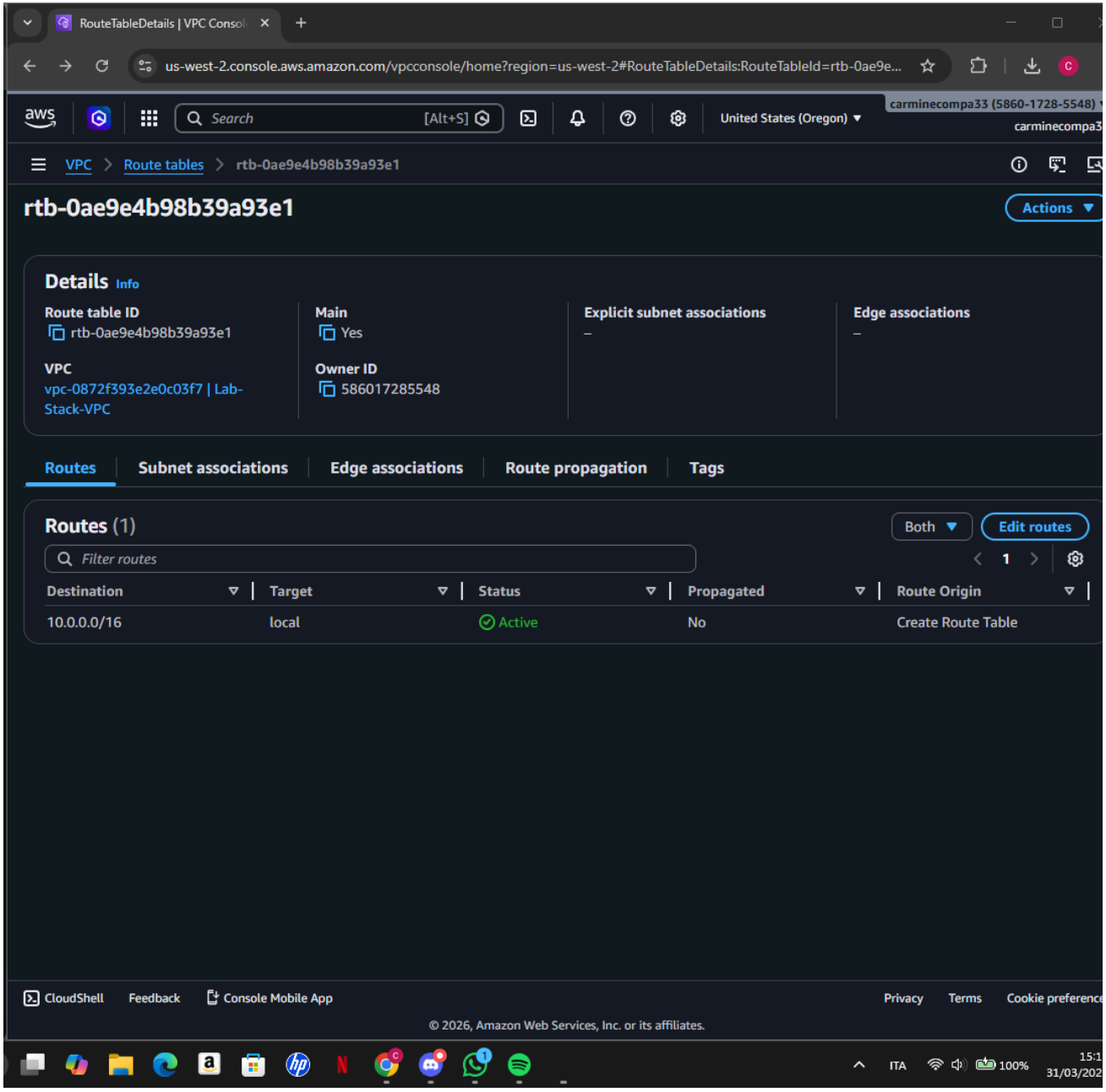
Step 1: Building the Infrastructure (The "Skeleton")

Create the VPC

Enable DNS hostnames on the VPC

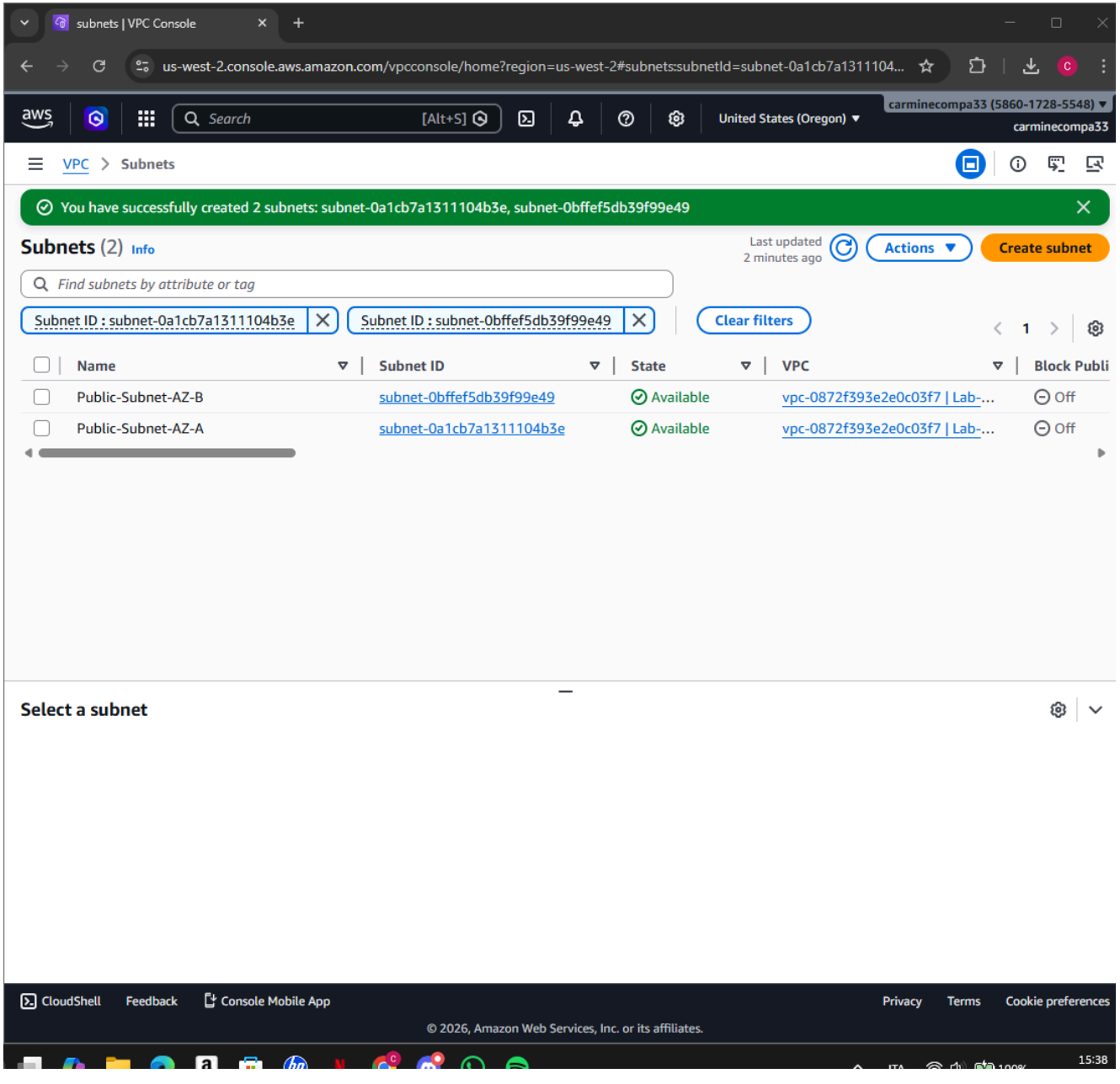


Route Table of the VPC

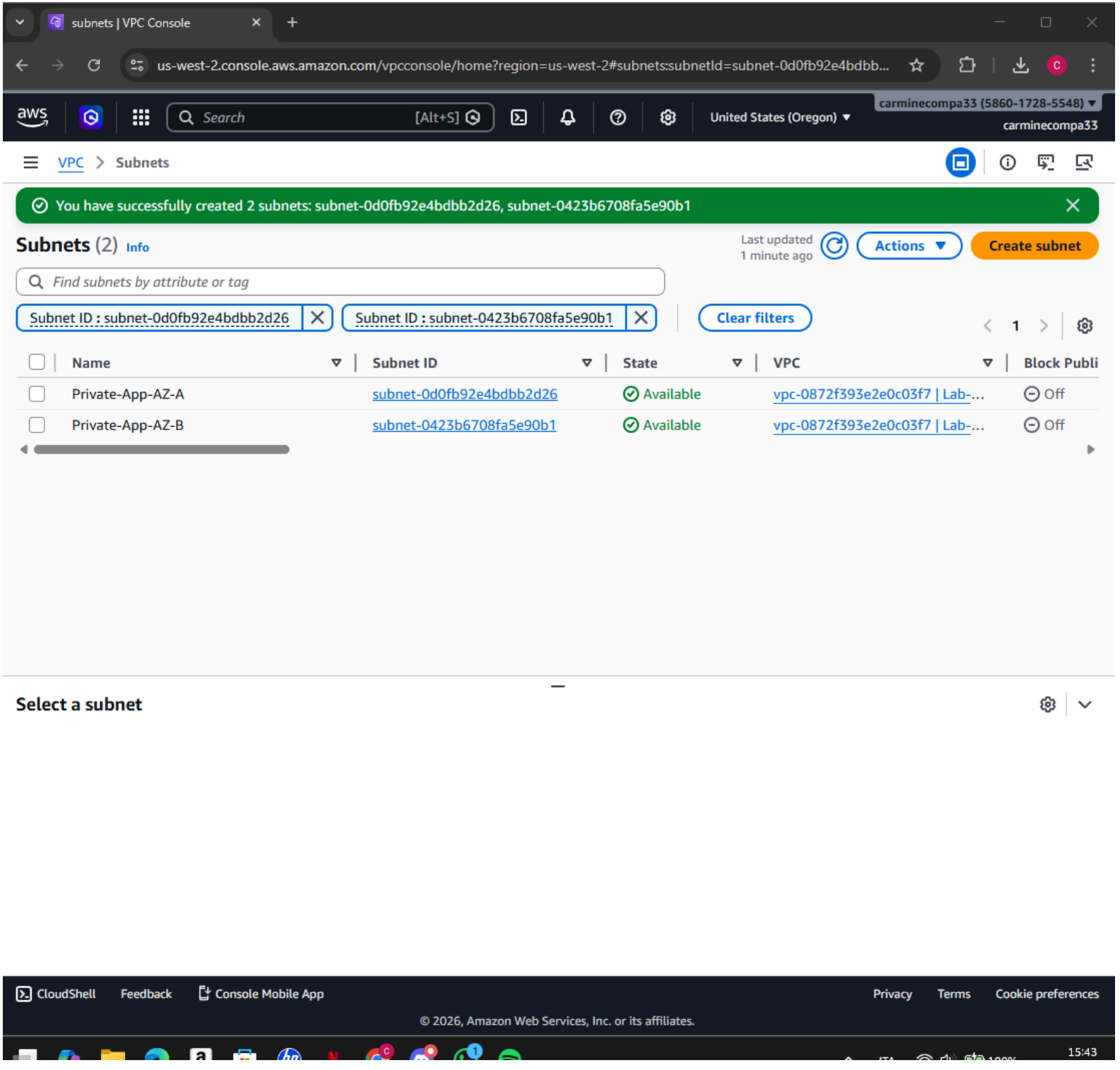


Create the Subnets

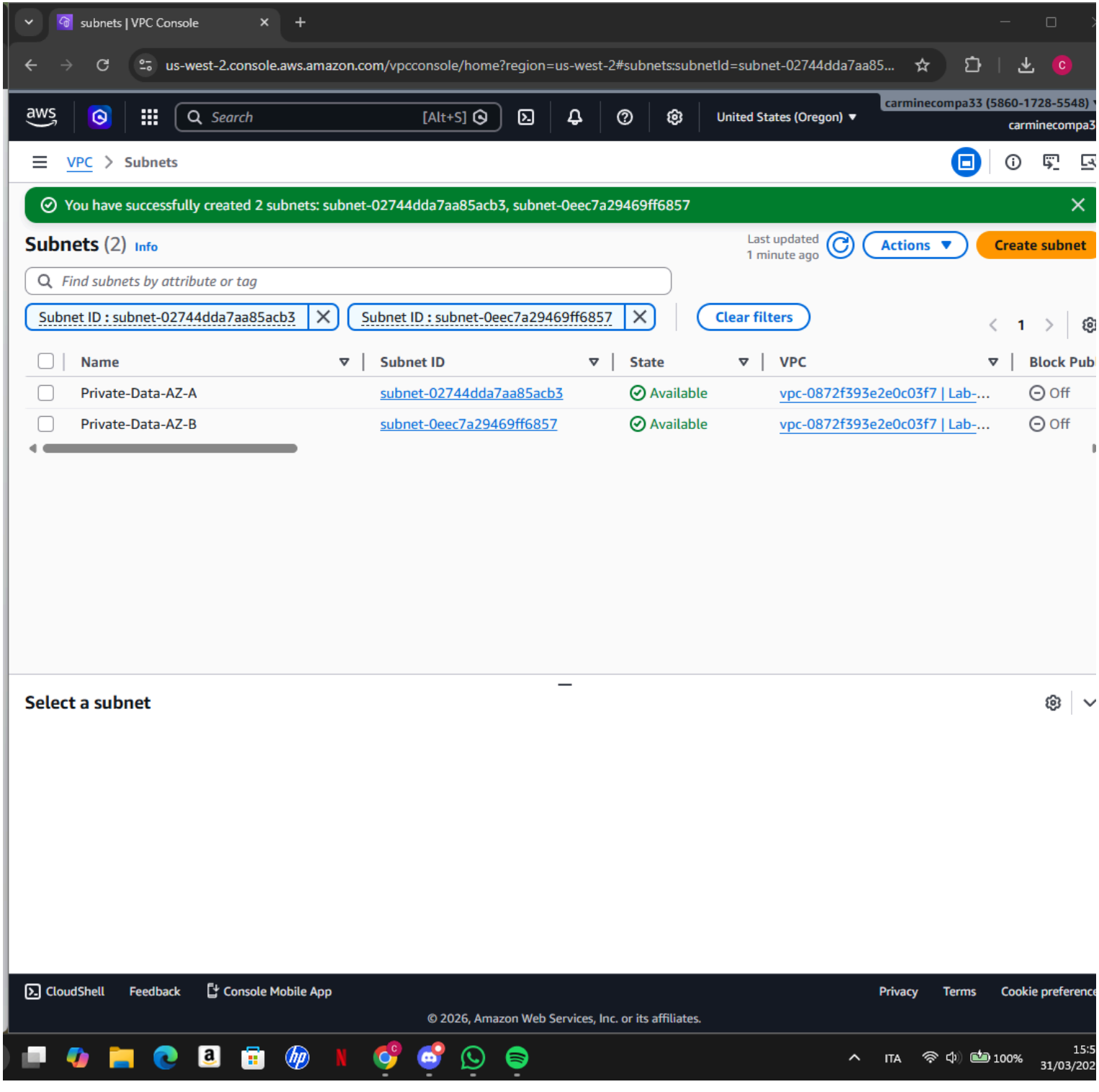
Public subnets (front door)



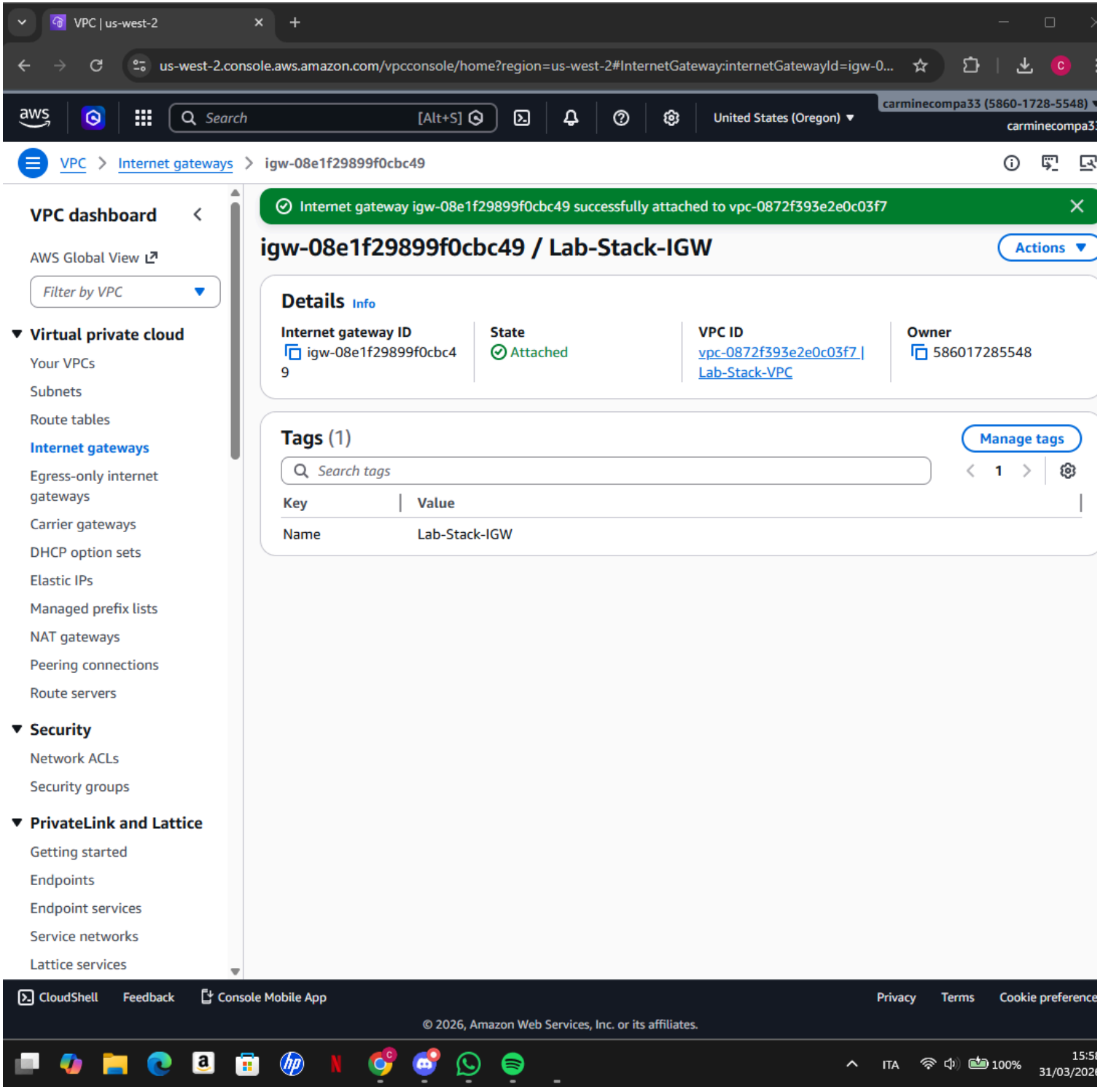
Private App Subnets (for the Servers)



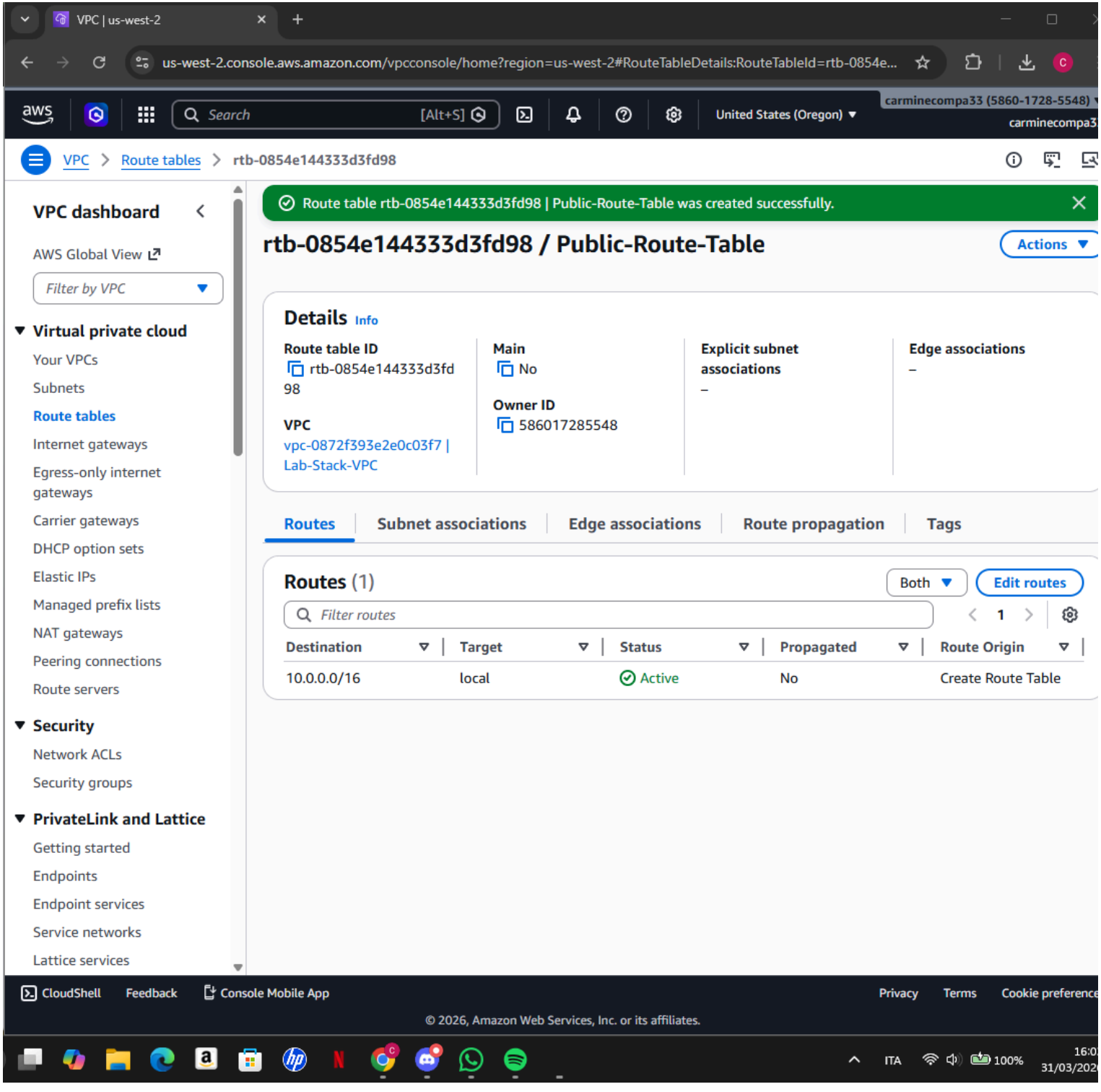
Private Data subnets (for the Databases)



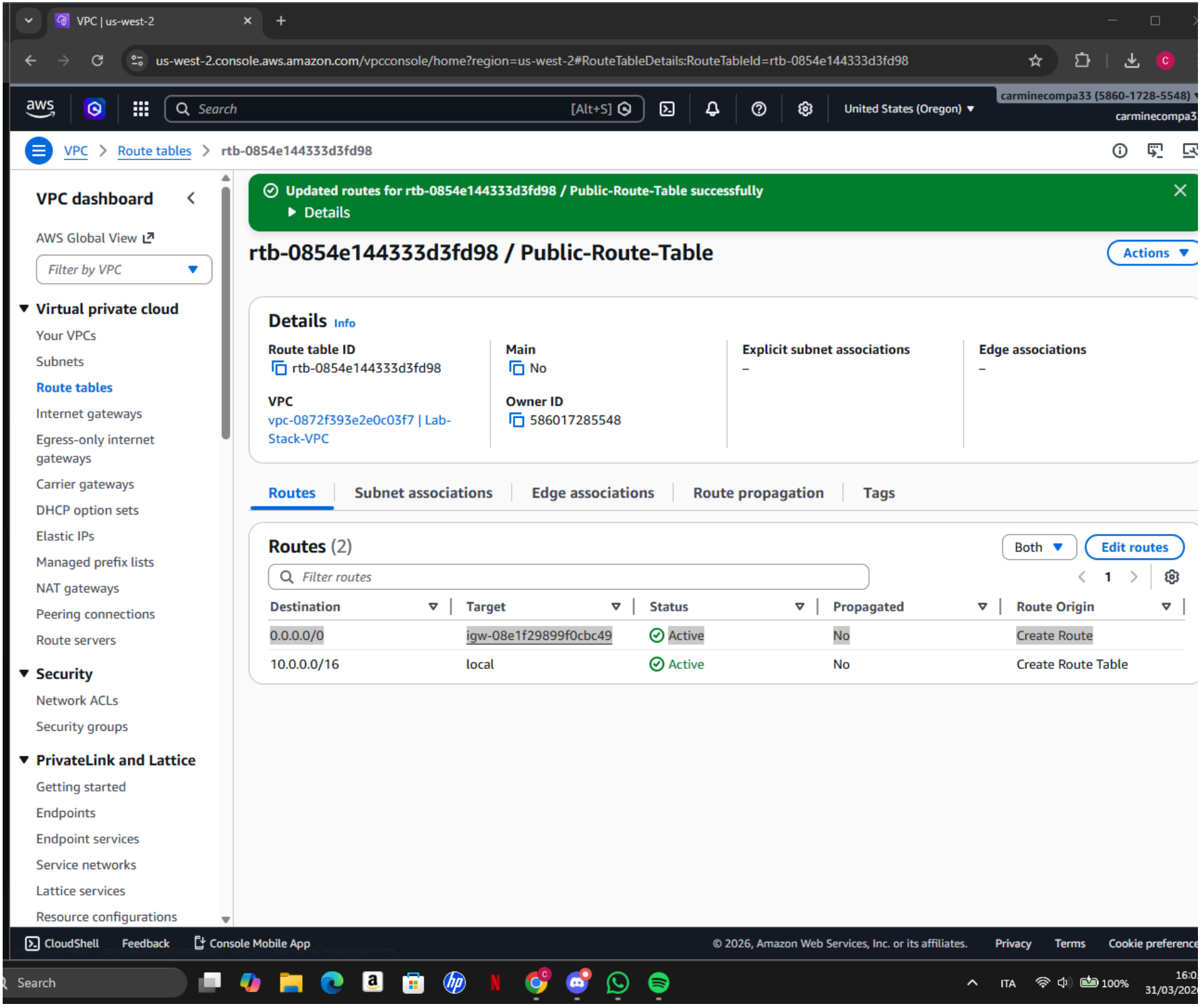
Create Internet Gateway



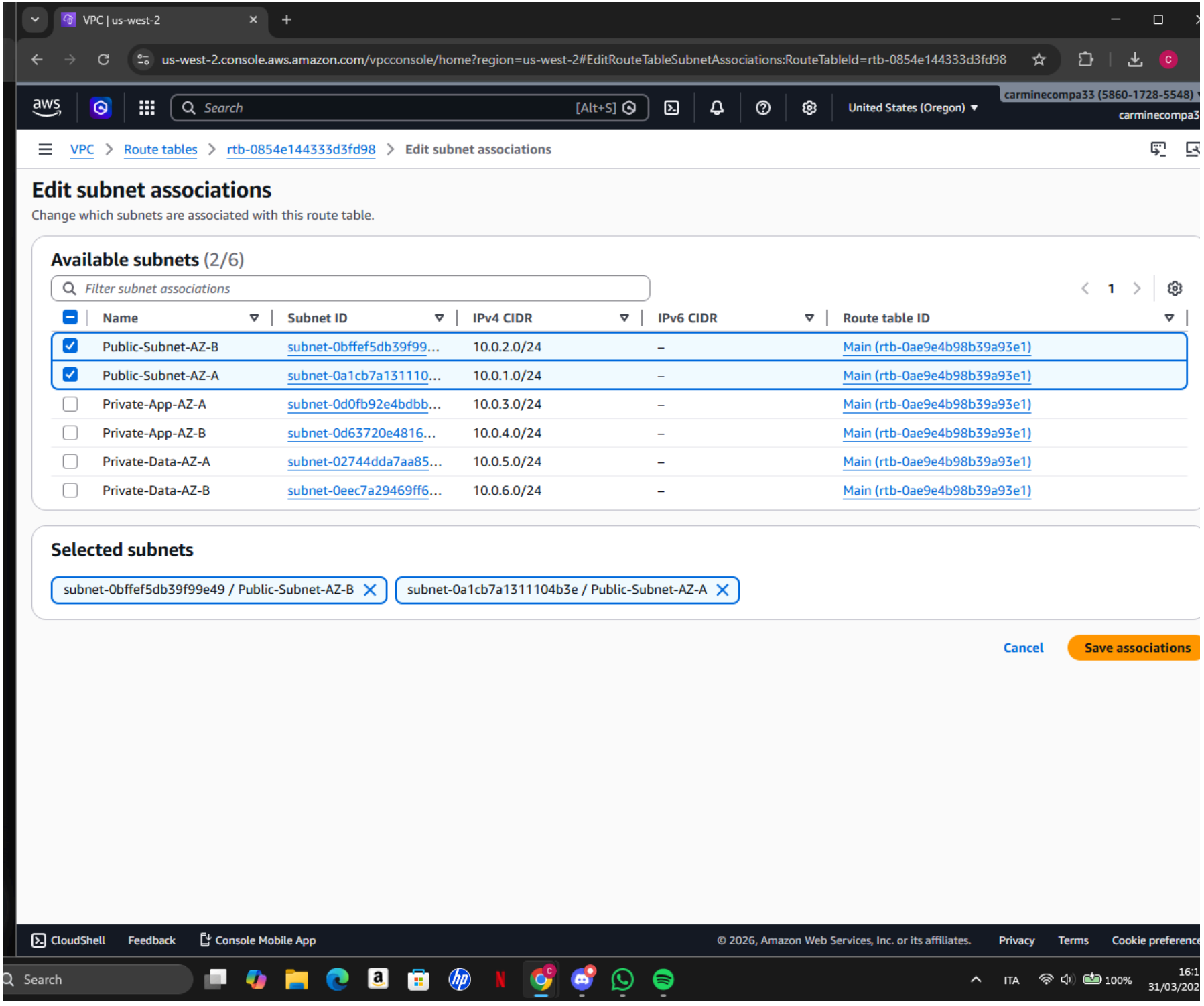
Create Route Table



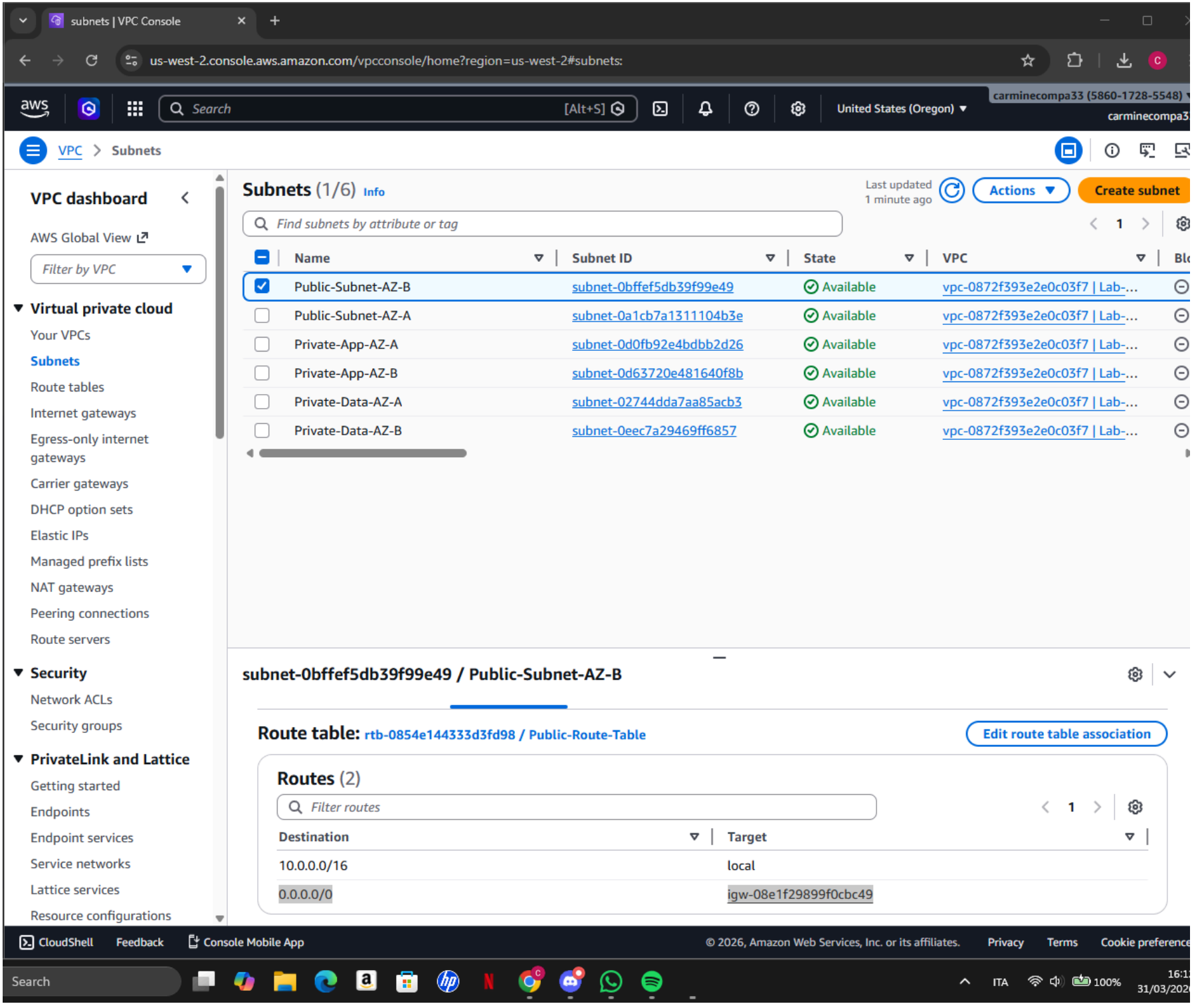
Add the Internet Route to the Route Table (OSI Layer 3: Network)



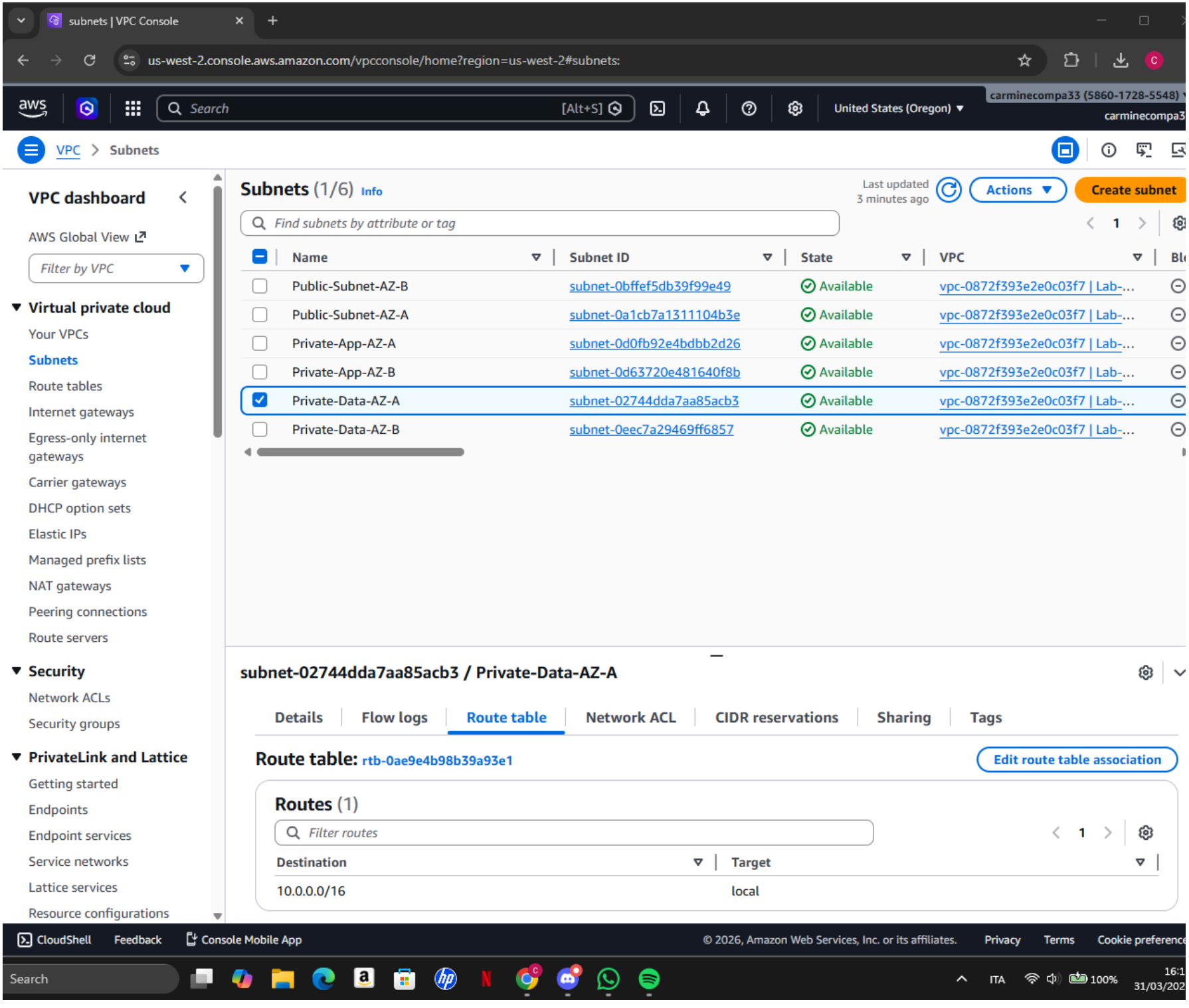
Associate Internet Gateway to the Public Subnets



Route Table route 0.0.0.0.for Public Subnets

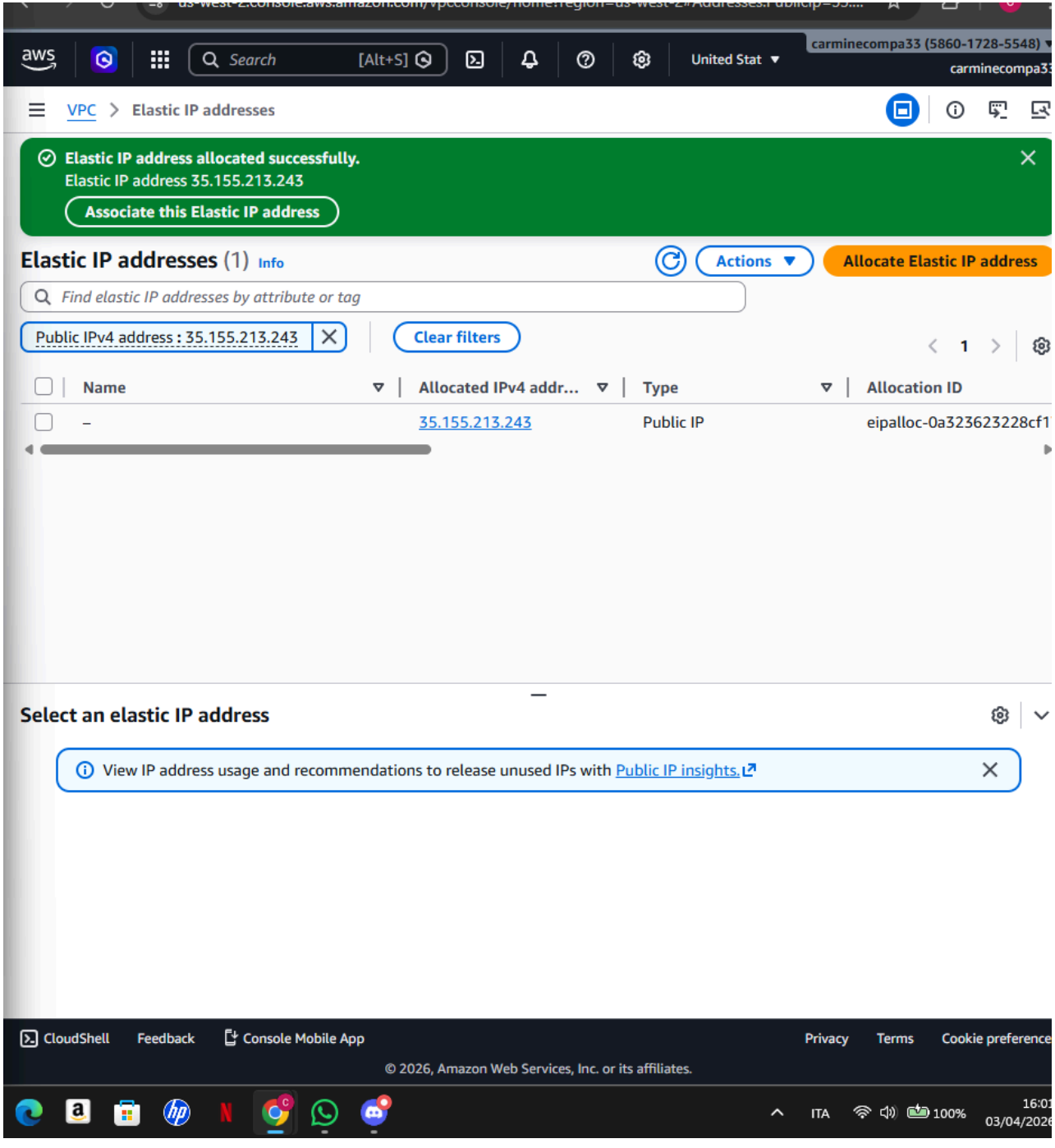


Route Table route WITHOUT 0.0.0.0.for Private Subnets

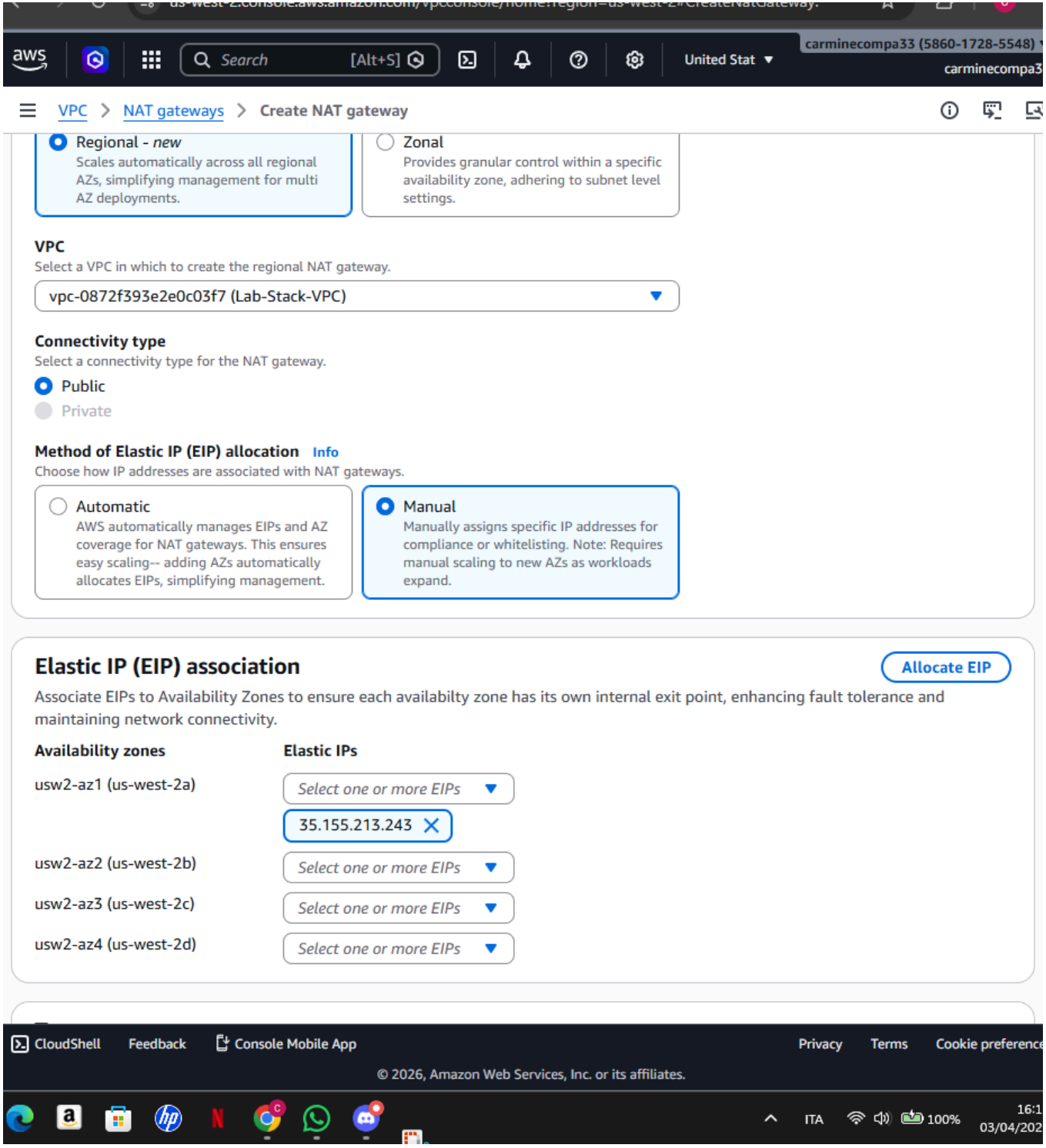


Step 2: Routing Logic

Allocate an Elastic IP



Create the NAT Gateway



VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
Carrier gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers

Security
Network ACLs
Security groups

PrivateLink and Lattice
Getting started
Endpoints
Endpoint services
Service networks
Lattice services

nat-1f916bdbbafdb4c01 / Project-NAT-Gateway

Actions

Details

NAT gateway ID
nat-1f916bdbbafdb4c01

NAT gateway ARN
arn:aws:ec2:us-west-2:586017285548:natgateway/nat-1f916bdbbafdb4c01

VPC
vpc-0872f393e2e0c03f7 / Lab-Stack-VPC

State
Available

Created
Friday, April 3, 2026 at 16:18:47 GMT+2

Availability mode
Regional

Connectivity type
Public

Method of EIP allocation
Manual

State message Info
–

Deleted
–

IP addresses | Monitoring | Flow logs | Tags

Associated IP addresses (1)

Edit IP address associations

Search

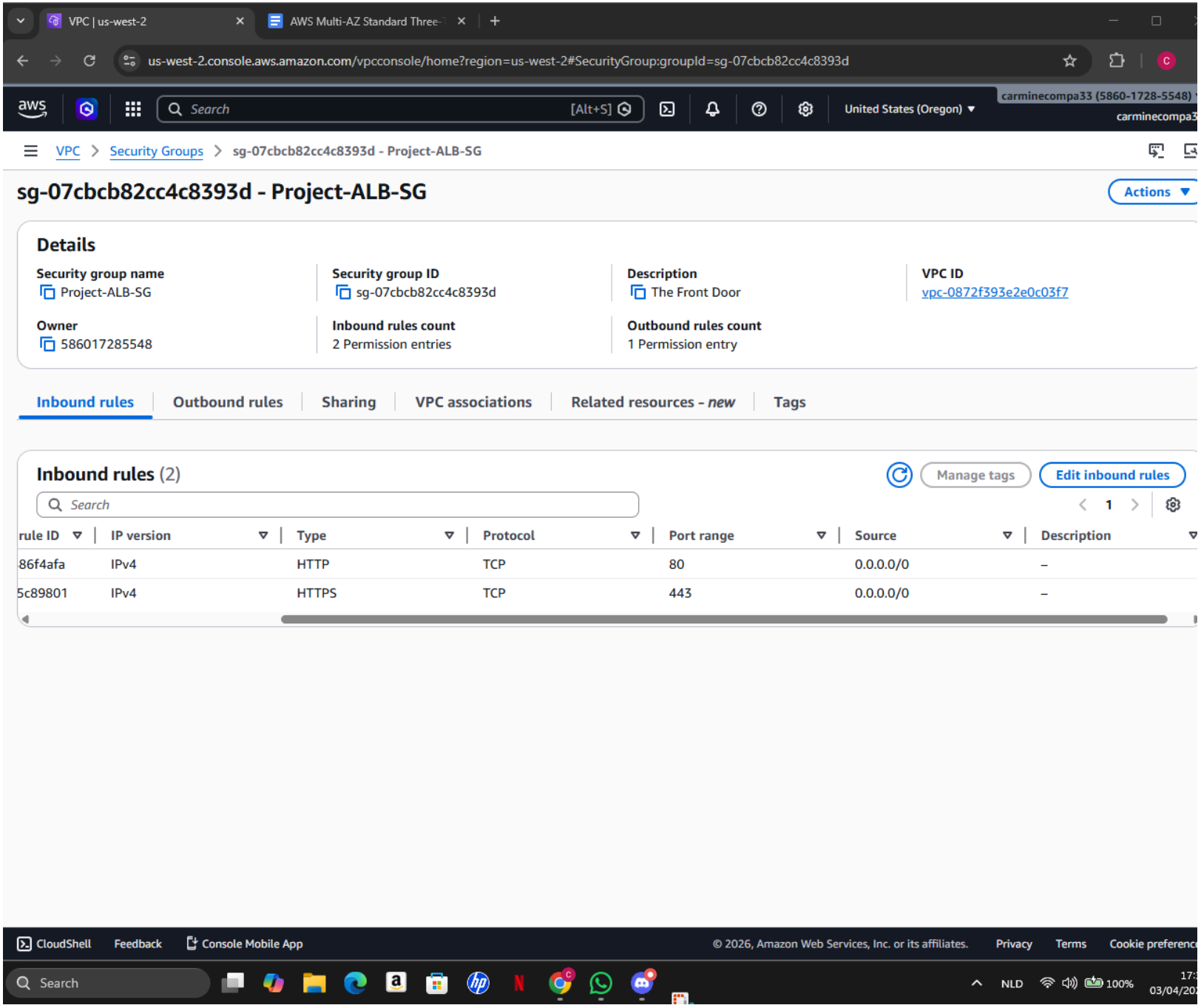
1

IP address	Status	Availa
35.155.213.243	Succeeded	usw2-

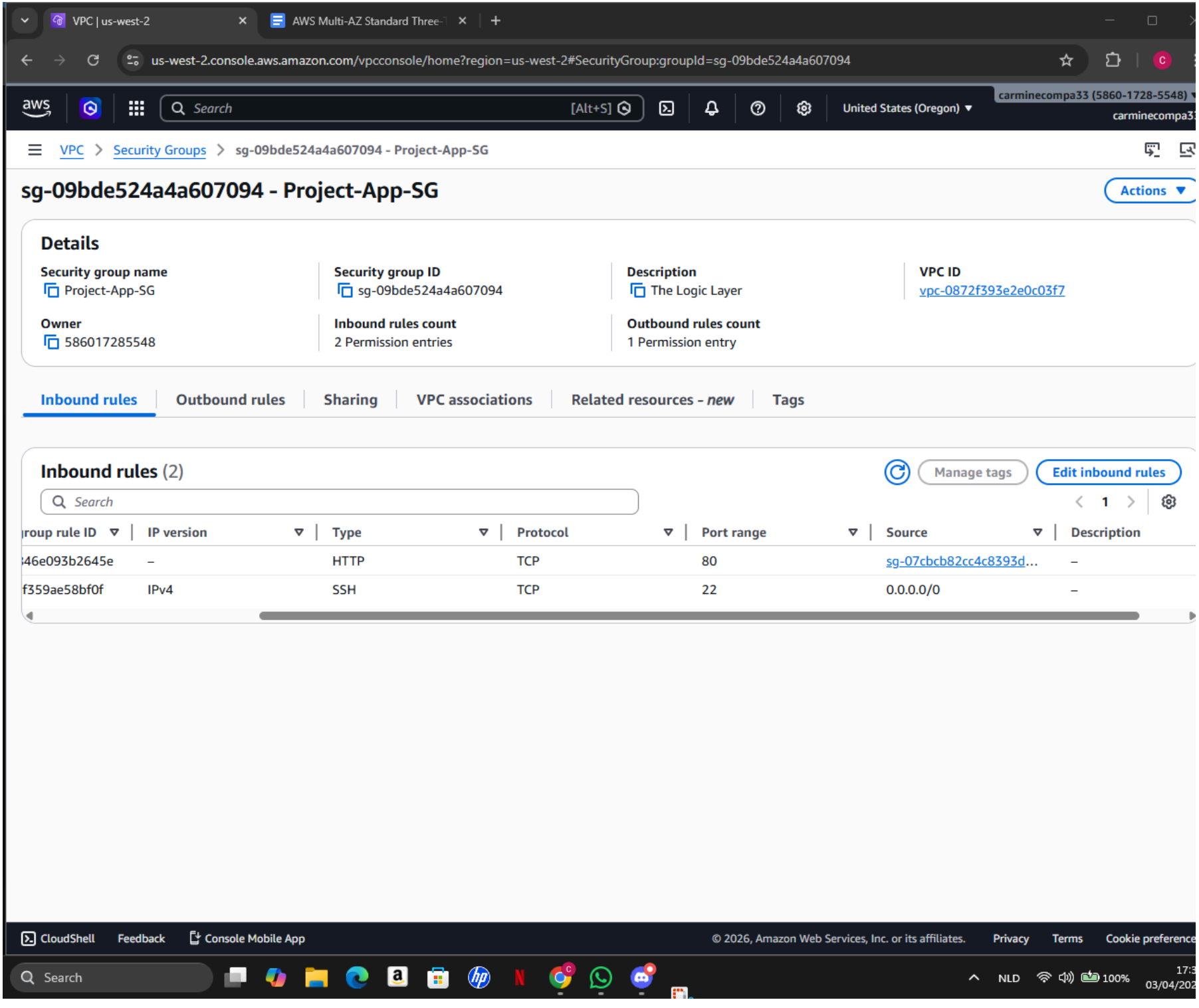
© 2026, Amazon Web Services, Inc. or its affiliates.

Step 3: Creating Security Groups

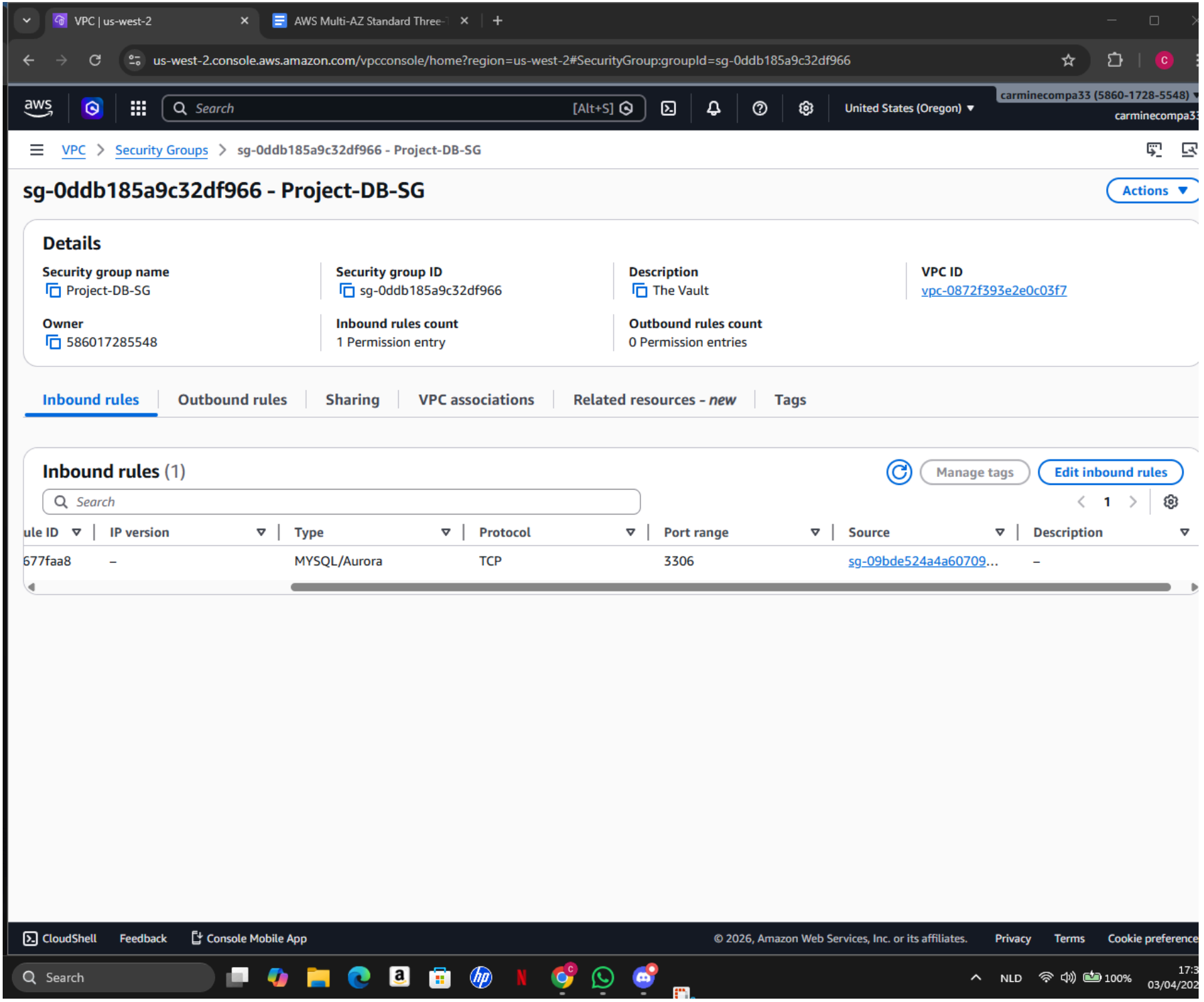
Public Load Balancer SG - The Front Door



Private App SG - The Logic Layer

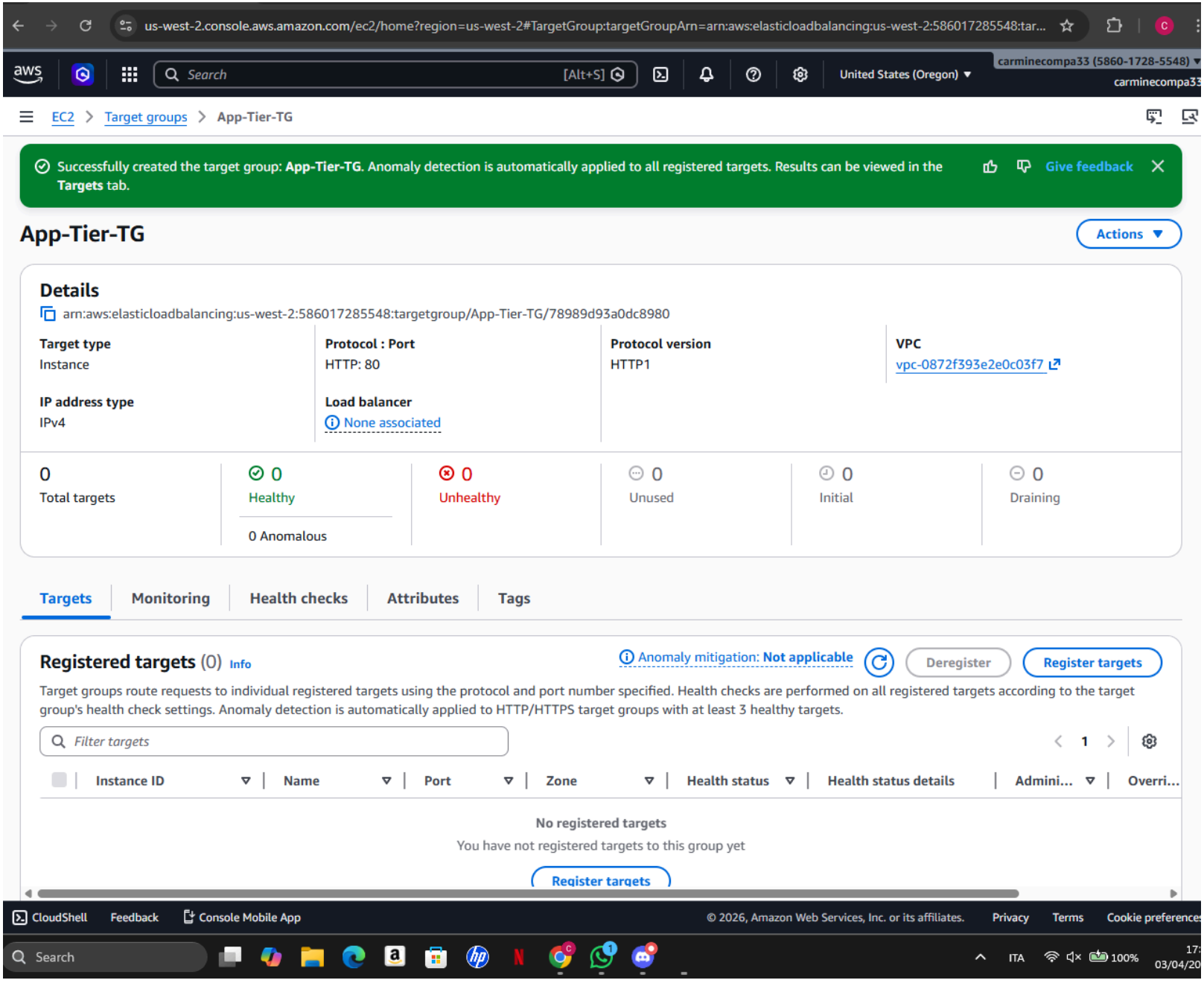


Private Database SG - The Vault

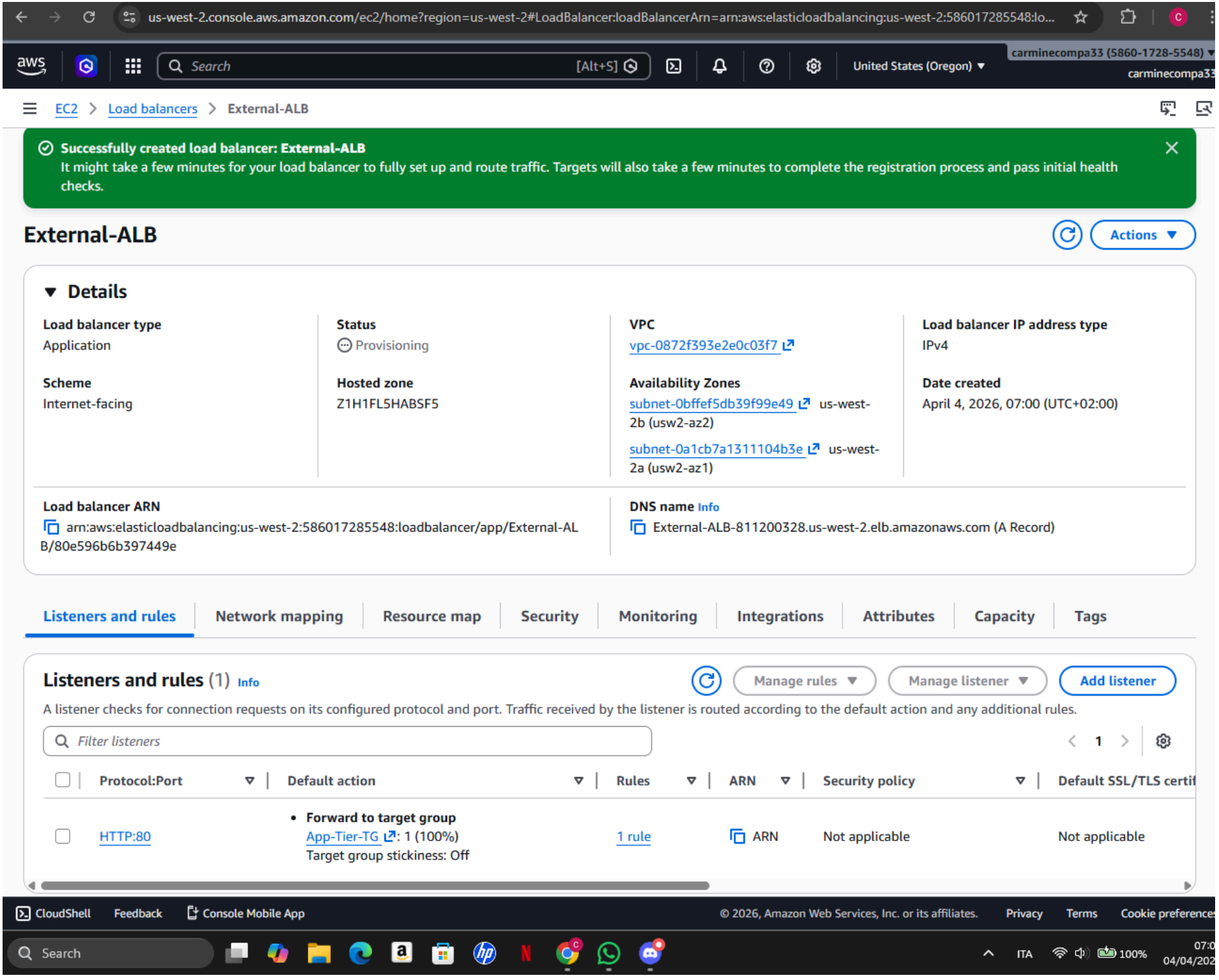


Step 4: Set up the Application Load Balancer (ALB)

Create a Target Group

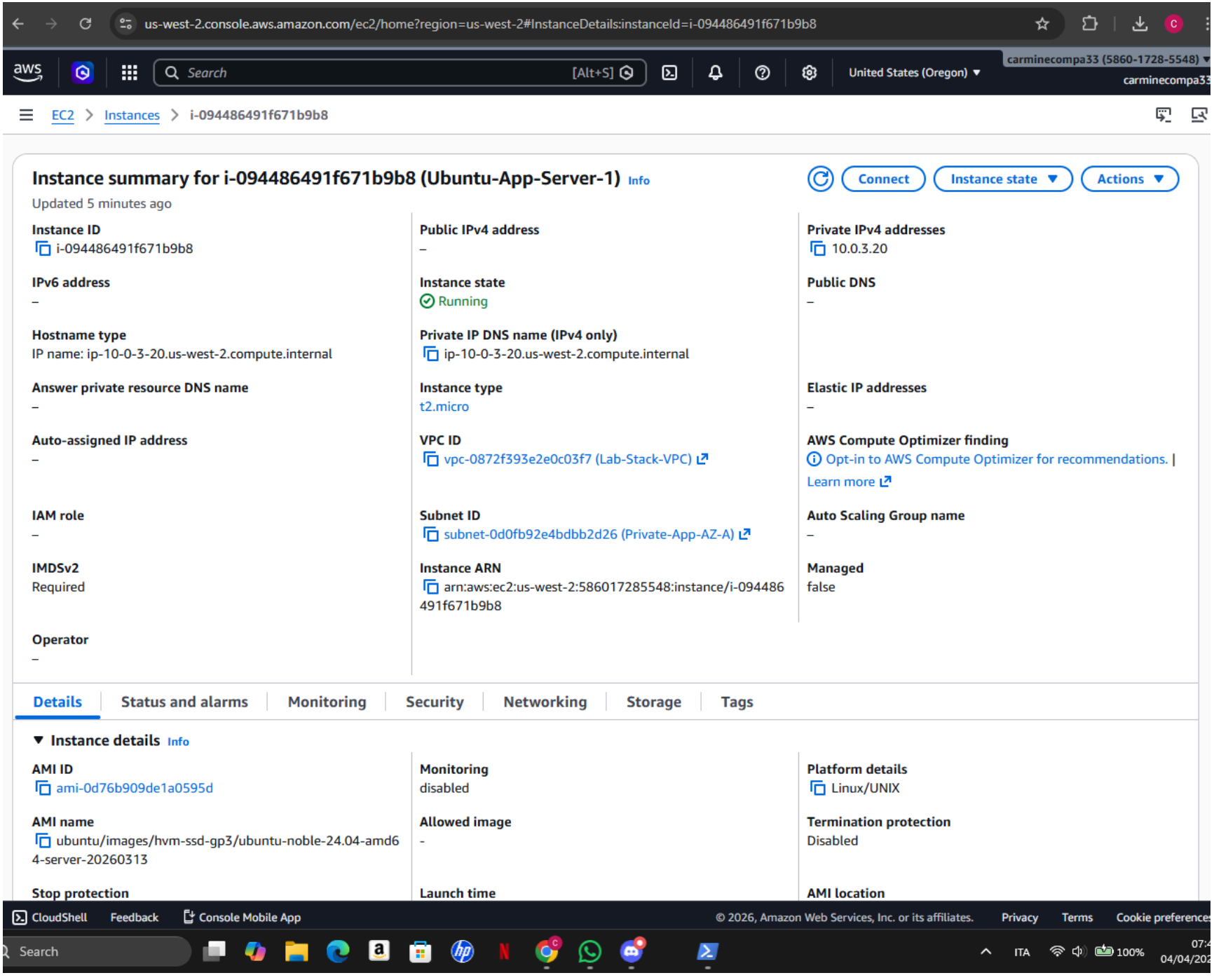


Create Application Load Balancer

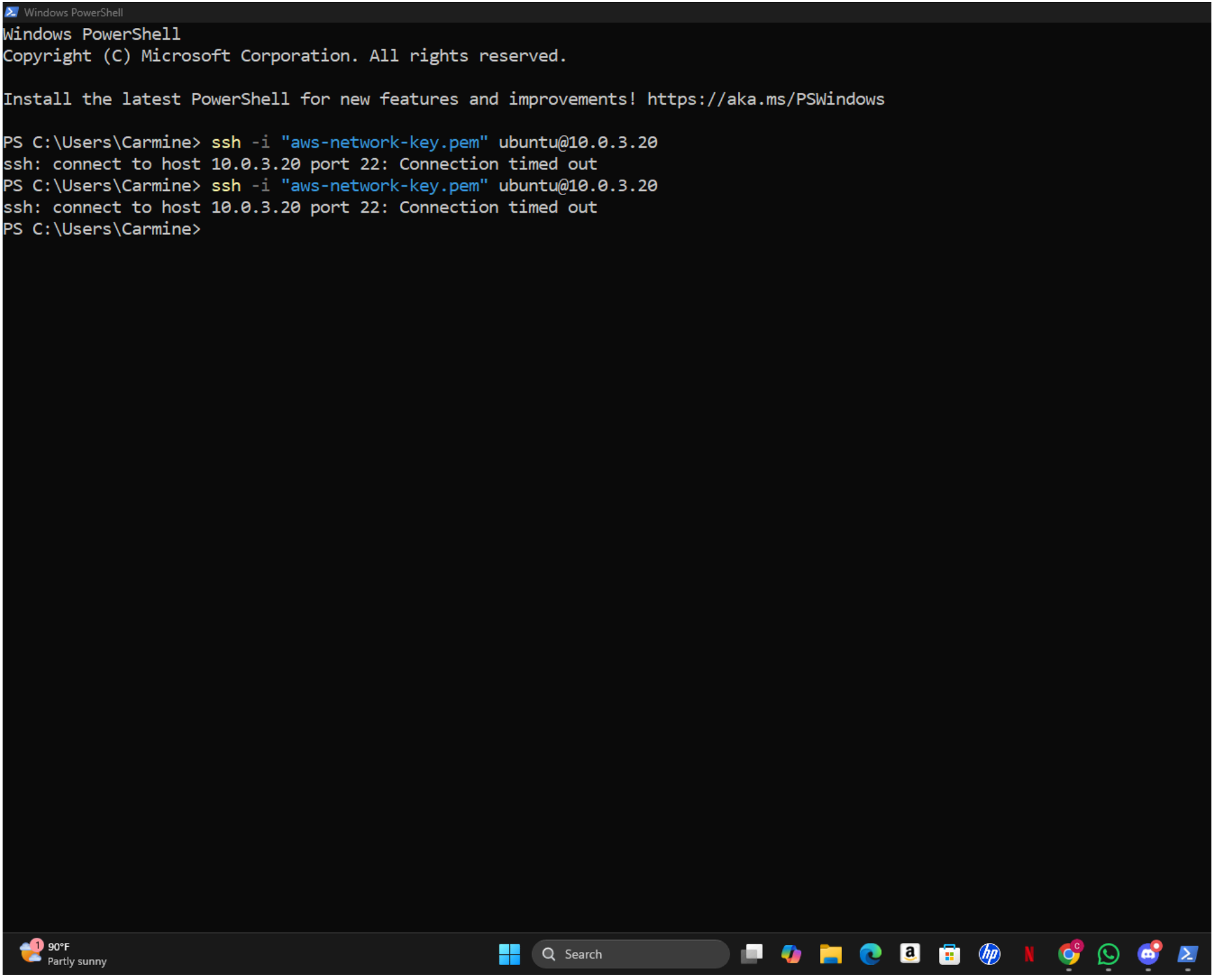


Launch the EC2 Instances (The “ App Tier”)

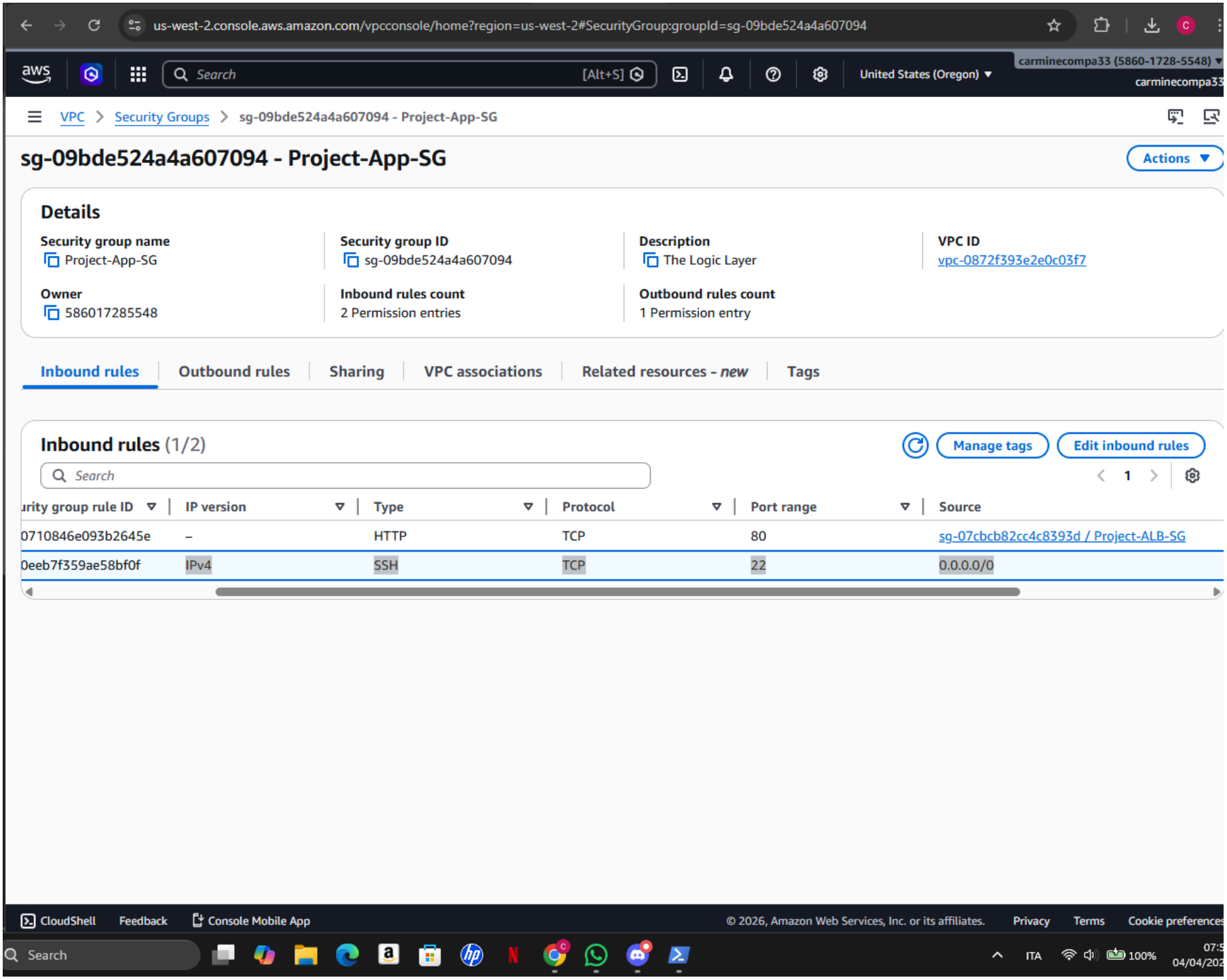
Create EC2 instance



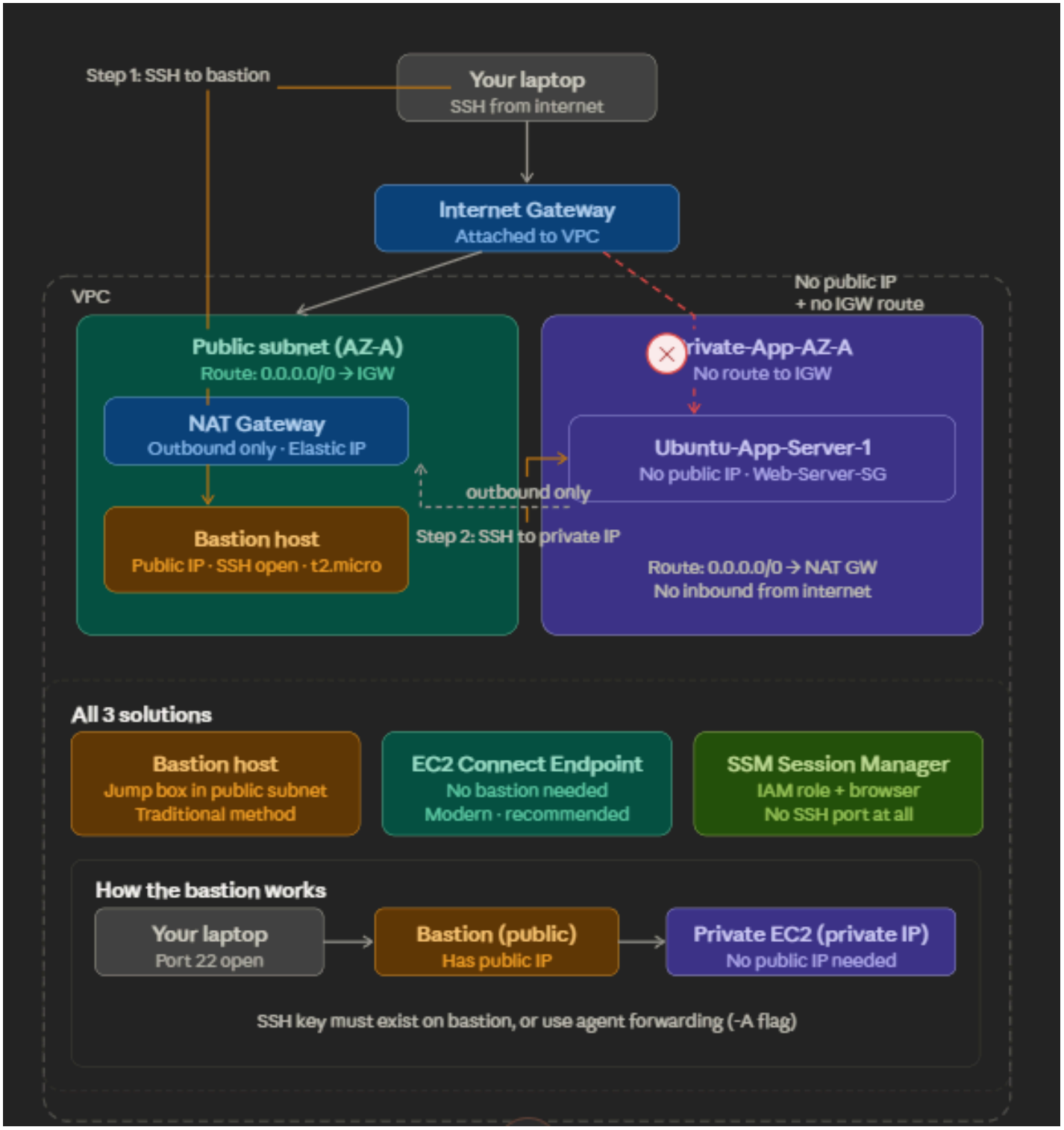
Cannot access via SSH



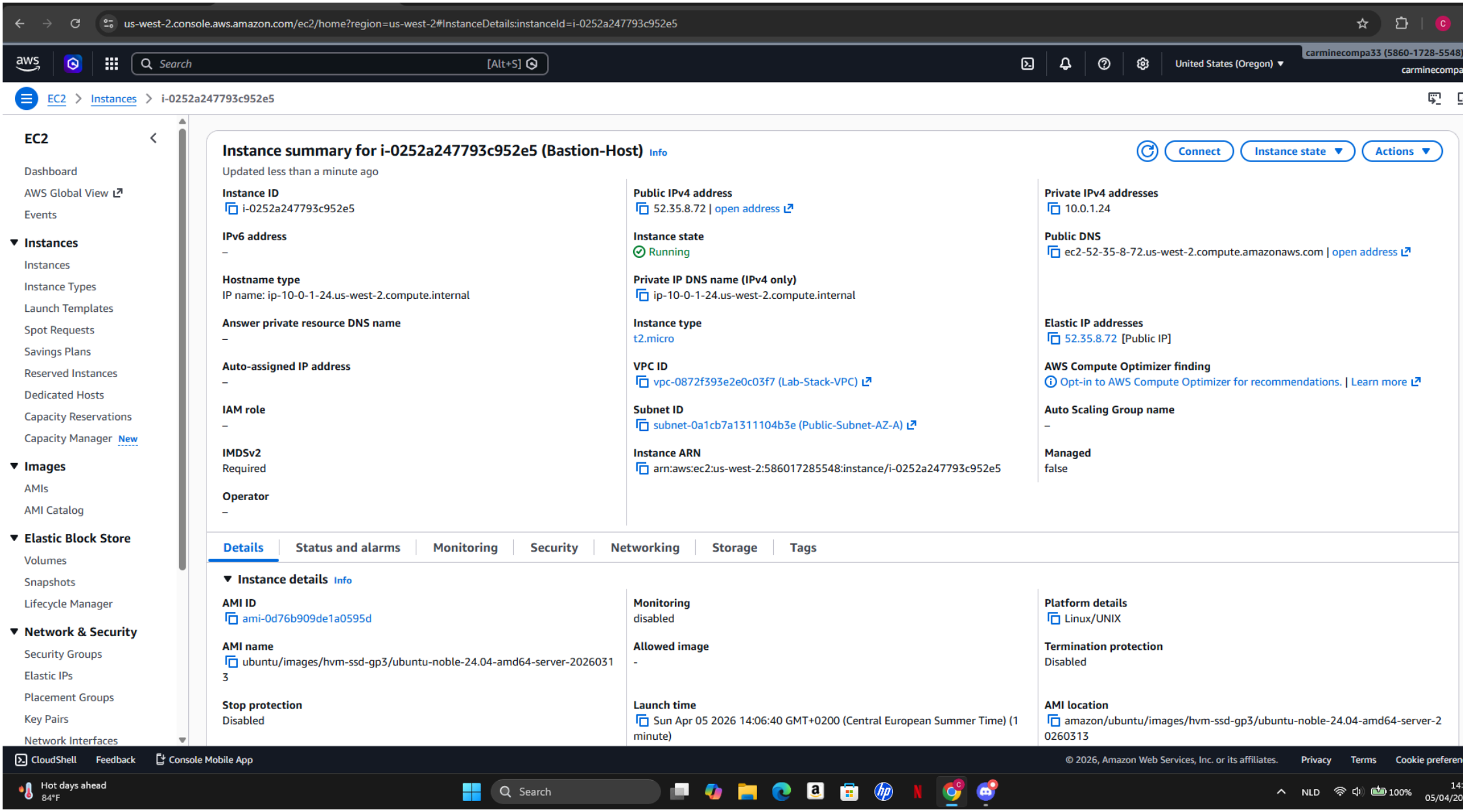
Security Group allows SSH



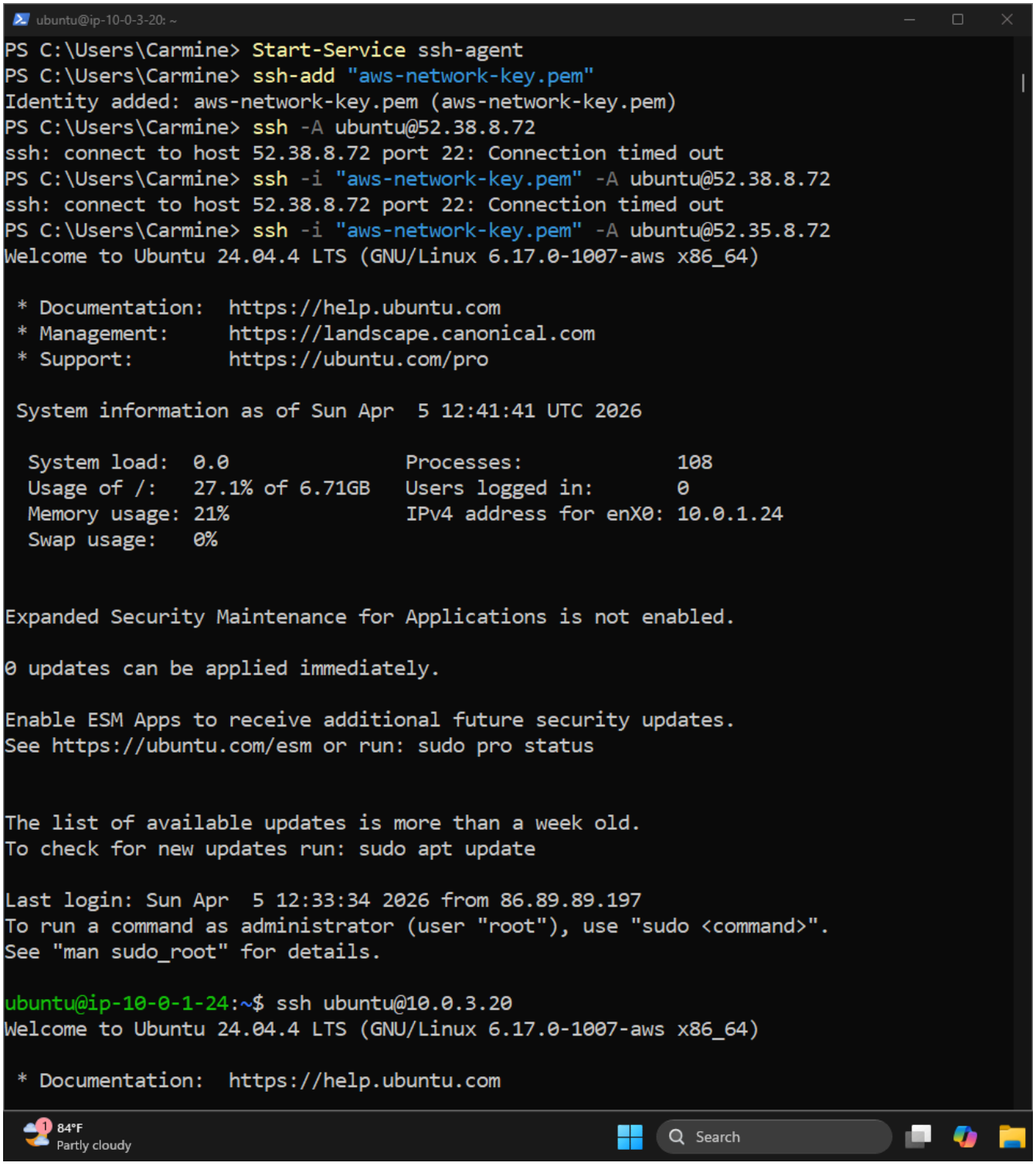
SSH Blocked: no public IP + no Internet Gateway route

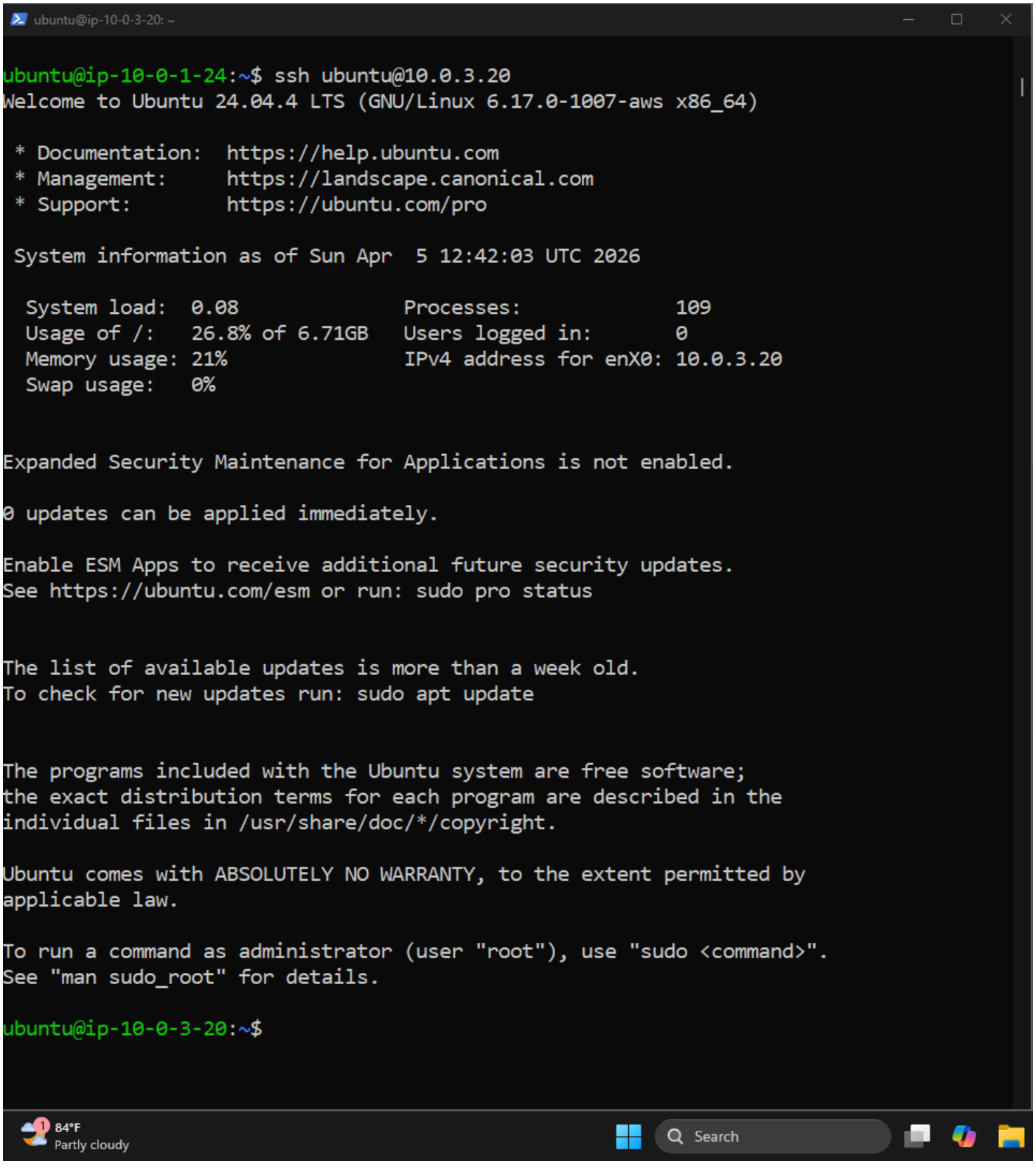


Created Bastion Host

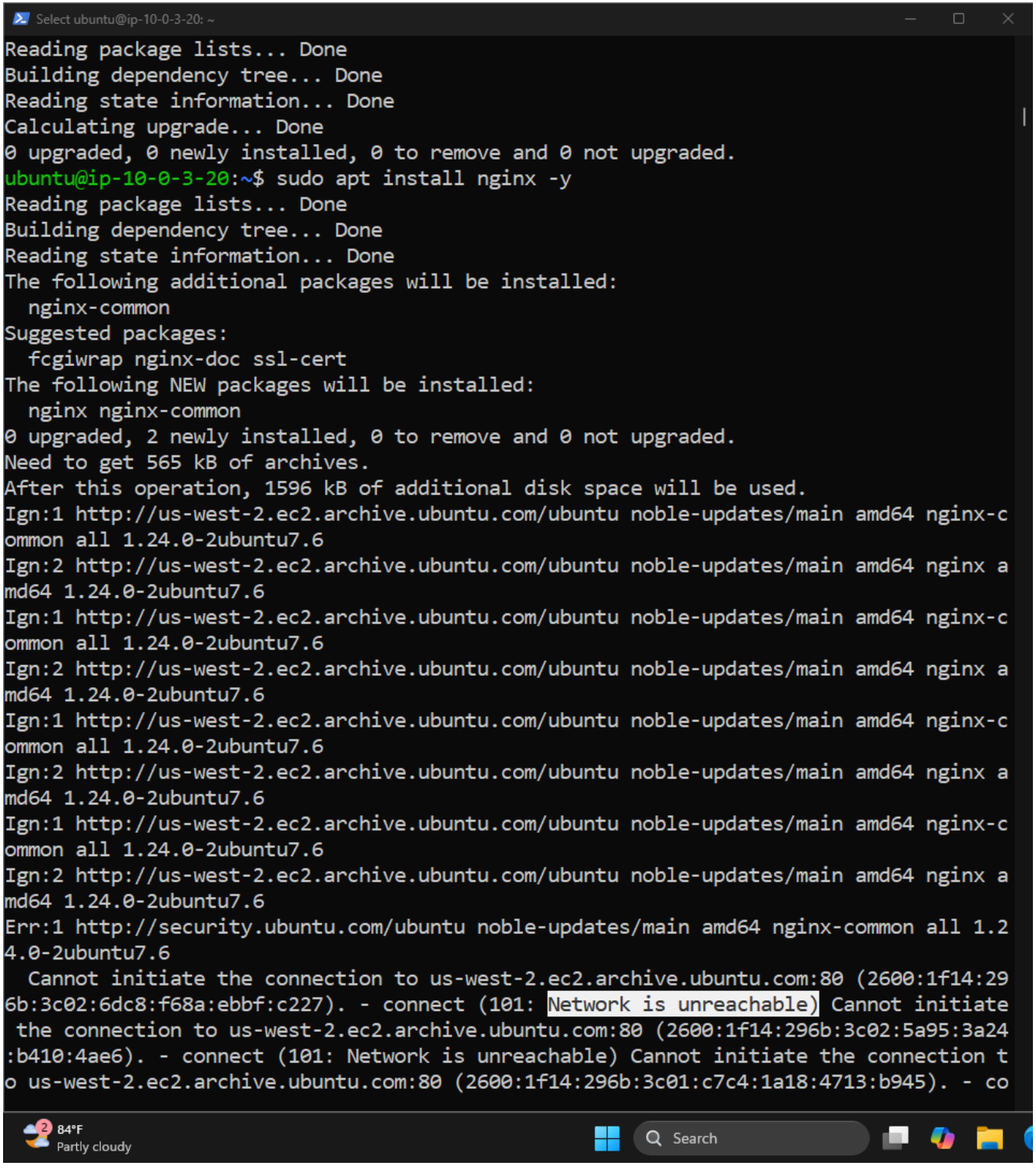


SSH into App EC2 server via Bastion Host

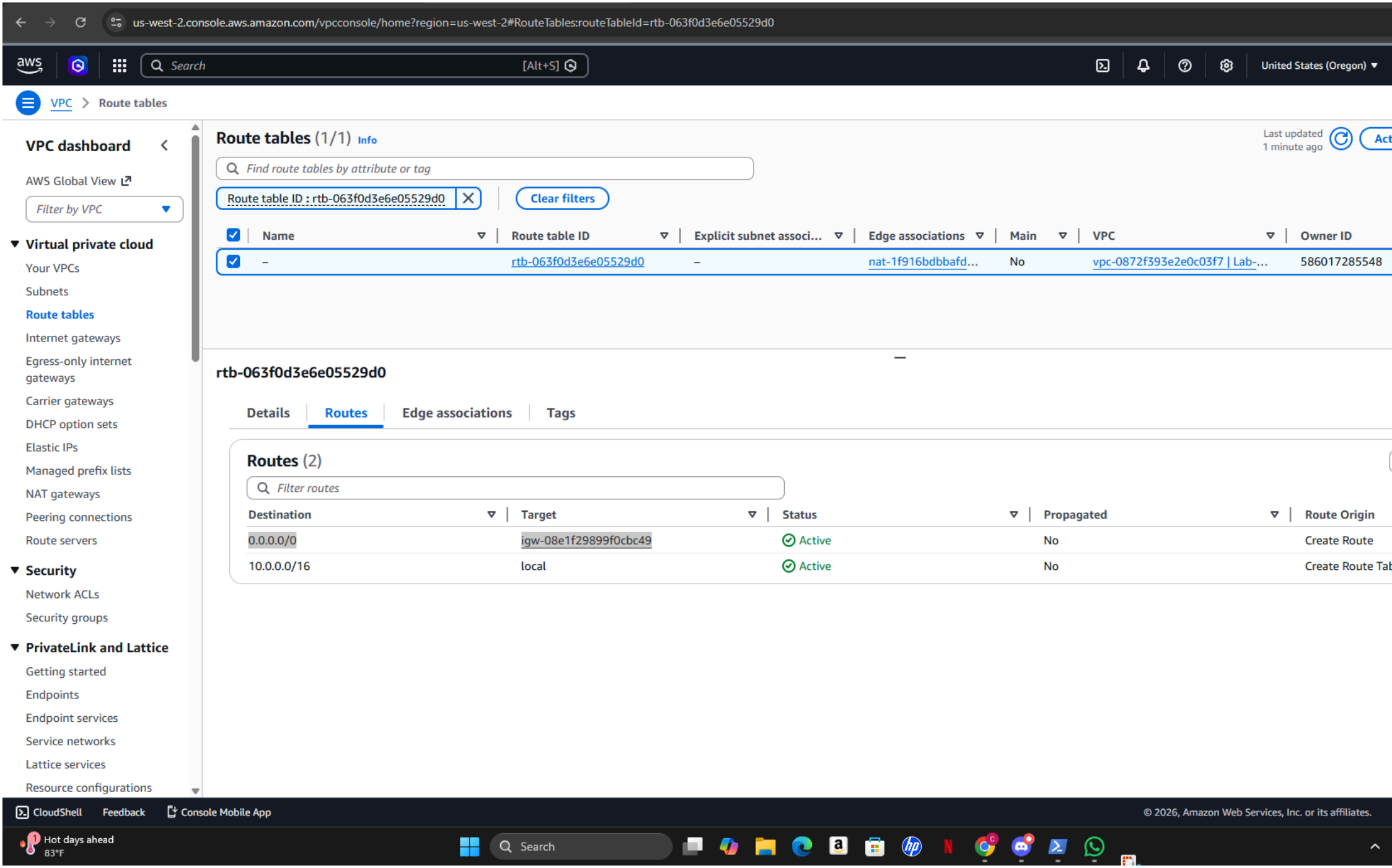



ubuntu@ip-10-0-3-20: ~
ubuntu@ip-10-0-1-24:~$ ssh ubuntu@10.0.3.20
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)
* Documentation: https://help.ubuntu.com
* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro
System information as of Sun Apr 5 12:42:03 UTC 2026
System load: 0.08
Usage of /: 26.8% of 6.71GB
Memory usage: 21%
Swap usage: 0%
Processes: 109
Users logged in: 0
IPv4 address for enX0: 10.0.3.20
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
The list of available updates is more than a week old.
To check for new updates run: sudo apt update
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
ubuntu@ip-10-0-3-20:~$
84°F
Partly cloudy
Search

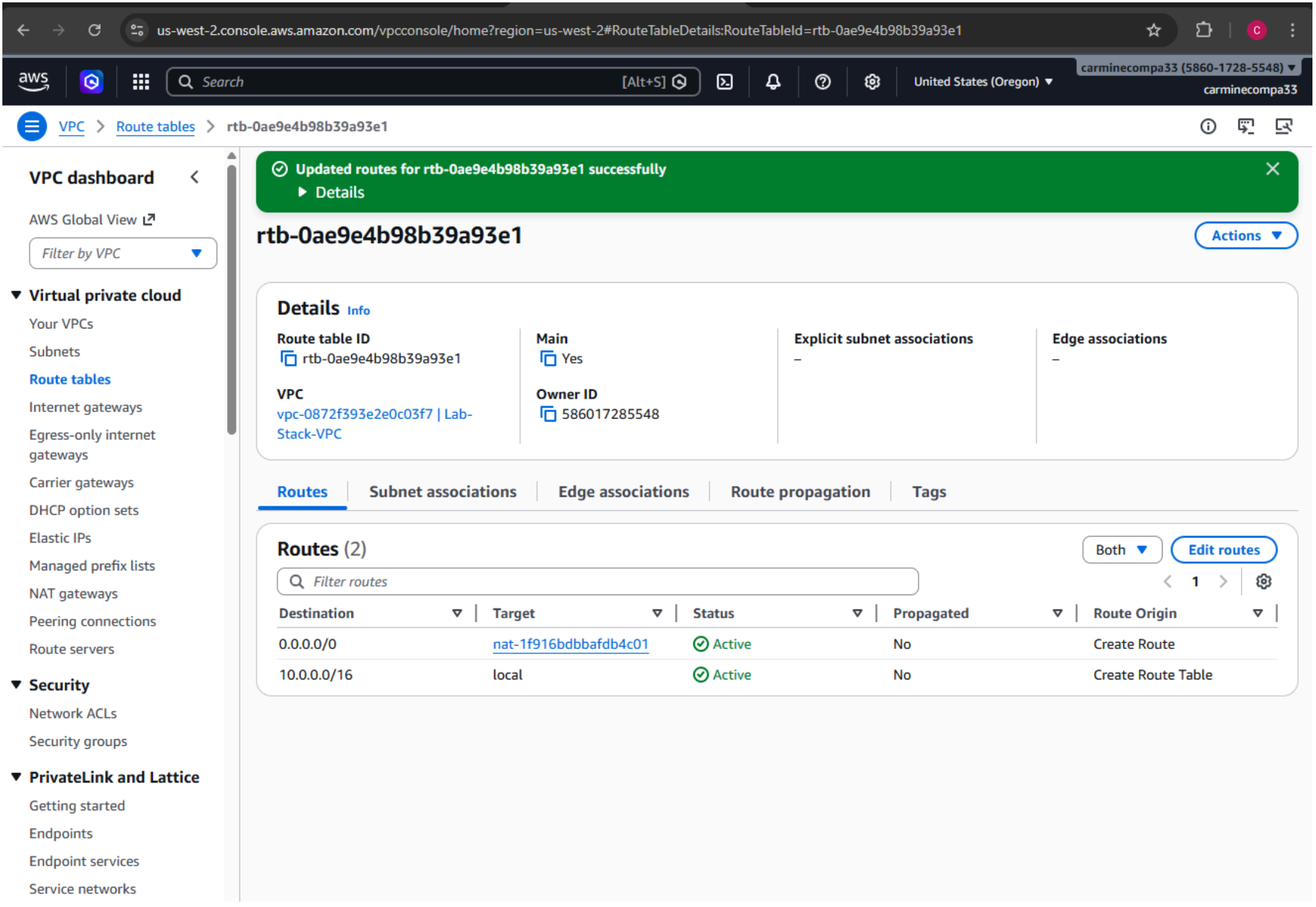
Cannot install nginx on the Ubuntu App EC2 (private subnet) - “Network in unreachable”



NAT Gateway is in a public subnet (0.0.0.0/0 to Internet Gateway)

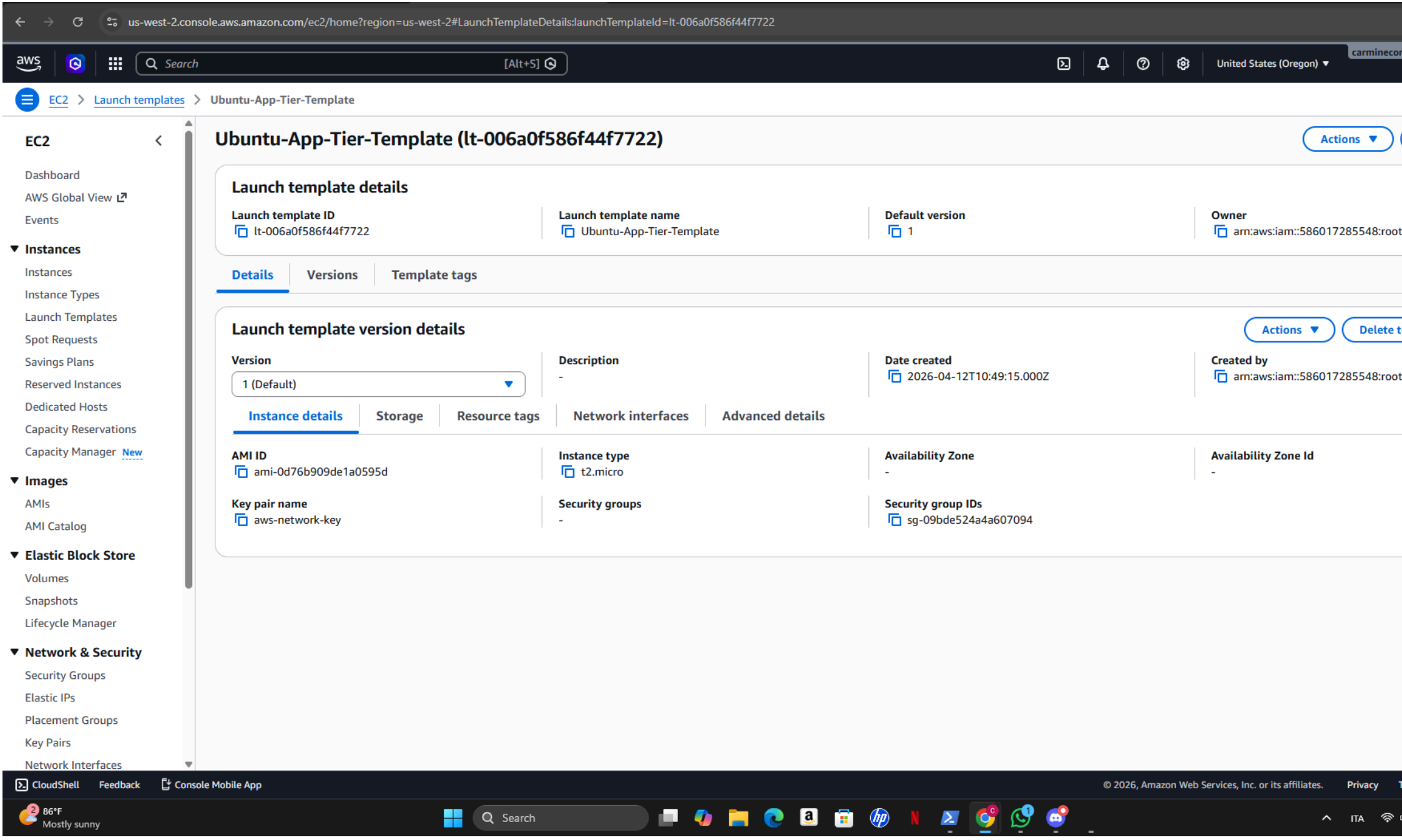


App EC2 is in a private subnet not connected to NAT Gateway > Link NAT Gateway

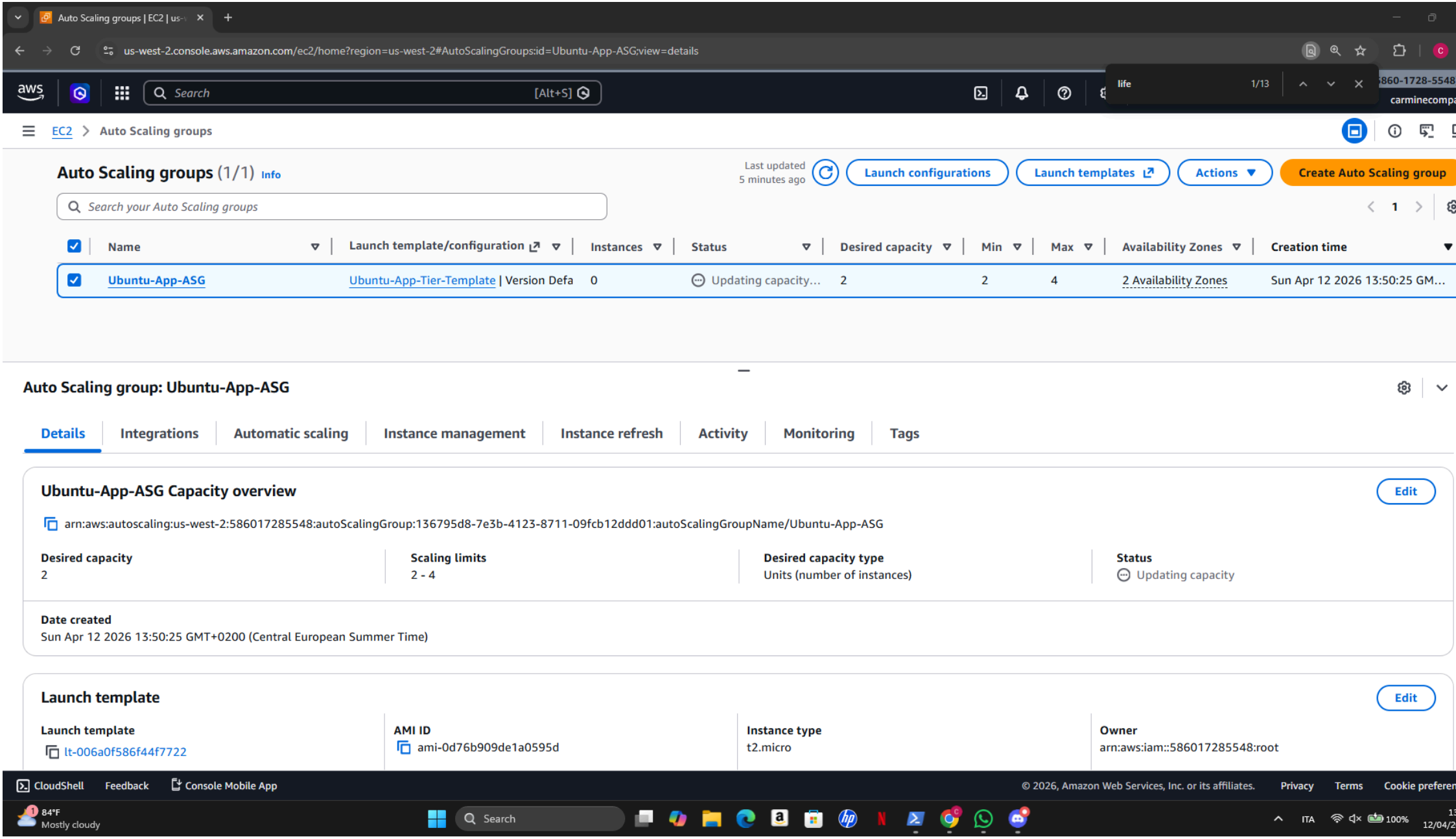


Step 5: The Resilience Upgrade

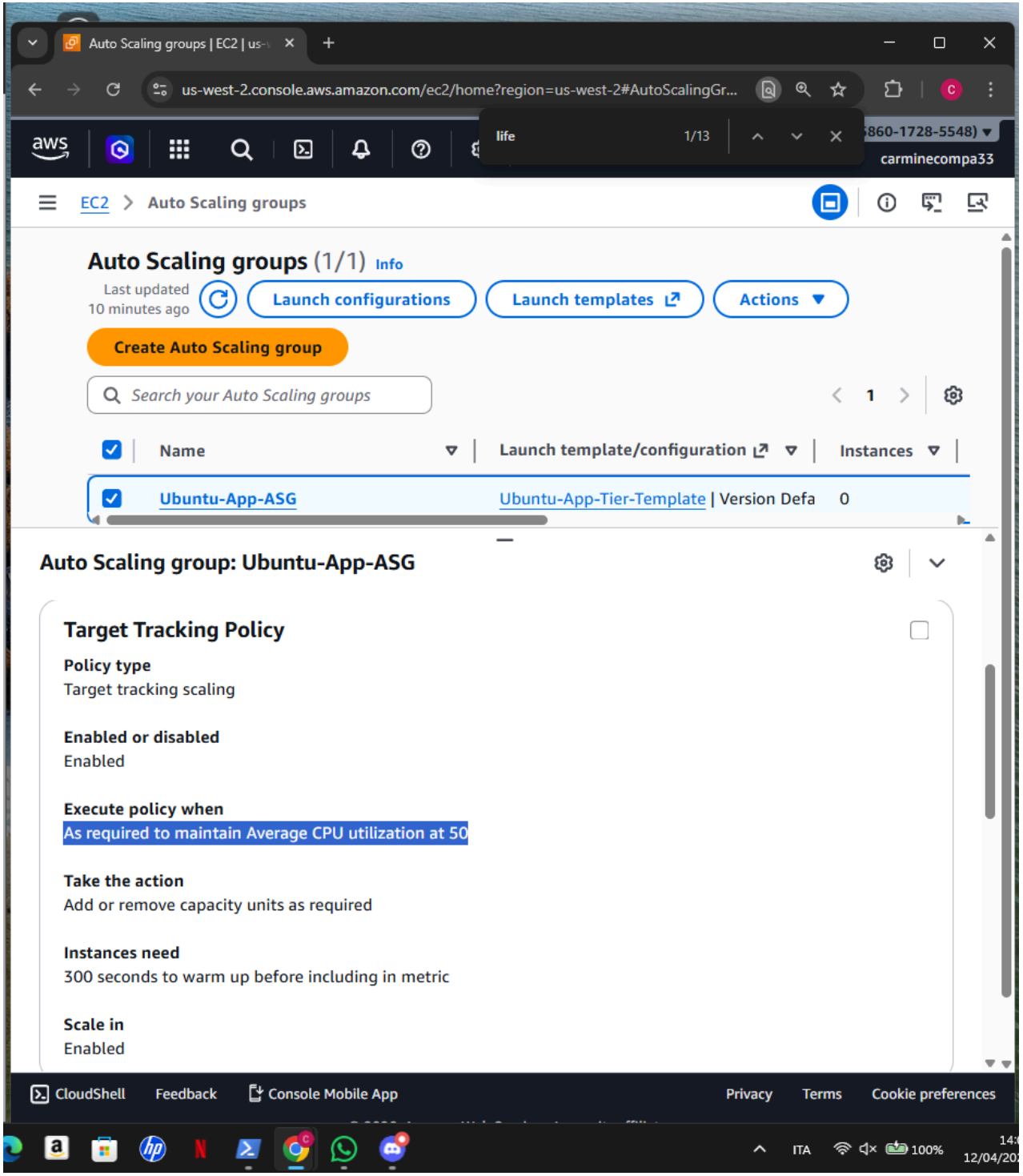
Create Launch template for the Auto scaling group



Create Auto-Scaling Group

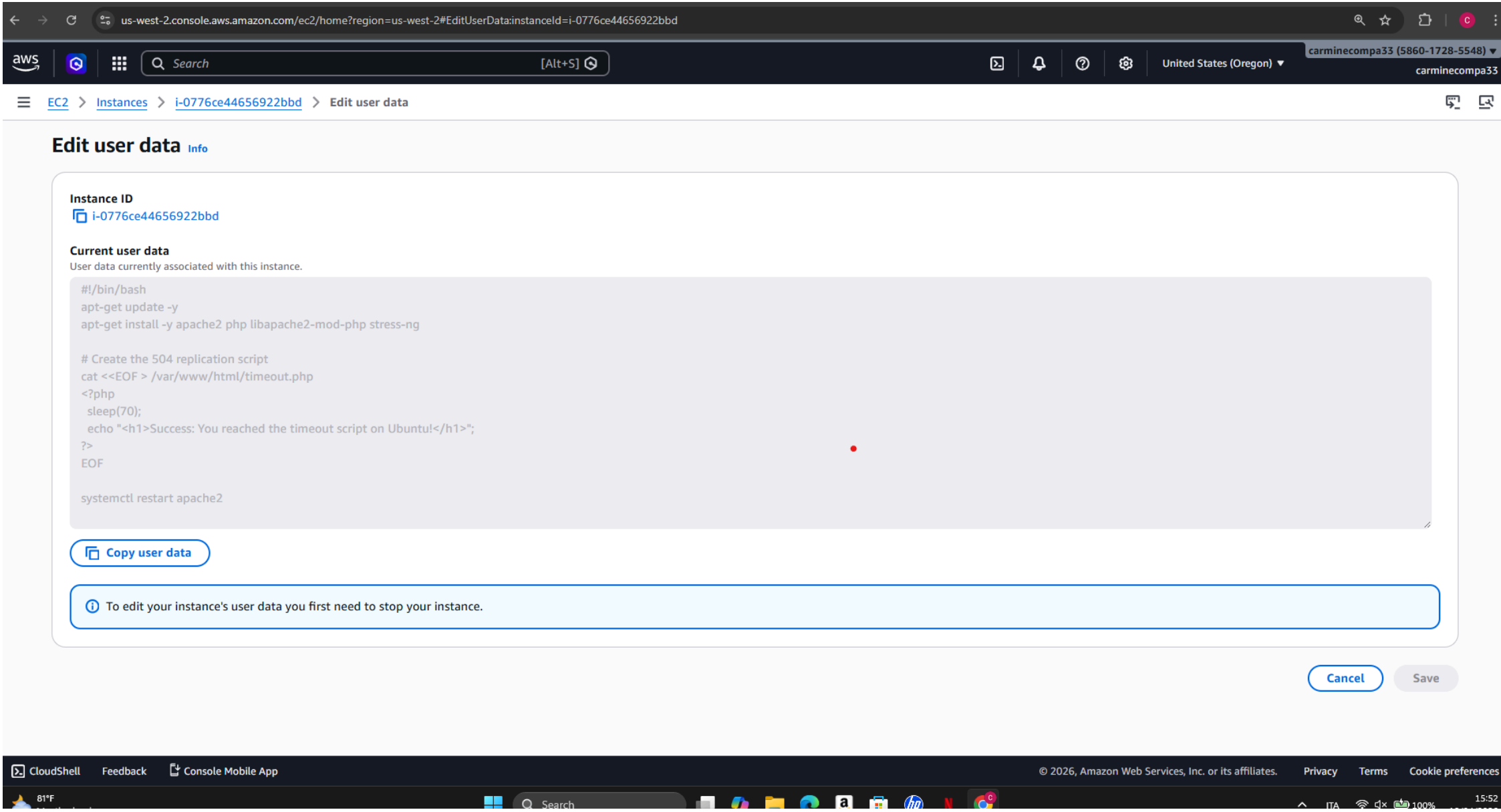


Set a Target Tracking Policy based on Average CPU Utilization of 50%

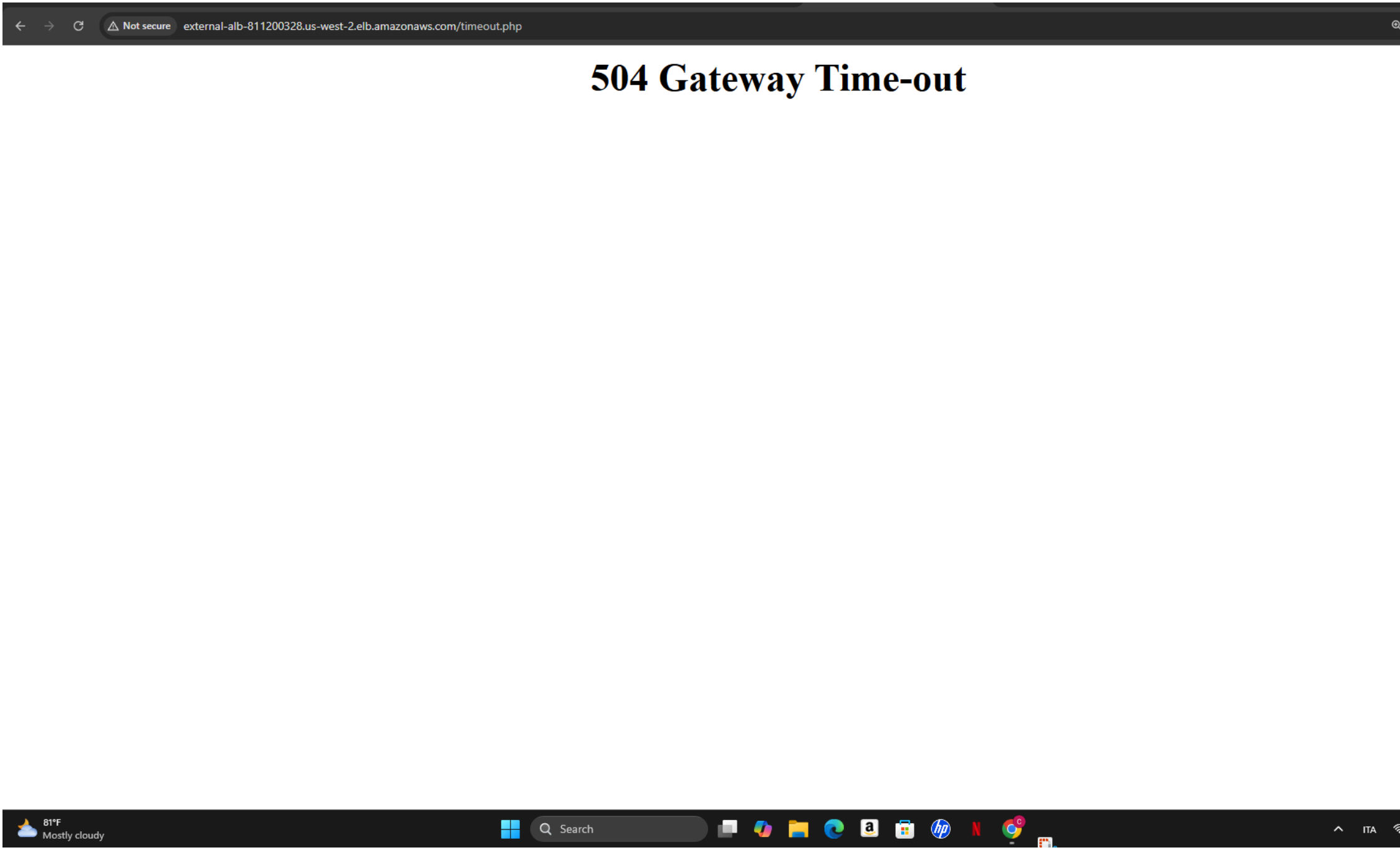


Replicate 504 Gateway error

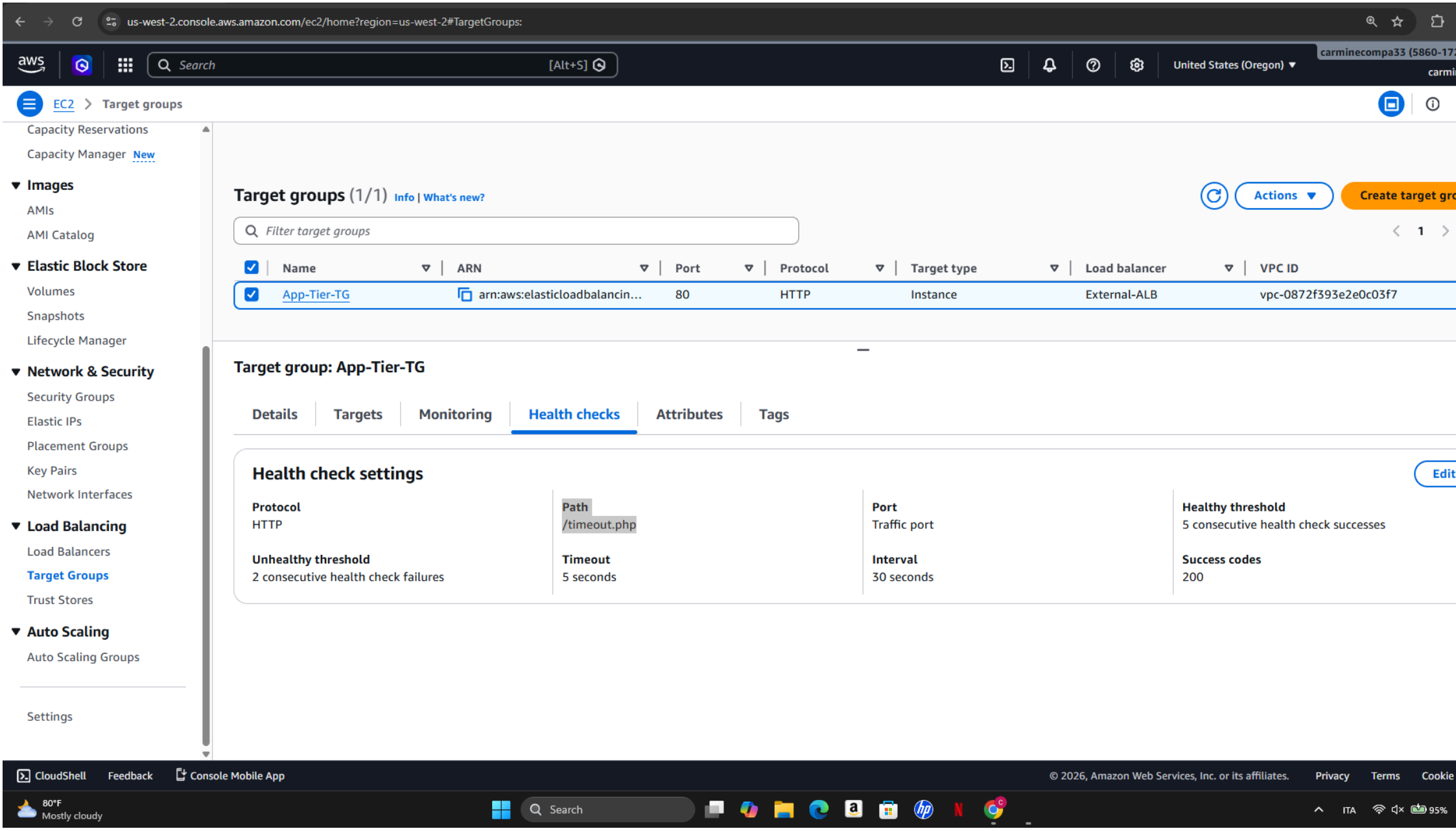
PHP webpage with 70 seconds of sleep



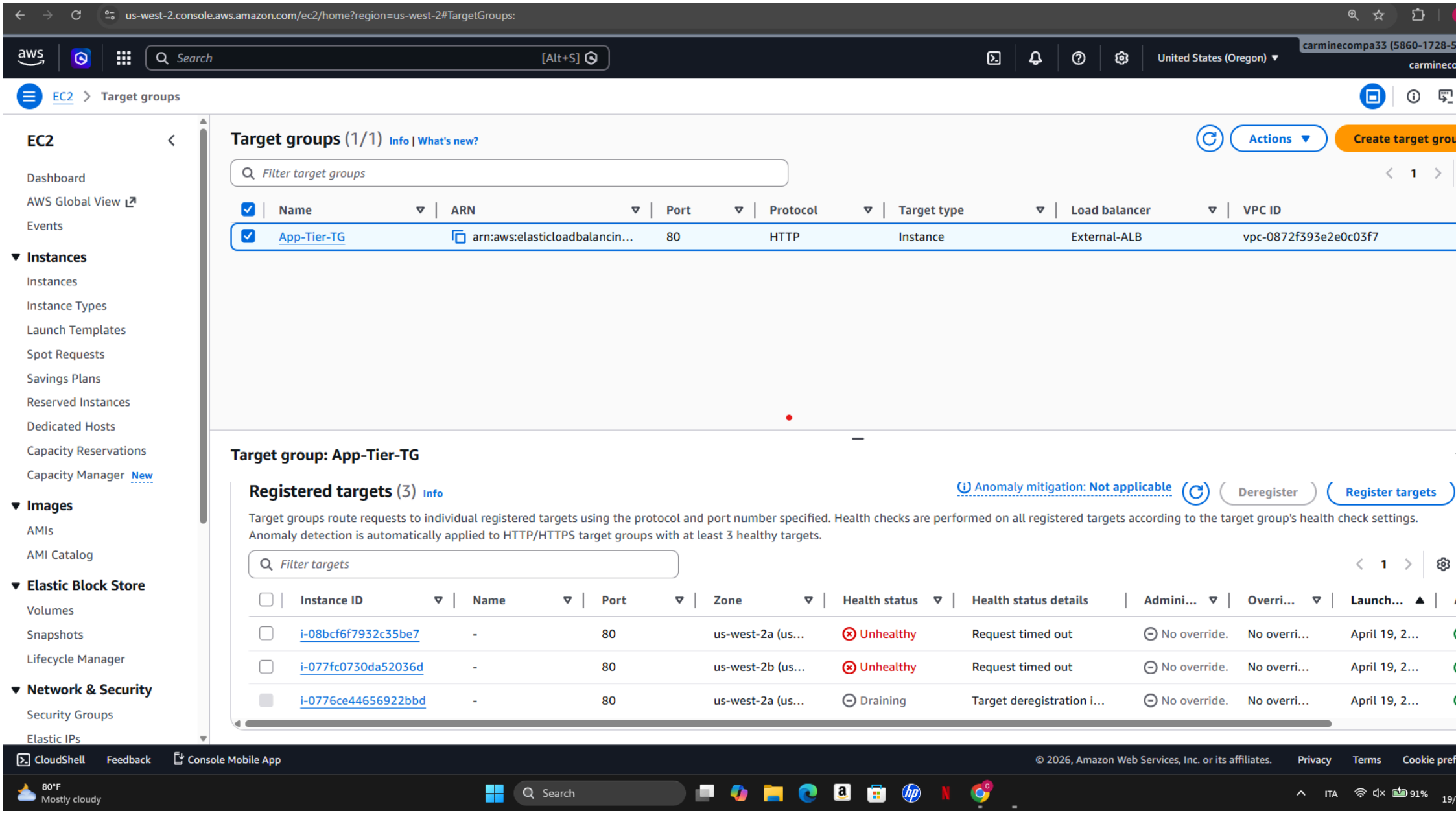
Application Load Balancer timeouts by default after 60 seconds

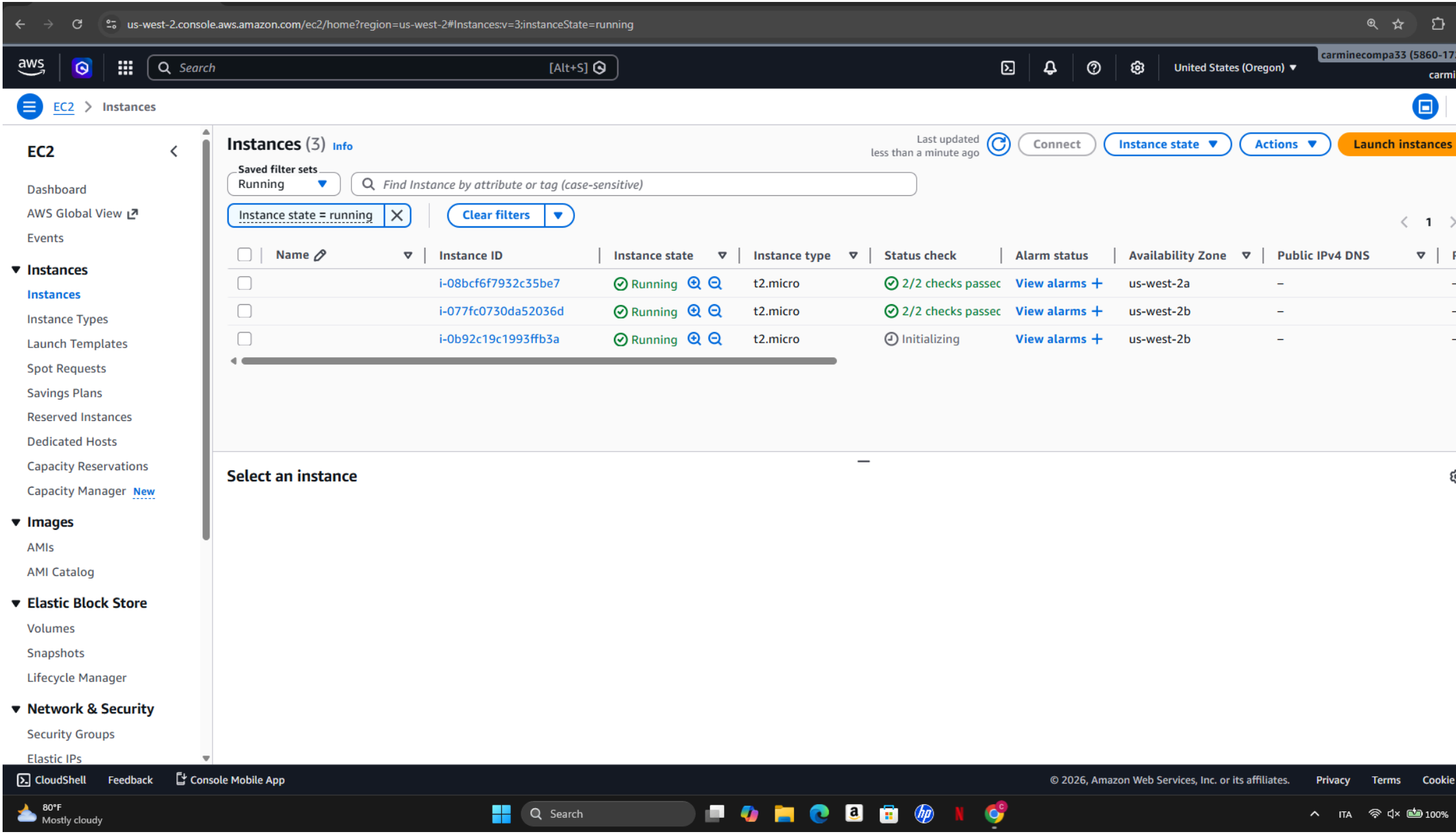


Target Group Health Check path changed to the PHP script location /timeout.php

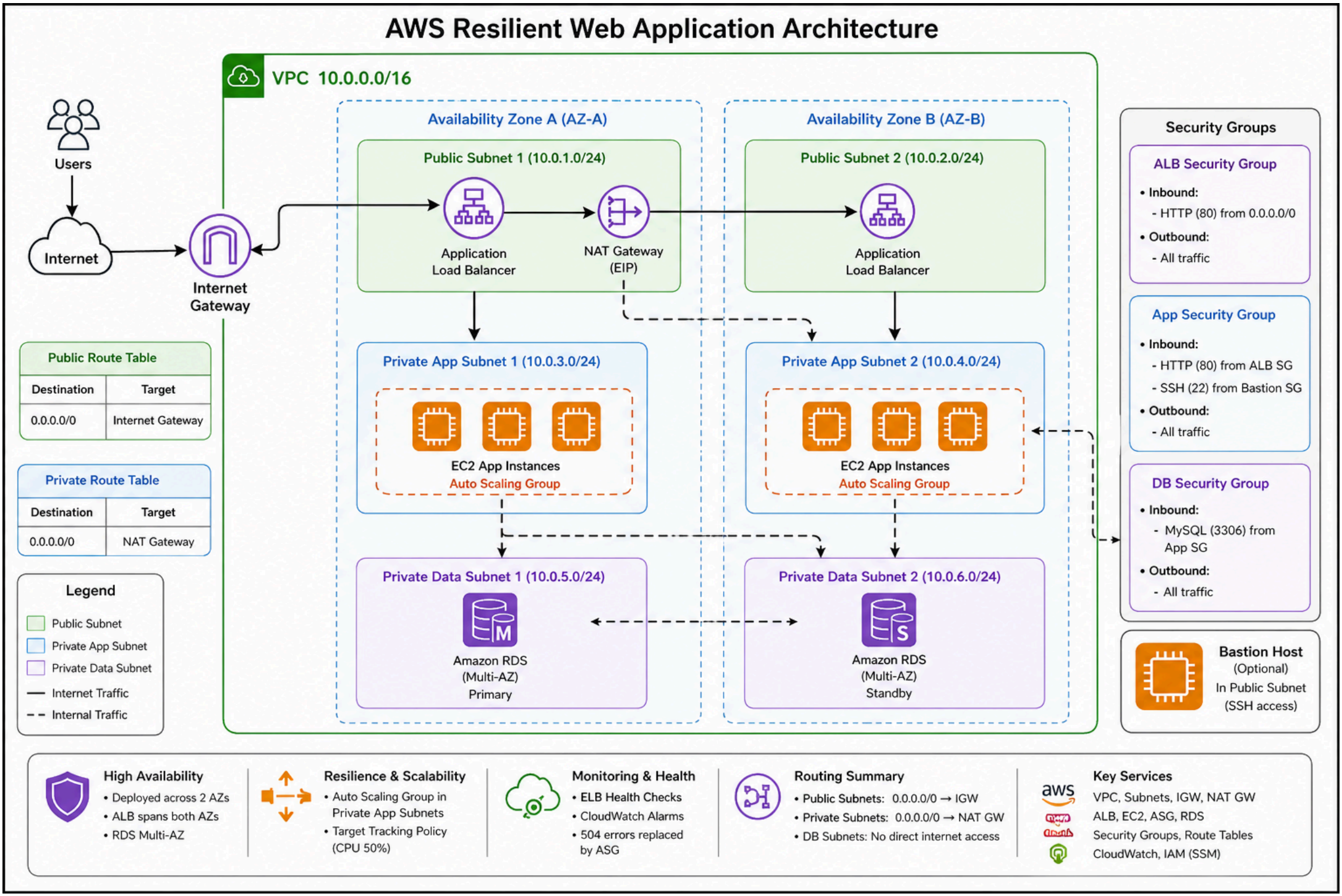


Target Group Health Check fails and third instance is created now



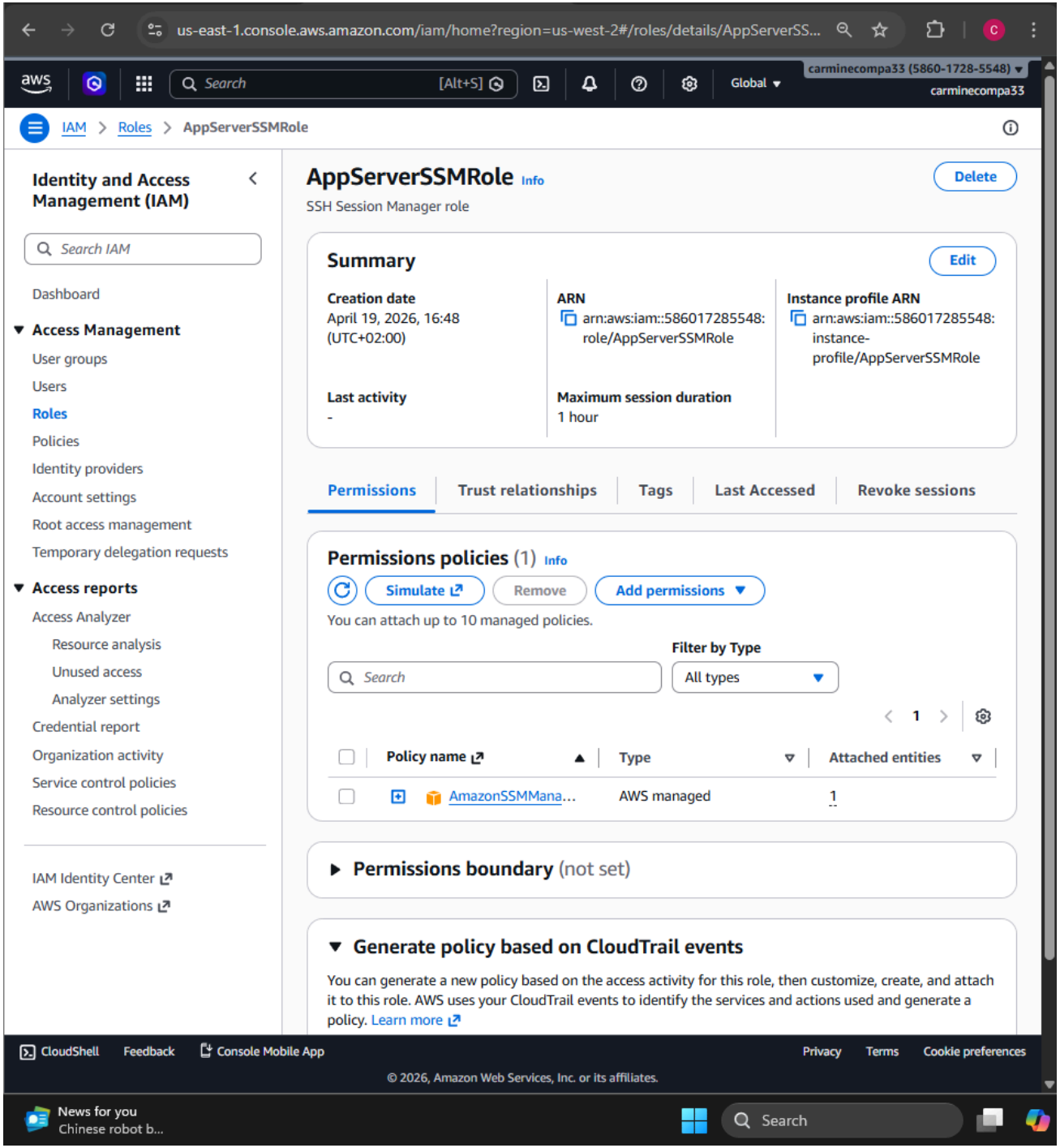

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#Instances:v=3;instanceState=running
Search
[Alt+S]
United States (Oregon)
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Instances (3) Info
Last updated less than a minute ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Running
Find Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
i-08bcf6f7932c35be7 Running t2.micro 2/2 checks passec View alarms us-west-2a
i-077fc0730da52036d Running t2.micro 2/2 checks passec View alarms us-west-2b
i-0b92c19c1993ffb3a Running t2.micro Initializing View alarms us-west-2b
Select an instance
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie
80°F
Mostly cloudy
Search
ITA
100%

Picture of the AWS architecture so far

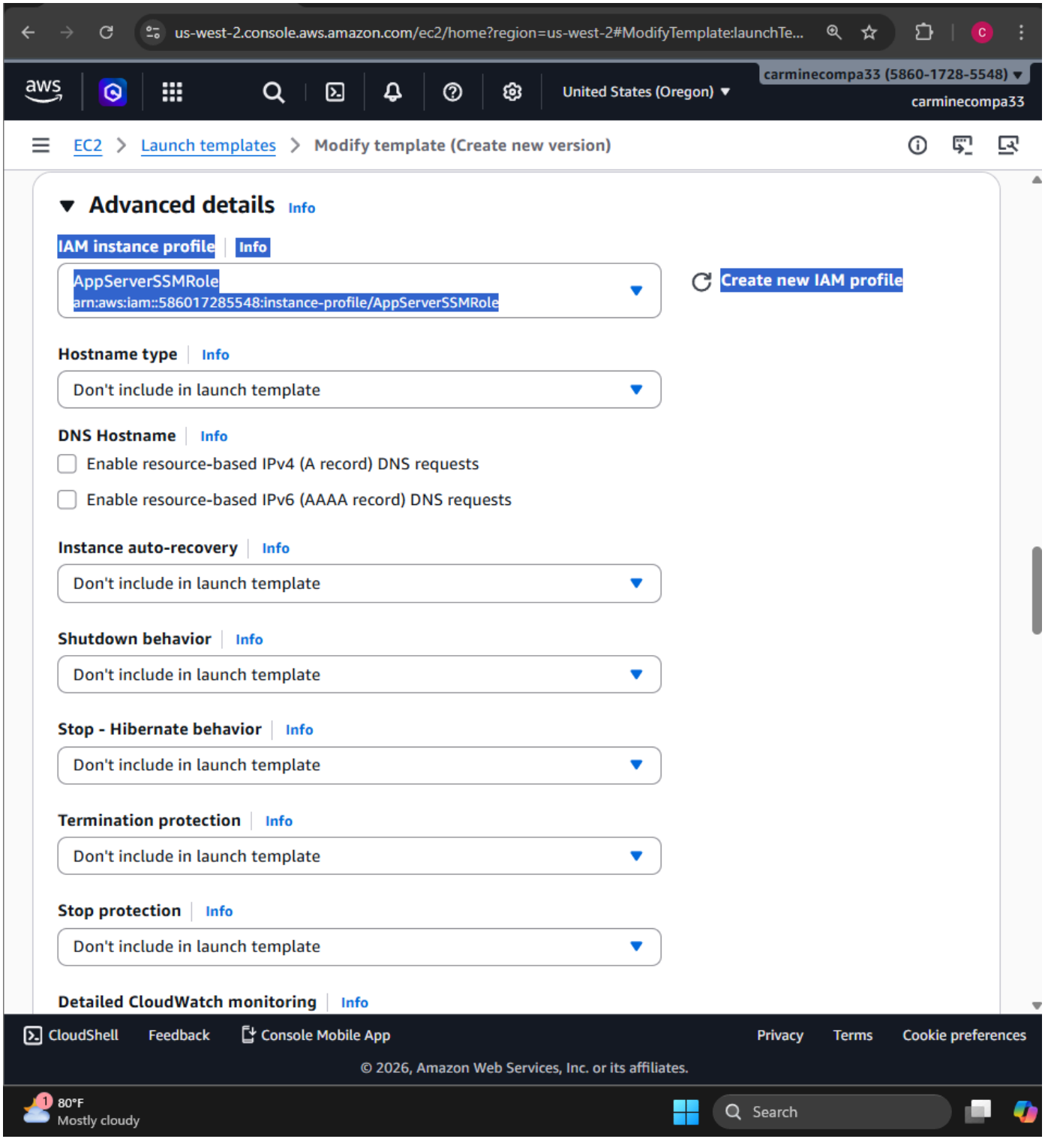


Scale-Out Stress Test

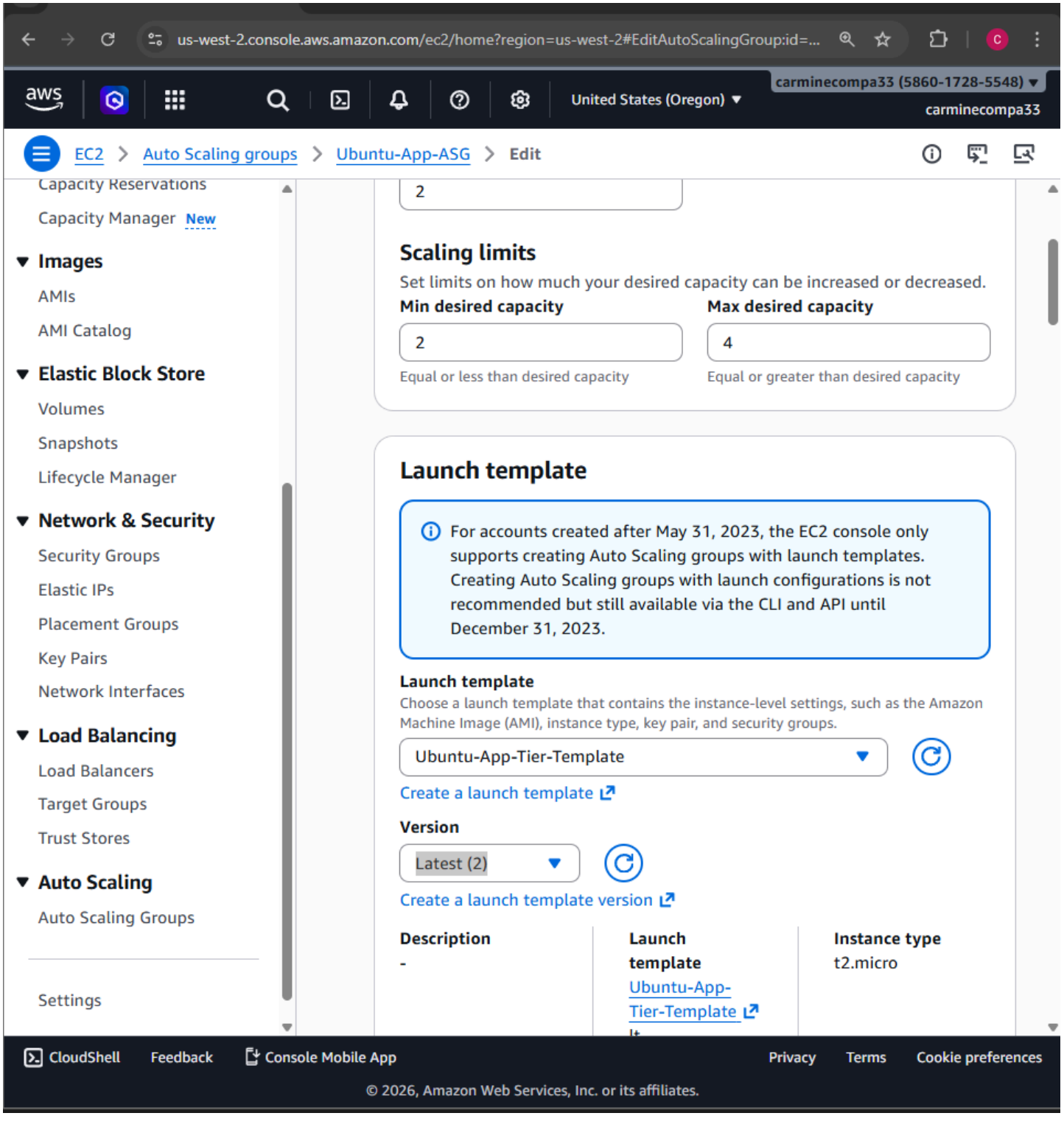
Create IAM role for SSH Session Manager



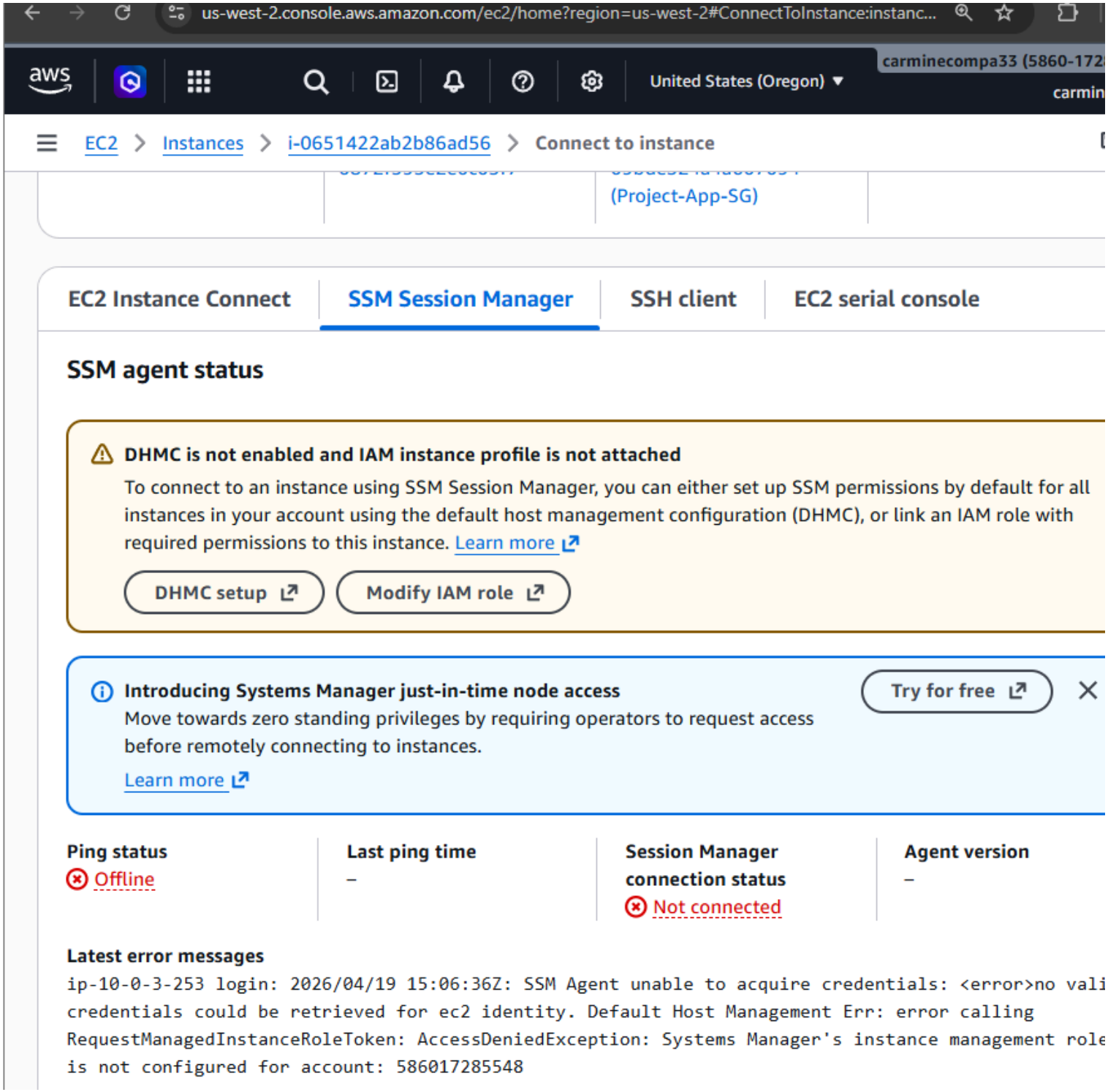
Update launch template with new IAM Profile



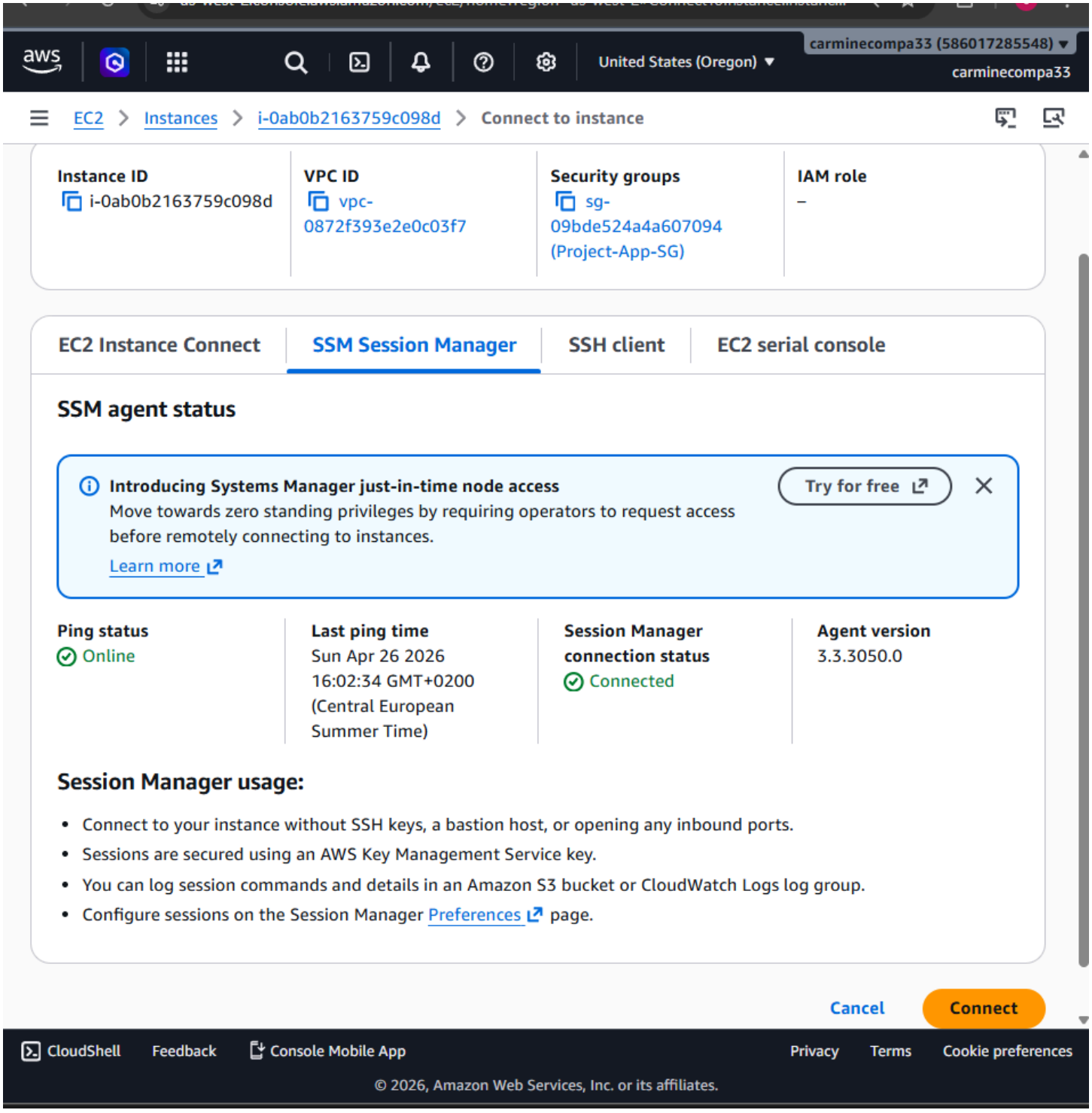
Update Launch Template in the Auto Scaling Group



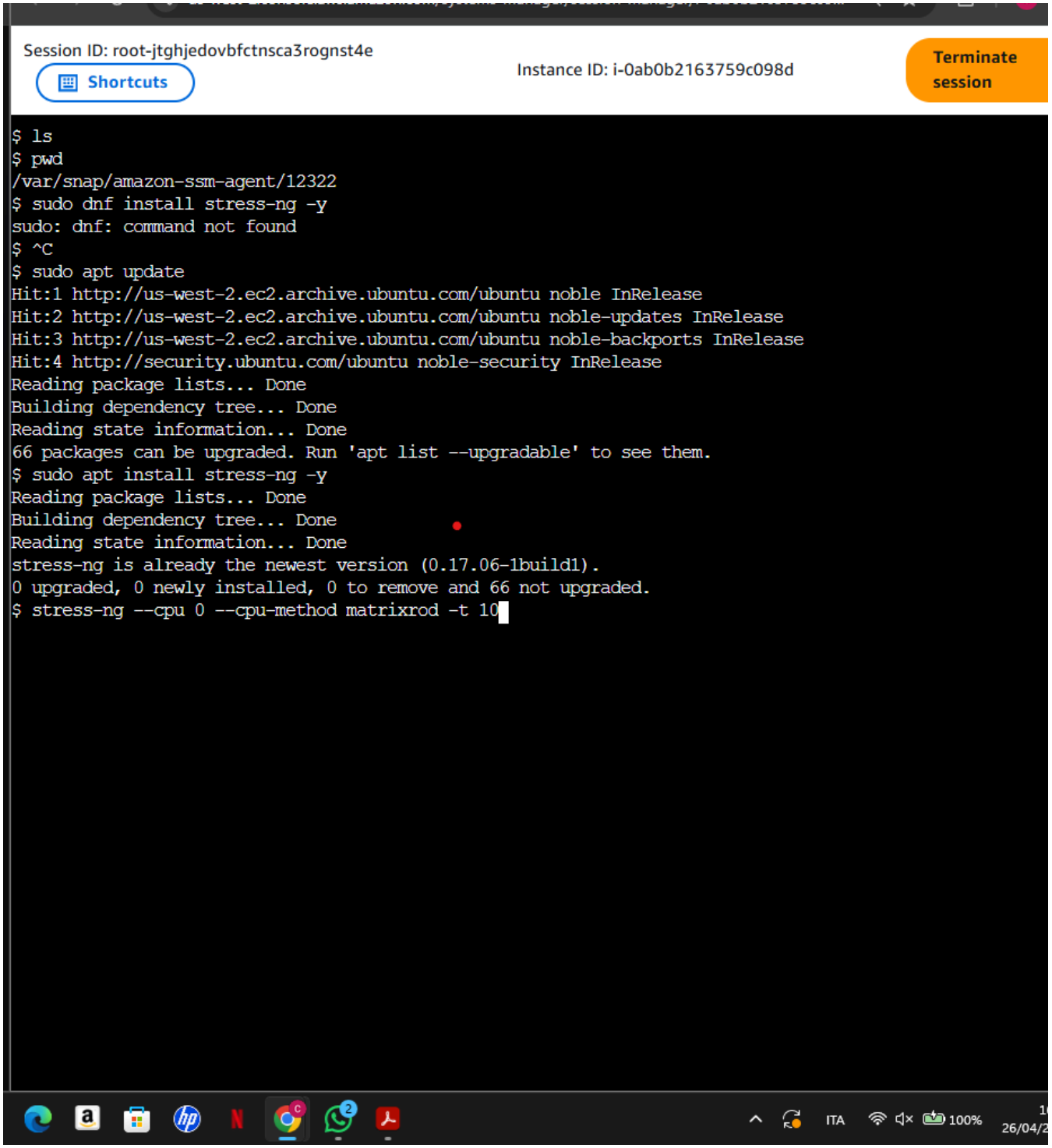
Error while launching the new EC2 instances with SSM Session Manager



SSM Session Manager is healthy now



Fire up the CPU spike



Session ID: root-jtghjedovbfctnsca3rognst4e

Shortcuts

Instance ID: i-0ab0b2163759c098d

Terminate session

```
$ ls
$ pwd
/var/snap/amazon-ssm-agent/12322
$ sudo dnf install stress-ng -y
sudo: dnf: command not found
$ ^C
$ sudo apt update
Hit:1 http://us-west-2.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://us-west-2.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://us-west-2.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
66 packages can be upgraded. Run 'apt list --upgradable' to see them.
$ sudo apt install stress-ng -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
stress-ng is already the newest version (0.17.06-1build1).
0 upgraded, 0 newly installed, 0 to remove and 66 not upgraded.
$ stress-ng --cpu 0 --cpu-method matrixrod -t 10
cpu-method must be one of: all ackermann apery bitops callfunc cdouble cfloat clongdouble collatz correl
ate crc16 dither div8 div16 div32 div64 div128 double euler explog factorial fibonacci fft fletcher16 fl
oat floatconversion gamma gcd gray hamming hanoi hyperbolic idct int128 int64 int32 int16 int8 int128flo
at int128double int128longdouble int64float int64double int64longdouble int32float int32double int32long
double intconversion ipv4checksum jmp lfsr32 ln2 logmap longdouble loop matrixprod nsqrt omega parity ph
i pi prime psi queens rand rand48 rgb sieve stats sqrt trig union zeta
$ stressng --cpu 0 --cpu-method matrixprod -t 10
sh: 7: stressng: not found
$ stress-ng --cpu 0 --cpu-method matrixprod -t 10
aborting: temp-path '.' must be readable and writeable
$ ^C
$ pwd
/var/snap/amazon-ssm-agent/12322
$ cd ~
$ pwd
/home/ssm-user
$ stress-ng --cpu 0 --cpu-method matrixprod -t 10m
stress-ng: info:  [9162] setting to a 10 mins, 0 secs run per stressor
stress-ng: info:  [9162] dispatching hogs: 15 cpu
```

ITA 100% 16:1 26/04/202

CPU usage is 100%

Session ID: root-xa3fdrxaqzchbnfdh87ssqe93u

Shortcuts

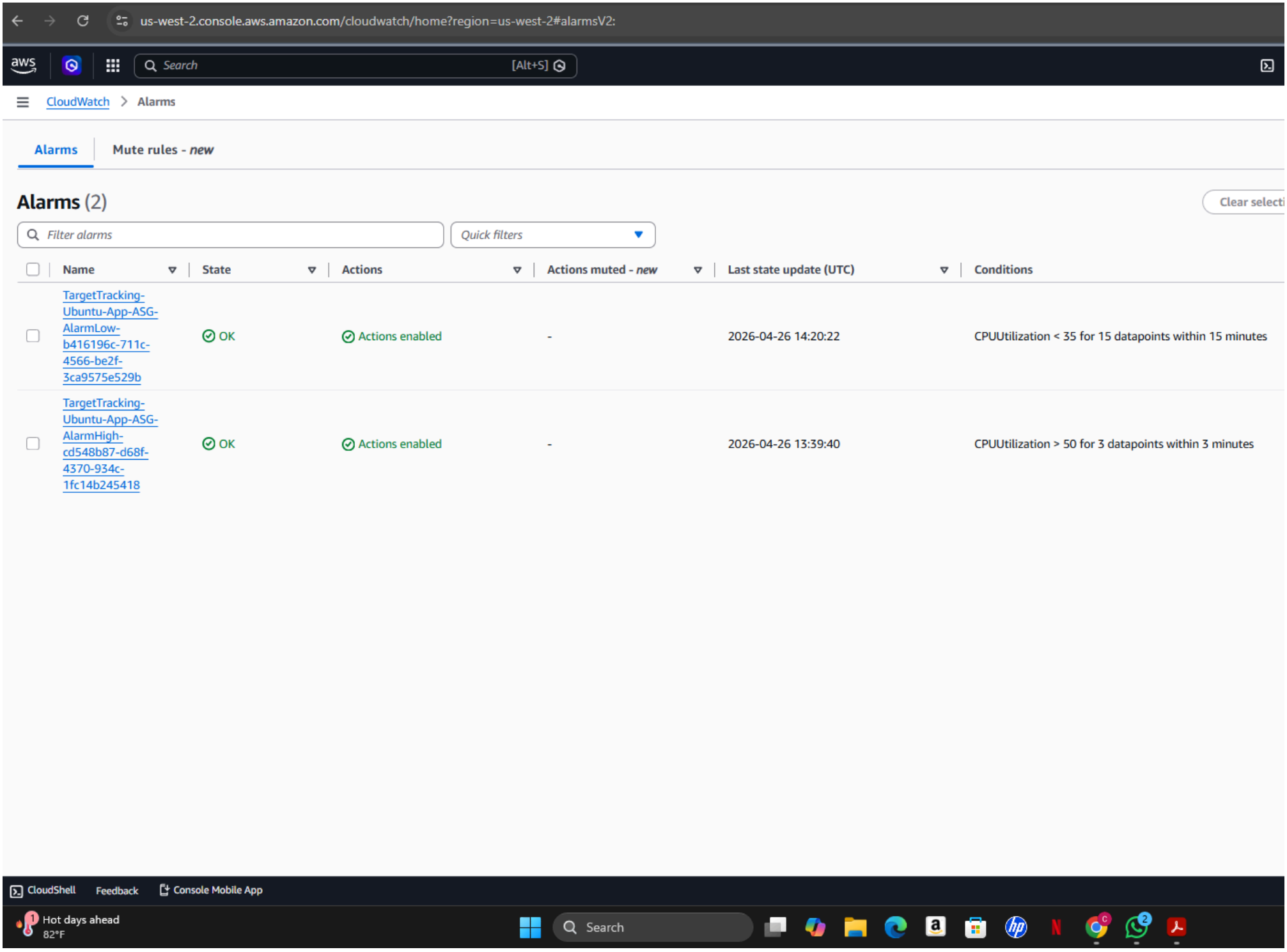
Instance ID: i-0ab0b2163759c098d

Terminate session

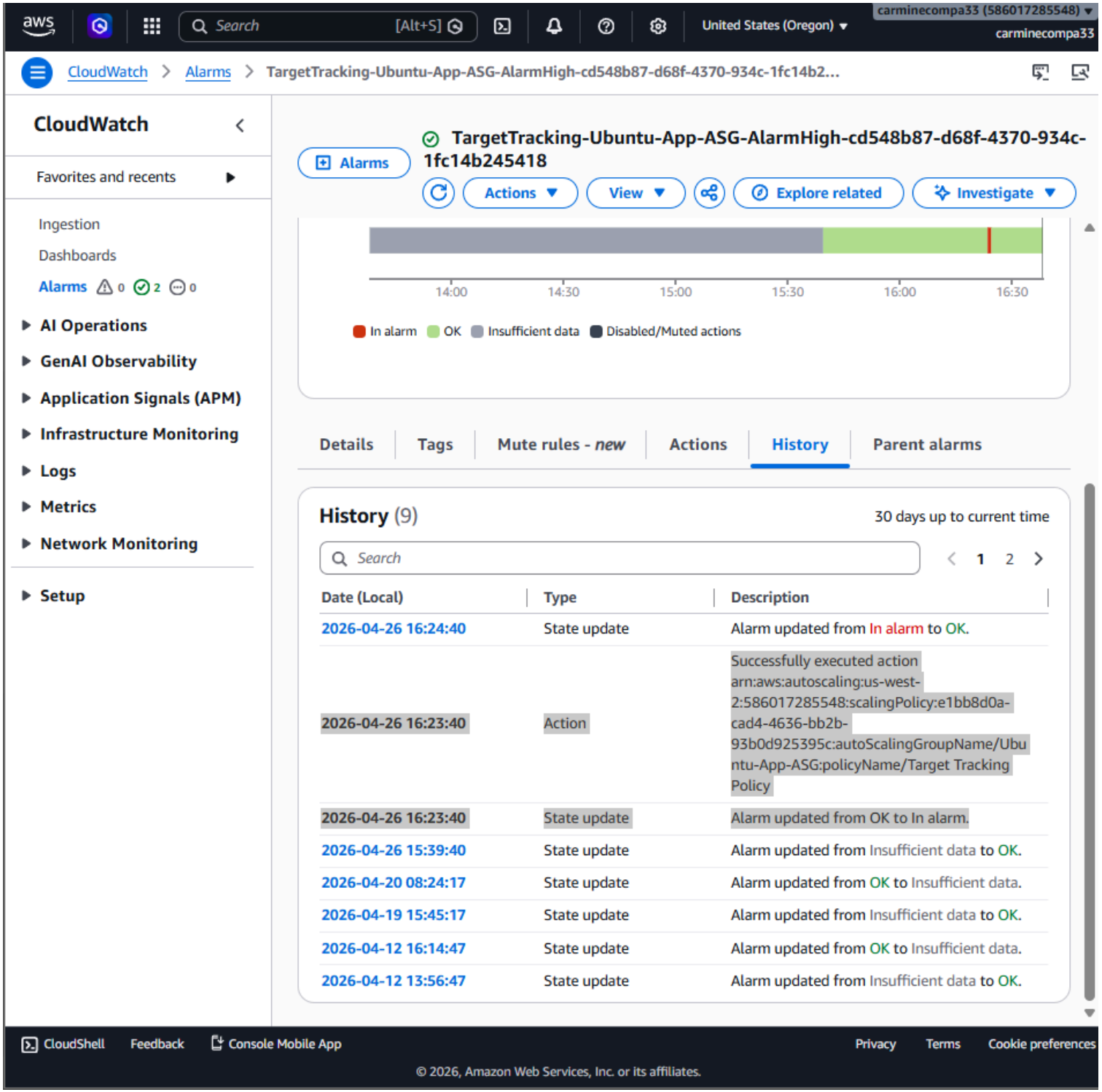
```
CPU[||||||||||||||||||||||||||||||||||100.0%] Tasks: 55, 68 thr, 75 kthr; 1 running
Mem[|||||||||||||||||||||||||282M/955M] Load average: 13.88 6.09 2.32
Swp[                          0K/0K] Uptime: 00:45:06

  Main   I/O
    PID USER       PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
   9171 ssm-user    20   0 90532  3584  1444 R   7.3  0.4  0:10.21 stress-ng-cpu [run]
   9176 ssm-user    20   0 90532  3584  1444 R   7.3  0.4  0:10.21 stress-ng-cpu [run]
   9177 ssm-user    20   0 90532  3584  1444 R   7.3  0.4  0:10.21 stress-ng-cpu [run]
   9163 ssm-user    20   0 90532  3584  1444 R   6.6  0.4  0:10.21 stress-ng-cpu [run]
   9164 ssm-user    20   0 90532  3584  1444 R   6.6  0.4  0:10.21 stress-ng-cpu [run]
   9165 ssm-user    20   0 90532  3584  1444 R   6.6  0.4  0:10.21 stress-ng-cpu [run]
   9166 ssm-user    20   0 90532  3584  1444 R   6.6  0.4  0:10.21 stress-ng-cpu [run]
   9168 ssm-user    20   0 90532  3584  1444 R   6.6  0.4  0:10.21 stress-ng-cpu [run]
   9169 ssm-user    20   0 90532  3584  1444 R   6.6  0.4  0:10.21 stress-ng-cpu [run]
   9170 ssm-user    20   0 90532  3584  1444 R   6.6  0.4  0:10.21 stress-ng-cpu [run]
   9173 ssm-user    20   0 90532  3584  1444 R   6.6  0.4  0:10.21 stress-ng-cpu [run]
   9174 ssm-user    20   0 90532  3584  1444 R   6.6  0.4  0:10.20 stress-ng-cpu [run]
   9167 ssm-user    20   0 90532  3584  1444 R   6.0  0.4  0:10.20 stress-ng-cpu [run]
   9172 ssm-user    20   0 90532  3584  1444 R   6.0  0.4  0:10.20 stress-ng-cpu [run]
   9175 ssm-user    20   0 90532  3584  1444 R   6.0  0.4  0:10.20 stress-ng-cpu [run]
   9205 ssm-user    20   0  8756  4636  3548 R   0.7  0.5  0:00.11 htop
      1 root        20   0 22508 13684  9448 S   0.0  1.4  0:05.49 /sbin/init
    118 root        19  -1 50500 14136 12948 S   0.0  1.4  0:02.88 /usr/lib/systemd/systemd-journald
    179 root        RT   0  282M 27244  8712 S   0.0  2.8  0:00.03 /sbin/multipathd -d -s
    190 root        20   0 26496  8080  4912 S   0.0  0.8  0:00.20 /usr/lib/systemd/systemd-udevd
    191 root        20   0  282M 27244  8712 S   0.0  2.8  0:00.00 /sbin/multipathd -d -s
    192 root        RT   0  282M 27244  8712 S   0.0  2.8  0:00.00 /sbin/multipathd -d -s
    193 root        RT   0  282M 27244  8712 S   0.0  2.8  0:00.00 /sbin/multipathd -d -s
    194 root        RT   0  282M 27244  8712 S   0.0  2.8  0:00.00 /sbin/multipathd -d -s
    195 root        RT   0  282M 27244  8712 S   0.0  2.8  0:00.12 /sbin/multipathd -d -s
    196 root        RT   0  282M 27244  8712 S   0.0  2.8  0:00.00 /sbin/multipathd -d -s
    335 systemd-re  20   0 21600 12896 10540 S   0.0  1.3  0:00.14 /usr/lib/systemd/systemd-resolved
    497 systemd-ne  20   0 22420  9836  8660 S   0.0  1.0  0:00.03 /usr/lib/systemd/systemd-networkd
    559 root        20   0  2728  1952  1808 S   0.0  0.2  0:00.00 /usr/sbin/acpid
    563 root        20   0  7232  2760  2500 S   0.0  0.3  0:00.00 /usr/sbin/cron -f -P
    564 messagebus  20   0  9796  5432  4564 S   0.0  0.6  0:00.29 @dbus-daemon --system --address=syste
    571 root        20   0 32424 20372 10148 S   0.0  2.1  0:00.08 /usr/bin/python3 /usr/bin/networkd-di
    574 polkitd     20   0  374M  9724  7296 S   0.0  1.0  0:00.05 /usr/lib/polkit-1/polkitd --no-debug
    602 root        20   0 1813M 36044 22984 S   0.0  3.7  0:00.07 /usr/lib/snapd/snapd
F1Help  F2Setup  F3SearchF4FilterF5Tree   F6SortByF7Nice -F8Nice +F9Kill   F10Quit
```

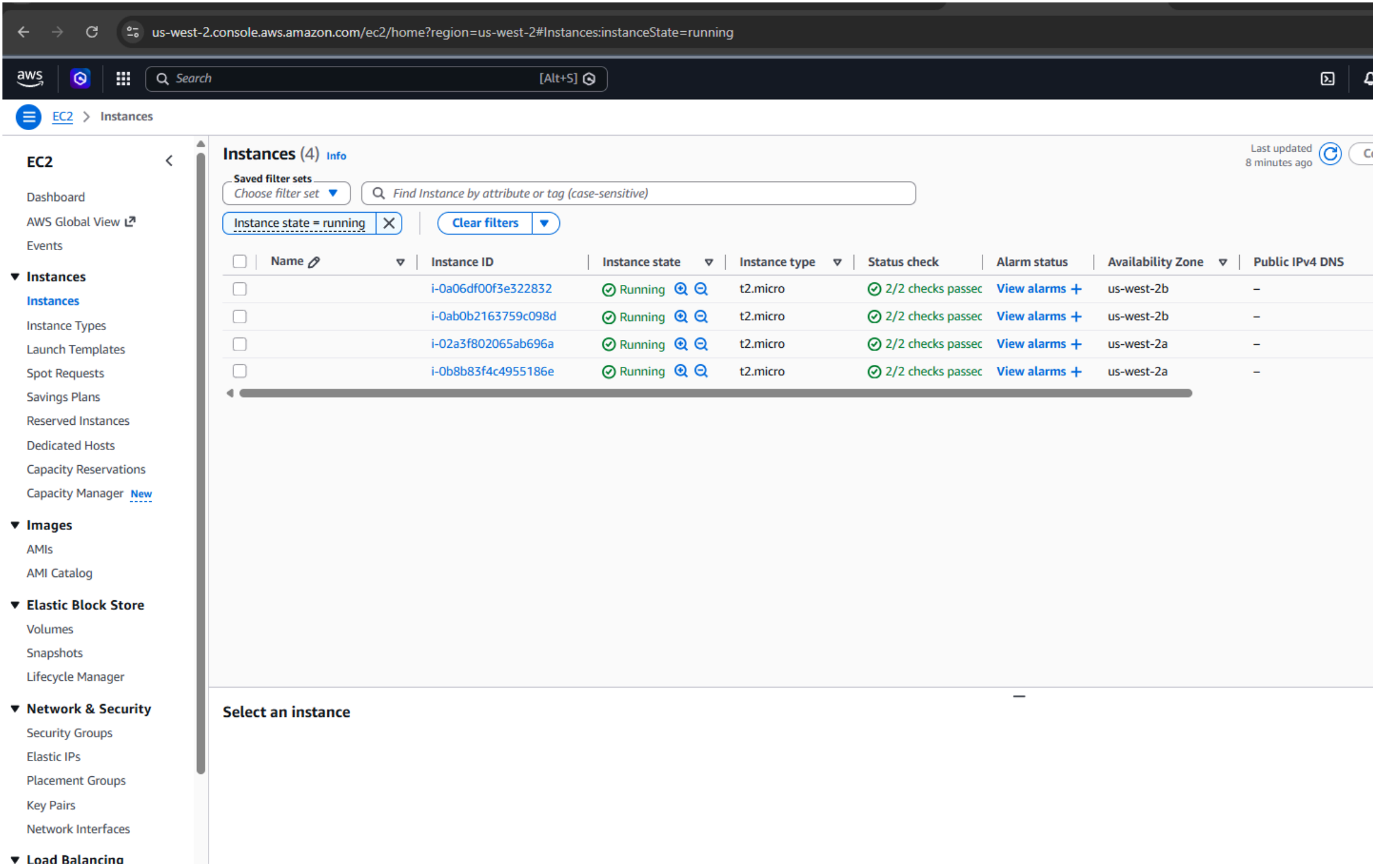
Cloud Watch will wait 3x60 seconds before triggering the alert



CloudWatch triggered in alert

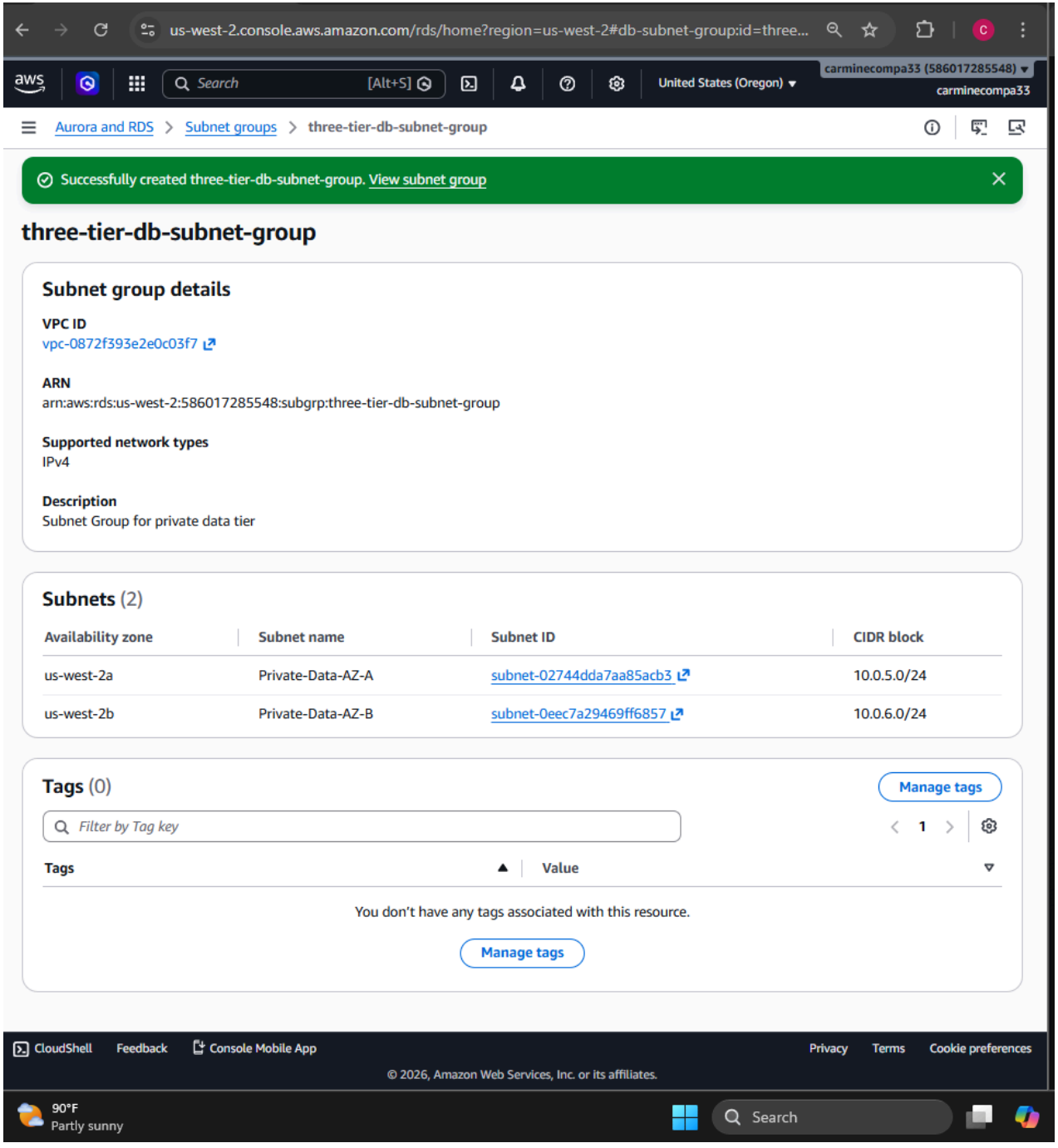


EC2 instances scaled out from 2 to 4



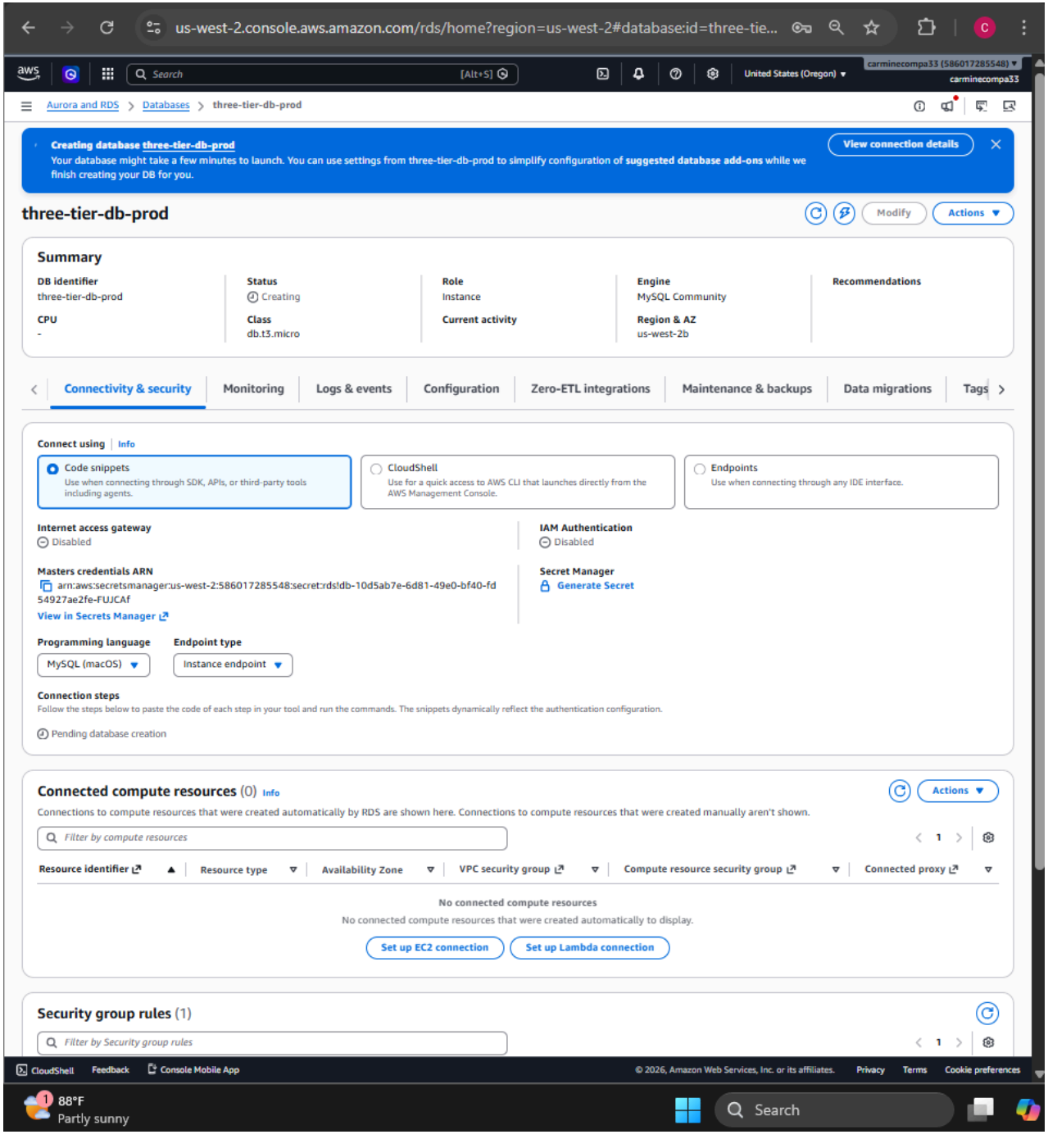
Step 6: The Data Tier (The "Vault")

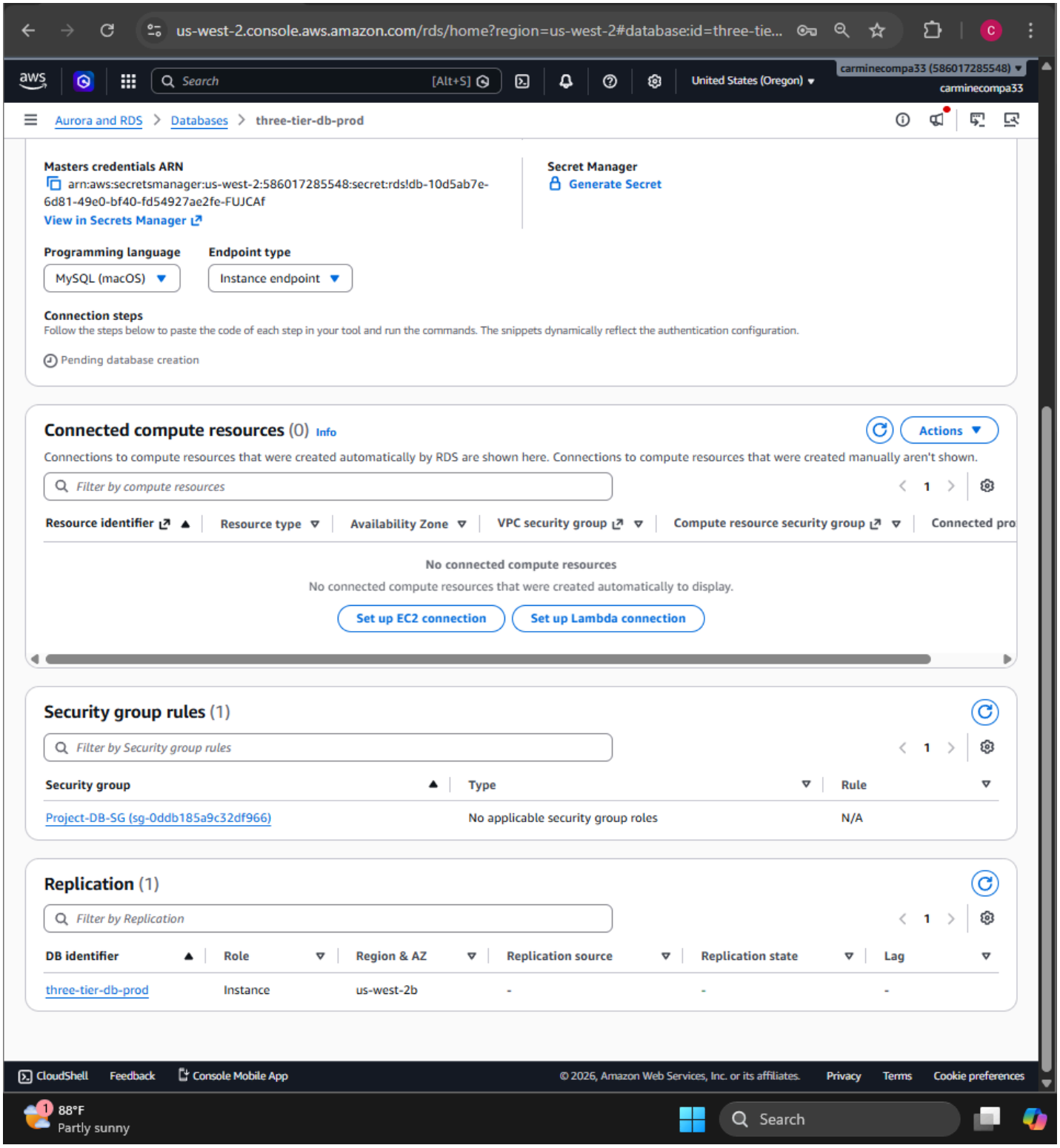
Create RDS Subnet Group



Creating the High-Availability Database

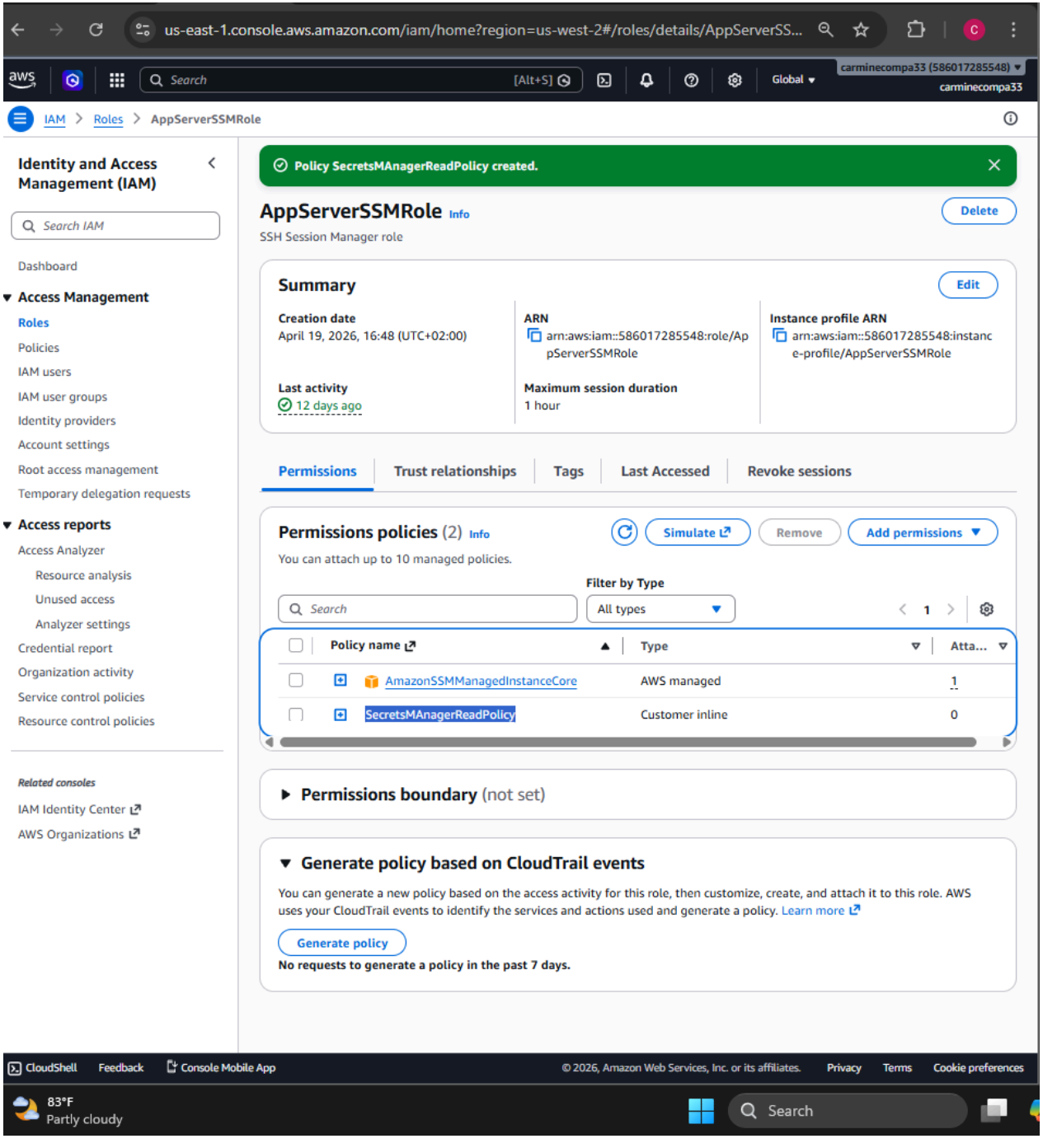
Deploy a MySQL RDS instance using Multi-AZ for high availability and automatic failover across Availability Zones. This step secures the Data Tier by disabling public access and isolating the database within private subnets. By integrating AWS Secrets Manager, we move away from hardcoded passwords toward a "Cloud-Native" security model, ensuring robust data protection.

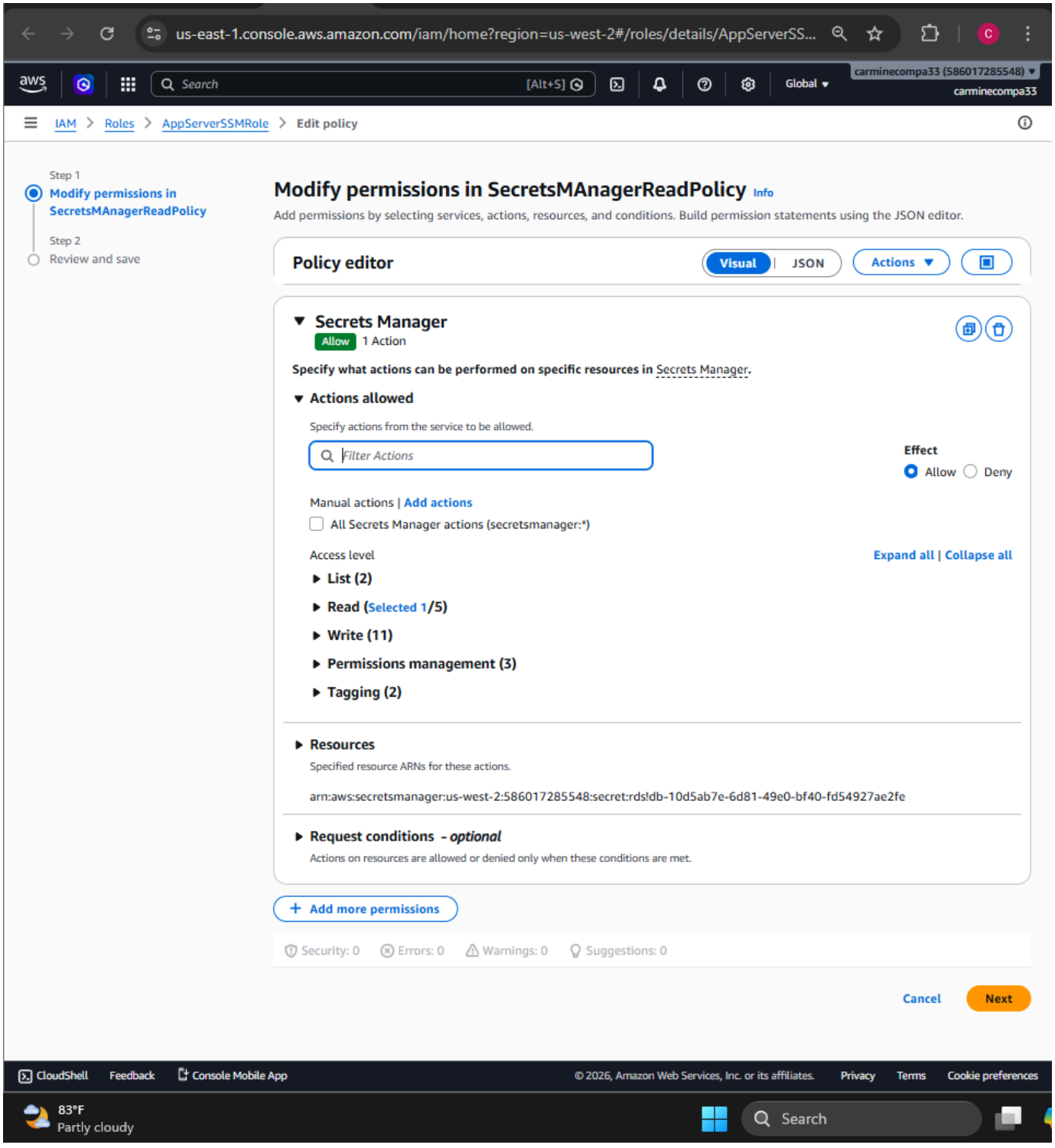



us-west-2.console.aws.amazon.com/rds/home?region=us-west-2#database:id=three-tie...
Search
[Alt+S]
United States (Oregon)
carminecompa33 (586017285548)
carminecompa33
Aurora and RDS
Databases
three-tier-db-prod
Masters credentials ARN
arn:aws:secretsmanager:us-west-2:586017285548:secret:rds!db-10d5ab7e-6d81-49e0-bf40-fd54927ae2fe-FUJCAf
View in Secrets Manager
Secret Manager
Generate Secret
Programming language
MySQL (macOS)
Endpoint type
Instance endpoint
Connection steps
Follow the steps below to paste the code of each step in your tool and run the commands. The snippets dynamically reflect the authentication configuration.
Pending database creation
Connected compute resources (0) Info
Actions
Connections to compute resources that were created automatically by RDS are shown here. Connections to compute resources that were created manually aren't shown.
Filter by compute resources
Resource identifier
Resource type
Availability Zone
VPC security group
Compute resource security group
Connected pro
No connected compute resources
No connected compute resources that were created automatically to display.
Set up EC2 connection
Set up Lambda connection
Security group rules (1)
Filter by Security group rules
Security group
Type
Rule
Project-DB-SG (sg-0ddb185a9c32df966)
No applicable security group roles
N/A
Replication (1)
Filter by Replication
DB identifier
Role
Region & AZ
Replication source
Replication state
Lag
three-tier-db-prod
Instance
us-west-2b
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
88°F
Partly sunny
Search

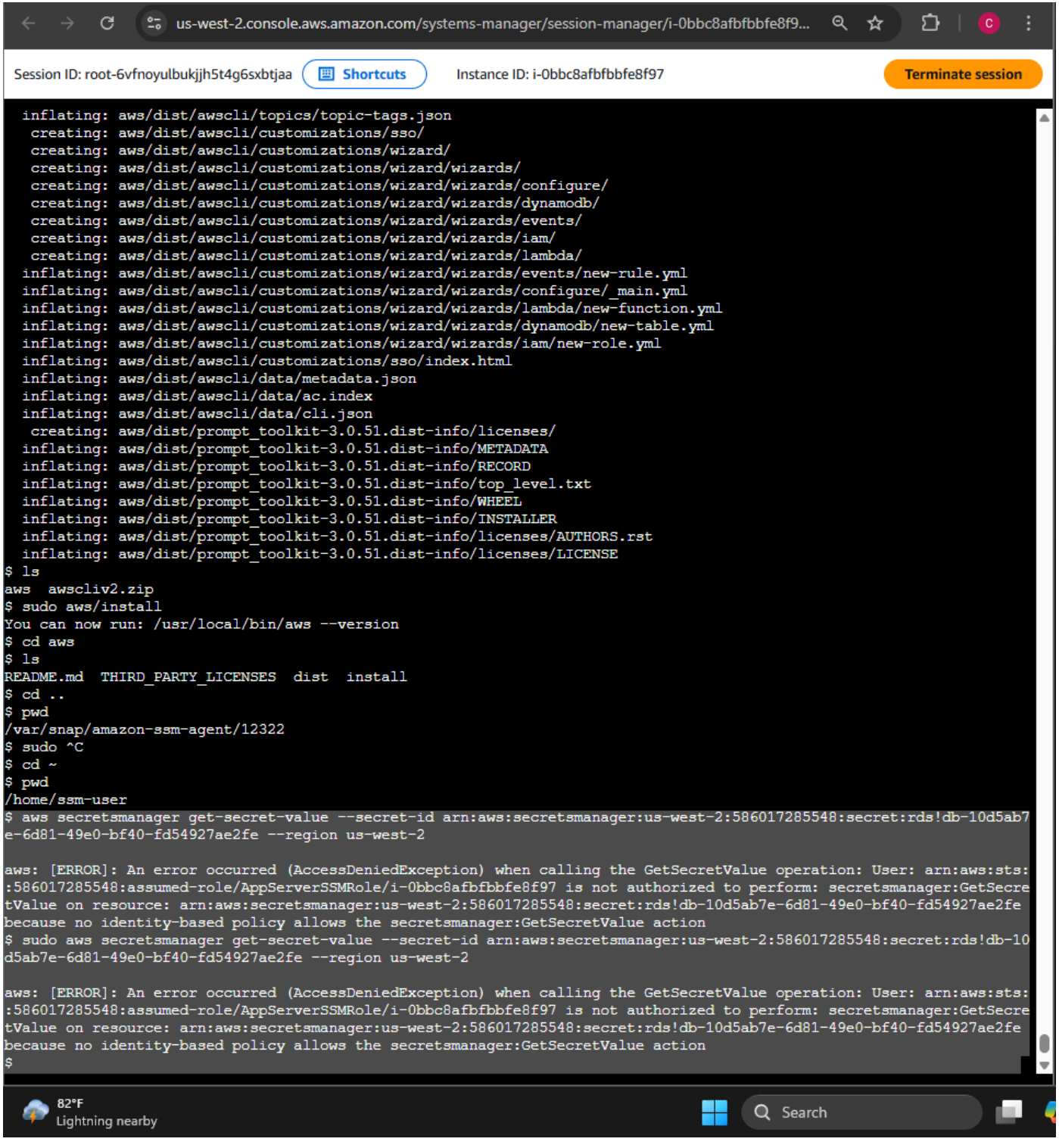
Step 7: Enhanced Security with Secrets Manager

Add AWS Secret Manager policy to IAM Role “AppServerSSMRole”

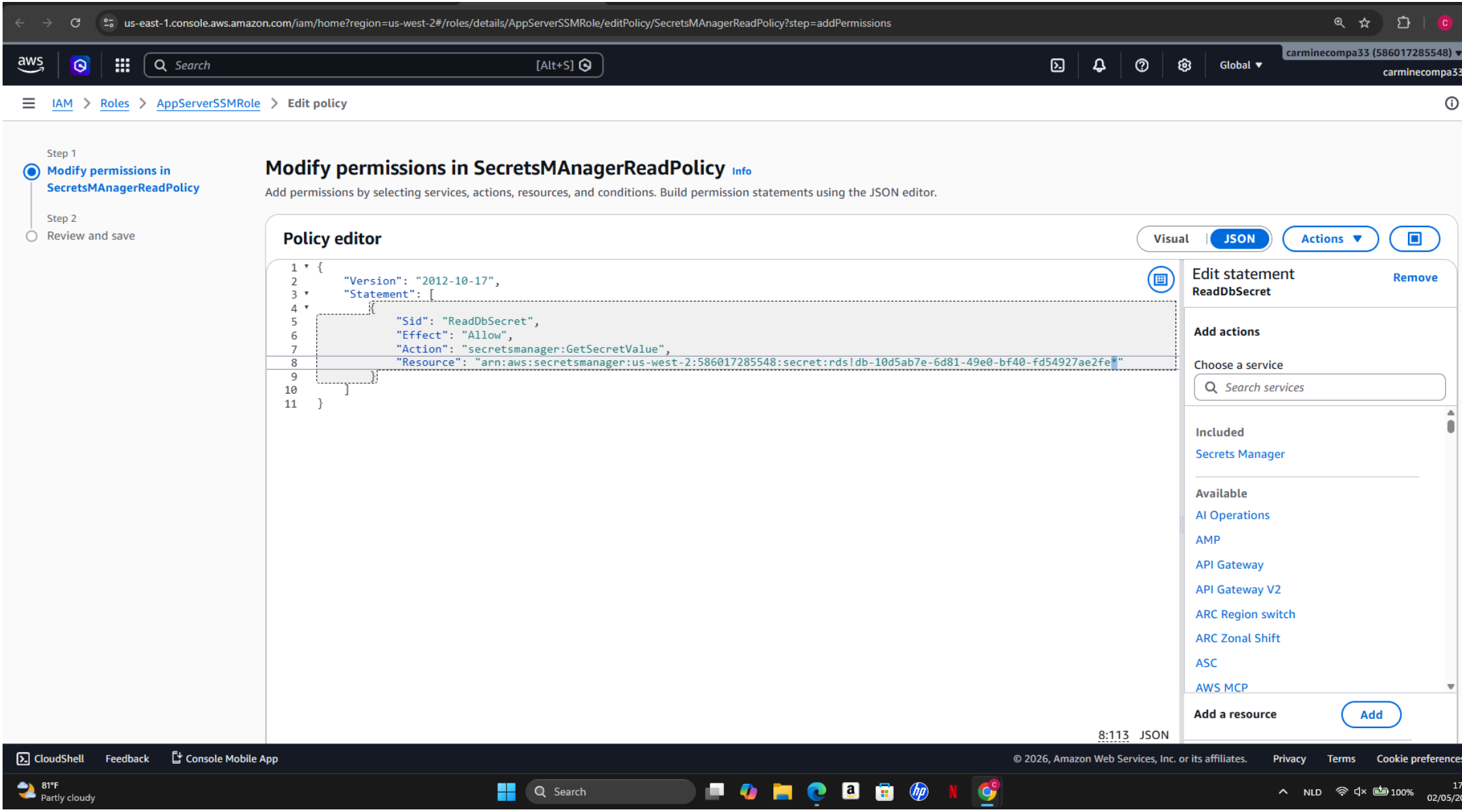



us-east-1.console.aws.amazon.com/iam/home?region=us-west-2#/roles/details/AppServerSS...
Search
[Alt+S]
Global
carminecompa33 (586017285548)
carminecompa33
IAM > Roles > AppServerSSMRole > Edit policy
Step 1
Modify permissions in SecretsMAnagerReadPolicy
Step 2
Review and save
Modify permissions in SecretsMAnagerReadPolicy Info
Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.
Policy editor
Visual
JSON
Actions
Secrets Manager
Allow 1 Action
Specify what actions can be performed on specific resources in Secrets Manager.
Actions allowed
Specify actions from the service to be allowed.
Filter Actions
Effect
Allow
Deny
Manual actions | Add actions
All Secrets Manager actions (secretsmanager:*)
Access level
Expand all | Collapse all
List (2)
Read (Selected 1/5)
Write (11)
Permissions management (3)
Tagging (2)
Resources
Specified resource ARNs for these actions.
arn:aws:secretsmanager:us-west-2:586017285548:secret:rds!db-10d5ab7e-6d81-49e0-bf40-fd54927ae2fe
Request conditions - optional
Actions on resources are allowed or denied only when these conditions are met.
Add more permissions
Security: 0
Errors: 0
Warnings: 0
Suggestions: 0
Cancel
Next
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences
83°F
Partly cloudy
Search

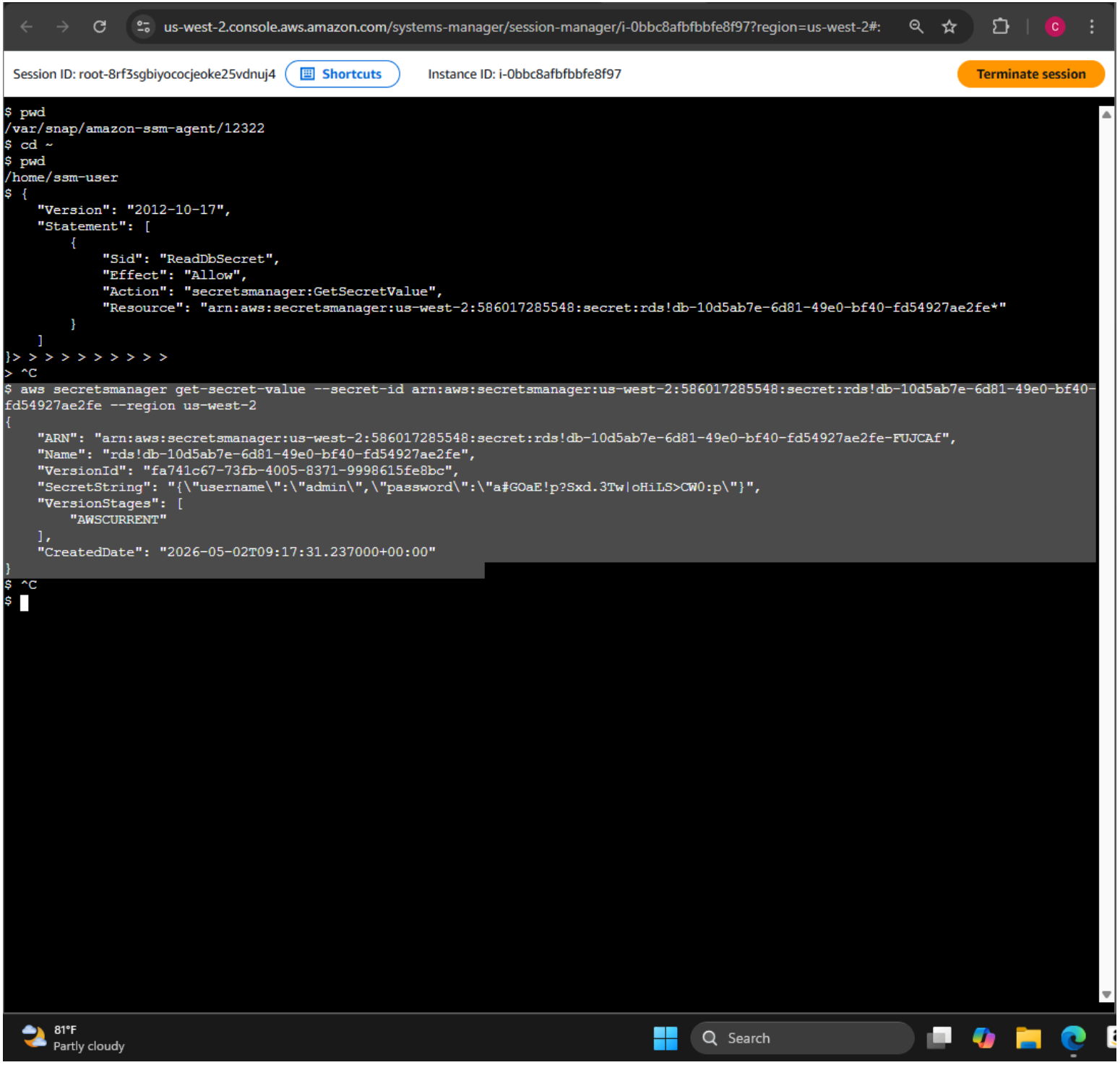
aws secretsmanager get-secret-value --secret-id <ARN-RESOURCE> --region us-west-2 is failing



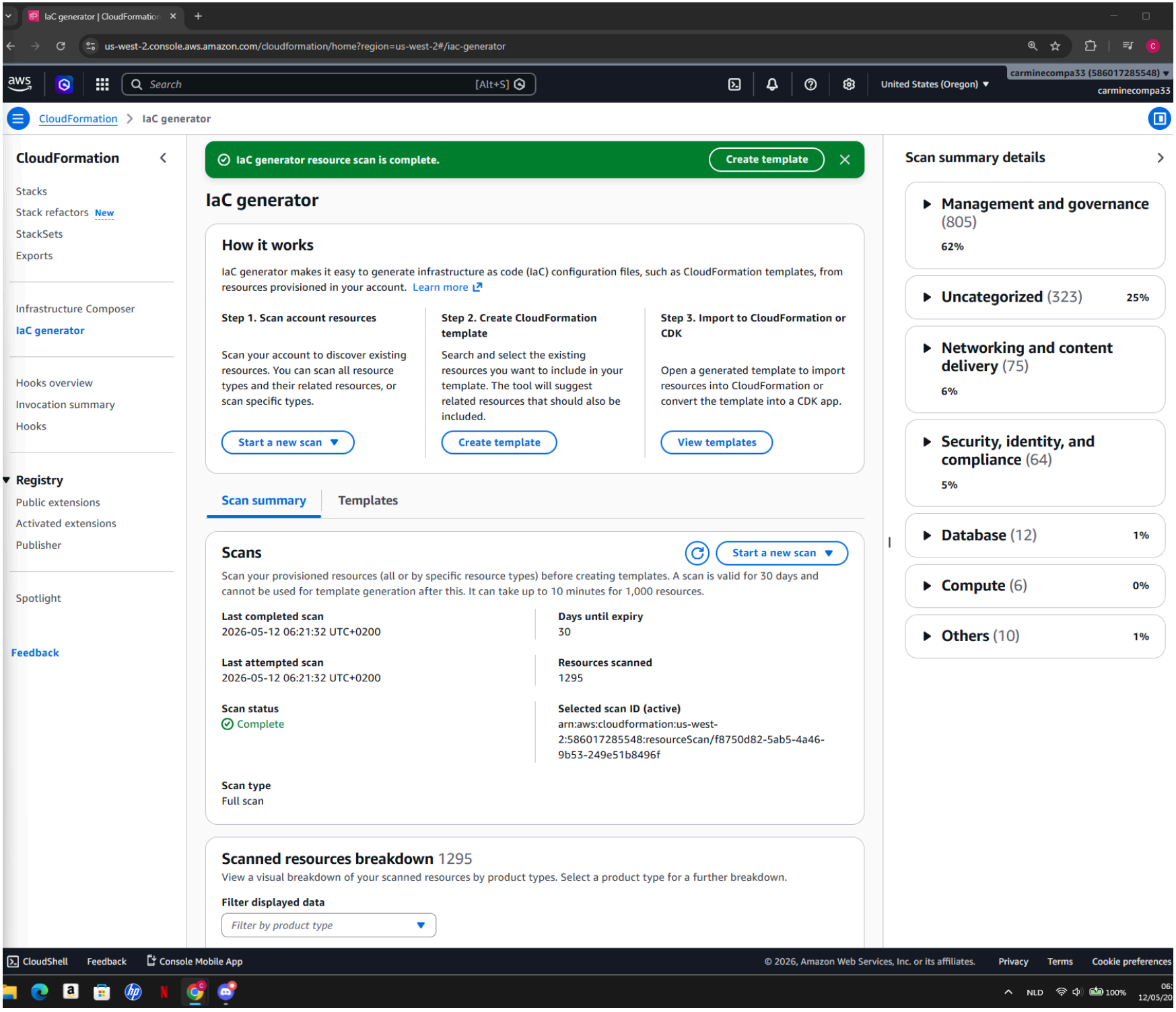
Added * to the role policy of the IAM role



aws secretsmanager get-secret-value --secret-id <ARN-RESOURCE> --region us-west-2 is now successful



Step 8: save AWS Infrastructure with CloudFormation - IaC generator



Step 1
Specify template details
Step 2
Add scanned resources
Step 3
Add related resources
Step 4
Review and create

Review and create

Step 1: Specify template details

Edit

Template name
MyResilientProject

Deletion policy
RETAIN

Update replace policy
RETAIN

Base stack name
-

Base stack ID
-

Step 2: Add scanned resources

Edit

Scanned resources

Filter resources by ID

1

Resource identifier	Resource type	Managed by stack
AppConfig.Canary10Percent20Minutes	AWS::AppConfig::DeploymentStrategy	No
AppConfig.Linear20PercentEvery6Minutes	AWS::AppConfig::DeploymentStrategy	No
AppConfig.Linear50PercentEvery30Seconds	AWS::AppConfig::DeploymentStrategy	No
AppConfig.AllAtOnce	AWS::AppConfig::DeploymentStrategy	No
AWS.Evidently.ProjectImporter	AWS::AppConfig::Extension	No
AWS.AppConfig.FeatureFlags	AWS::AppConfig::Extension	No
AWS.AppConfig.DeploymentNotificationsToEventBridge	AWS::AppConfig::Extension	No
AWS.AppConfig.DeploymentNotificationsToSqs	AWS::AppConfig::Extension	No
AWS.AppConfig.DeploymentNotificationsToSns	AWS::AppConfig::Extension	No
AWS.AppConfig.JiraIntegration	AWS::AppConfig::Extension	No
arn:aws:apprunner:us-west-		

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You have successfully created the template. Next, you can migrate to CDK or import to a stack.

MyResilientProject

Edit Delete

Template details

Template generation status
Complete

Template ID
arn:aws:cloudformation:us-west-2:586017285548:generatedTemplate/0288cc46-f73b-45b8-a5c3-5e88b92d06d3

Creation time
2026-05-12 06:27:57 UTC+0200

Updated time
2026-05-12 06:28:02 UTC+0200

Configurations

Deletion policy
RETAIN

Update replace policy
RETAIN

Built off existing stack
-

Share your feedback on the generated template. Give feedback

Template definition | Template resources | AWS CDK

Template

YAML Download Copy template Import to stack

The template contains properties that can be replaced with other mutually exclusive properties. Additionally, some optional properties were not included. You can download the template and edit it to modify the properties to suit your needs. Learn more

View warning details Import edited template

Template | Canvas

Copy

```
---
Metadata:
  AWSToolsMetrics:
    IaC_Generator: "arn:aws:cloudformation:us-west-2:586017285548:generatedTemplate/0288cc46-f73b-45b8-a5c3-5e88b92d06d3"
Parameters:
  EC2LaunchTemplateLaunchTemplateDataInstanceRequirementsMemoryGiBPerVCpuMinFfISH:
    NoEcho: "true"
    Type: "Number"
    Description: "The minimum amount of memory per vCPU, in GiB. To specify no minimum\
      \ limit, omit this parameter."
```


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